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43665347

2MVAa
Multivariable & Vector Analysis 1

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Outline of syllabus

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2MVAa is a general module, containing both "pure" and "applied" materials. It includes formal definitions, theorems, proofs, and problem-solving techniques. The syllabus is outlined as follows:

- Partial derivatives; differentiability of functions; the chain rule; implicit function theorem; higher-order partial derivatives; gradient vectors; directional derivatives; and the theorem of steepest ascent.
- Double and triple integrals in Cartesian coordinates; change of variables using the Jacobian; double integrals in polar coordinates; and triple integrals in cylindrical and spherical coordinates.
- Multivariable Taylor series. Stationary points; classification of maxima and minima for multivariable functions; the leading principal minor test; and the method of Lagrange multipliers.

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Learning materials

- Recorded lectures, notes, exercise sheets and solutions are provided online.
- If you find any errors in the notes, exercise sheets, or solutions, please email corrections to: q.x.wang@bham.ac.uk.
- Recommended reference book for 2MVAa is

  Adams, R. A. Calculus: a complete course, Addison Wesley.

  The following two books also cover most of the material in 2MVAa and are worth having a look at.

  Thomas, R. L. Calculus. Addison Wesley.

  Kreyszig, E. Advanced Engineering Mathematics. Wiley.
- Past exam papers can be accessed at

  https://canvas.bham.ac.uk/courses/66707/files

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Assessment

- The end term exam 80% + In-course Assessment 20%
- In-course Assessment for the first half (10%) consists of Problem Sheets 1 & 2 (each 5%).
- Deadline for Problem Sheet 1 is: 1700, Wed. 23 Oct.
- Deadline for Problem Sheet 2 is: 1700, Wed. 6 Nov.
- Submit your answers online using Canvas.
- Exercise sheets 1- 4 are provided with solutions for practice and self assessment.

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Study Suggestions for 2MVAa

- Read the lecture notes carefully to gain a thorough understanding of the material.
- Work through all examples in the notes independently — ensure you understand and can reproduce each step.
- Complete the exercise sheets:

  Focus on selected questions from Sheets 1 & 2

  Attempt all questions in Sheets 3 & 4 for full practice
- Make use of available support opportunities:

  Ask questions during guided study sessions

  Attend office hours for individual help and clarification

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I. Differential calculus of multivariable functions

1.1 Functions of several real variables (Adams, Section 12.1)

1.2 Partial derivatives (Adams, 12.3-4)

1.3 Differentiable functions (Adams, 12.5)

1.4 The chain rule (Adams, 12.6)

1.5 Vectors and vector geometry (Adams 10.2-4)

1.6 Gradient vectors and directional derivatives (Adams 12.7)

Adams, R. A. Calculus: a complete course, Addison Wesley. 7^{th} edition

This chapter is across 8 lectures for Weeks 1 & 2.

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1.1 Functions of several real variables

A function is a **rule** that assigns each element in a domain D to **exactly** one element in a codomain C.

A function of a single variable: $f: D \subseteq \mathbb{R} \to C \subseteq \mathbb{R}$.

A function f of n variables, for $n \in \mathbb{N}$, is denoted as

$$f: D \subseteq \mathbb{R}^n \to C \subseteq \mathbb{R}.$$

Here $\mathbb{R}^n$ is the n dimensional real space: $\mathbb{R}^n \equiv \{(x_1, x_2, \ldots, x_n): x_i \in \mathbb{R}\}$.

For convenience, we often just write it as

$$f: \mathbb{R}^n \to \mathbb{R}.$$

We will mainly study functions with two and three variables:

$$f(x, y): D \subseteq \mathbb{R}^2 \to C \subseteq \mathbb{R},$$

where $\mathbb{R}^2$ is the real plane: $\mathbb{R}^2 \equiv \{(x, y): x, y \in \mathbb{R}\}$.

$$f(x, y, z): D \subseteq \mathbb{R}^3 \to C \subseteq \mathbb{R}.$$

where $\mathbb{R}^3$ is the real space: $\mathbb{R}^3 \equiv \{(x, y, z): x, y, z \in \mathbb{R}\}$.

Example 1

Determine the domain D of the function,

$$f = \ln\left(4 - x^2 - y^2\right) + \sqrt{x^2 + 2x + y^2}$$

Soln. If the domain is not explicitly specified, it is understood to be the set of all points where the function is definable. From the properties of the logarithm, we have

$$4 - x^2 - y^2 > 0 \quad \Leftrightarrow \quad x^2 + y^2 < 2^2.$$

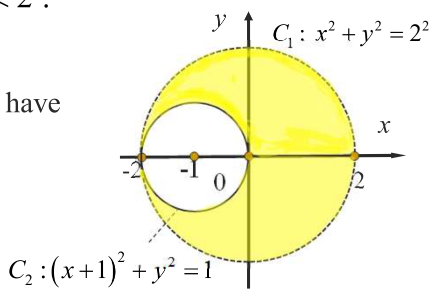
D is inside $C_1: x^2 + y^2 = 2^2$.

From the domain of square root, we have

$$x^2 + 2x + y^2 \geq 0$$

$$x^2 + 2x + 1 + y^2 \geq 1$$

$$(x+1)^2 + y^2 \geq 1$$



D is external and includes $C_2: (x+1)^2 + y^2 = 1$.

Graphs of functions

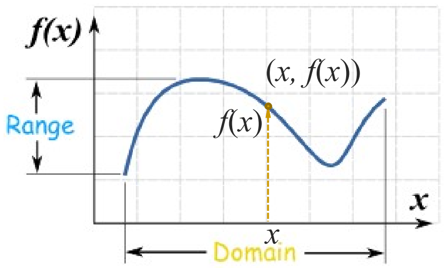
A real function of a single variable is denoted as

$$y = f(x),\ \ x \in D \subseteq \mathbb{R},\ \ \ y \in C \subseteq \mathbb{R}.$$

The graph of $y = f(x)$ is a curve in the real plane $\mathbb{R}^2$, given by:

For each x in the domain D, we draw a vertical line whose height is $f(x)$. We thus determine a point $(x, f(x))$ in the real plane. As x runs through the entire domain D, the corresponding points $(x, f(x))$ form a curve.

The graph of a single-variable function is a curve in the real plane.



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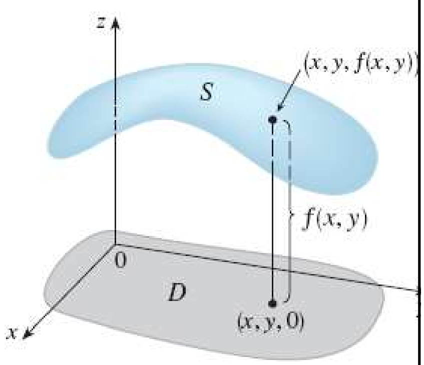
Graph for real functions of two variables: $z = f(x, y)$

For each point (x, y) in the domain D, we draw a vertical line with its height equal to $f(x, y)$. We thus determine a point $(x, y, f(x, y))$ in the real space.

As the point (x, y) runs through the entire domain D, the corresponding points $(x, y, f(x, y))$ form a surface in space. This is the graph of the function $z = f(x, y)$.

A function $z = f(x, y)$, $(x, y) \in D$ is a surface in $\mathbb{R}^3$.

Similarly, a function with three variables defines a hypersurface in 4D real space, which cannot be directly visualized.



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Example 2

Sketch a graph for the function,

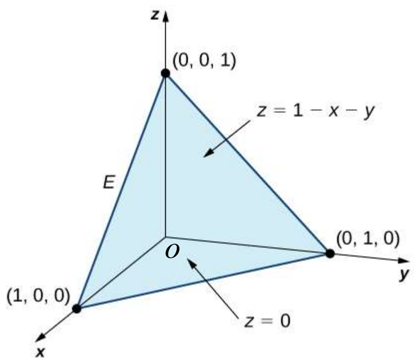
$z = 1\text{-}x\text{-}y.$

Soln. To sketch a plane in 3D, one common method is to find its intercepts with the x-, y-, and z-axes—like how lines are drawn in 2D.

Its intersection with the x- y- and z- axes are

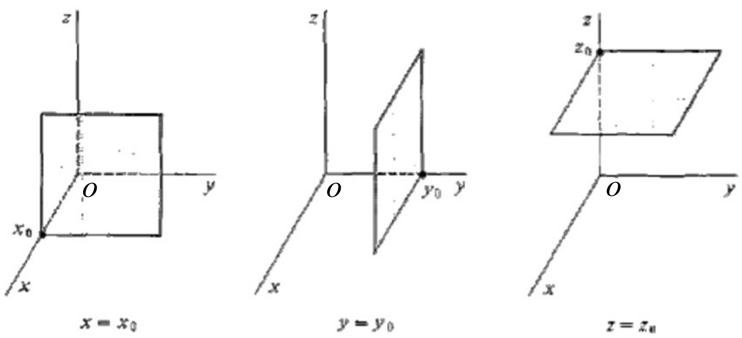
(1, 0, 0), (0, 1, 0) and (0, 0, 1).

Linking the three points displays the plane.



Example 3.

Sketch graphs for the functions: $x = x_0$, $y = y_0$, $z = z_0$.



Instead of using $x = x_0$, we often write $x = x$ for a plane parallel to the yz-plane that intersects the x-axis at a specific value of x.

$y = y$ and $z = z$ are are interpreted in the same way for planes parallel to the corresponding coordinate planes.

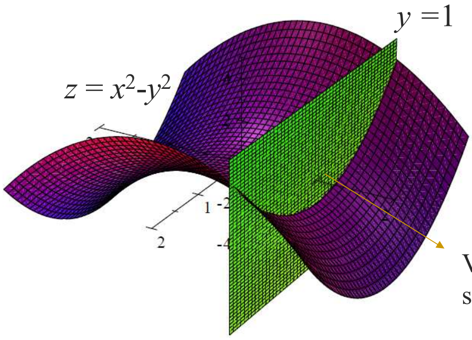
Vertical sections

For a function $z = f(x, y)$: A **vertical section** $z = f(x, c)$ is

$$\text{Intersection of } \begin{cases} z = f(x, y) & \text{(the surface)} \\ y = c & \text{(a vertical plane)} \end{cases}$$

$z = f(x, c)$ is a curve in the xz plane.

c is constant and its values should be chosen appropriately depending on the nature of functions.



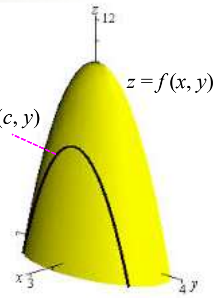
Level curves

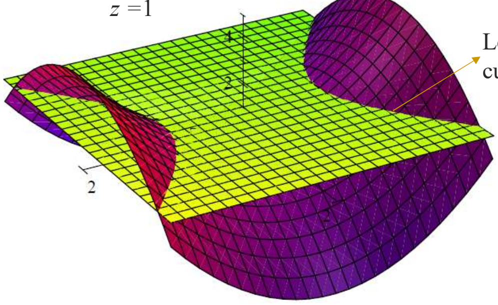
Vertical sections $z = f(c, y)$, (graphs in the yz plane), is

$$\text{Intersection of } \begin{cases} z = f(x, y) & \text{(the surface)} \\ x = c & \text{(a vertical plane)} \end{cases}$$

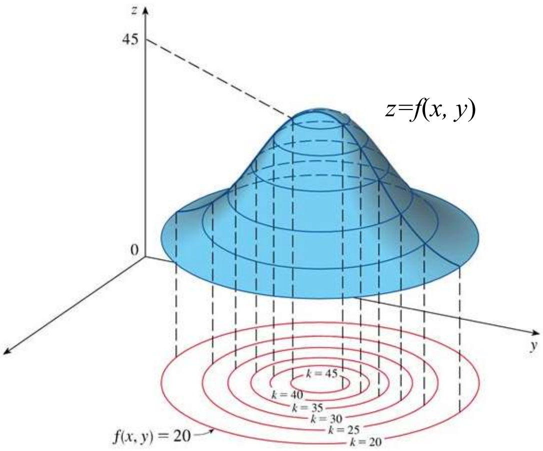
A level curve: $\boldsymbol{c = f(x, y)}$, (graphs in the xy plane), is

$$\text{Intersection of } \begin{cases} z = f(x, y) & \text{(the surface)} \\ z = c & \text{(a horizontal plane)} \end{cases}$$





Level curves



Level curves, or contours, are curves along which a surface has constant height.

They lie on the surface and are often projected onto the *xy*-plane. The height difference between neighbouring curves is constant.

Level curves are also termed contours, since all points on a given curve have the same height.

We know from the level curves: The function increases monotonically toward the center, reaching its maximum there.

The function increases more rapidly, or equivalently, the surface becomes steeper, as we approach the center.

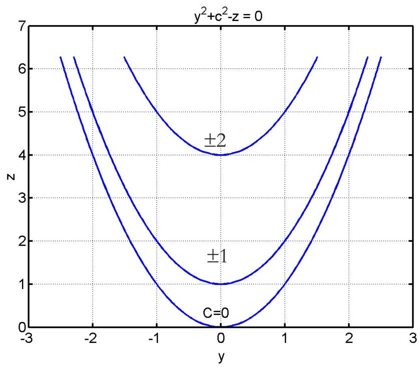
Example 4.

Plot the vertical sections and level curves for $z = x^2+y^2$.

Solution. The vertical sections in the *yz* plane

$$\begin{cases} z = x^2 + y^2 \\ x = c \end{cases} \Leftrightarrow \begin{cases} z = c^2 + y^2 \\ x = c \end{cases}$$

We can draw $z = c^2+y^2$ in the *yz* plane.

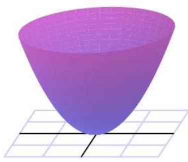


Similarly, one can draw the vertical sections in the *zx* plane. 16

Example 4 (cont.)

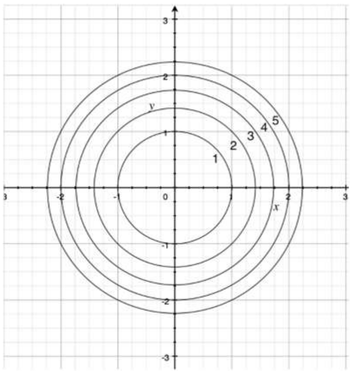
Plot the vertical sections and level curves for $z = x^2+y^2$.

Solution. The level curves



$$\begin{cases} z = x^2 + y^2 \\ z = c \end{cases} \Leftrightarrow \begin{cases} c = x^2 + y^2 \\ z = c \end{cases}$$

We can draw $c = x^2+y^2$ in the xy plane.



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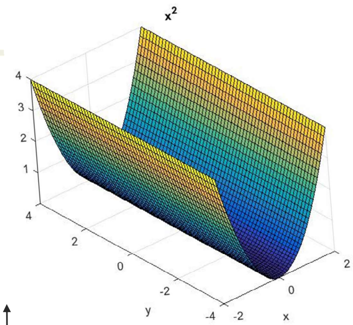
Example 5.

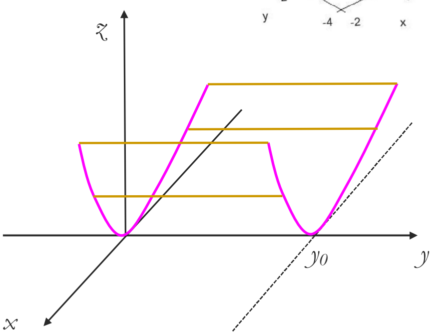
Sketch the graph in real space for the function $z = x^2$.

Soln. Its intersections with any vertical planes $y = y_0$ are the same curve

$$z = x^2.$$

It is thus a cylindrical surface parallel to the y-axis.





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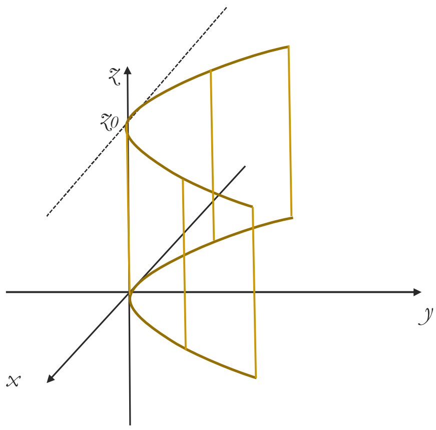
Example 6.

Sketch the graph for the function $y = x^2$ in the real space.

Soln. Its intersections with any horizontal planes $z = z_0$ are the same curve

$$y = x^2.$$

It is thus a cylindrical surface parallel to the z-axis.



Example 7.

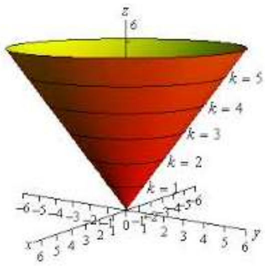
Sketch the graph for the function

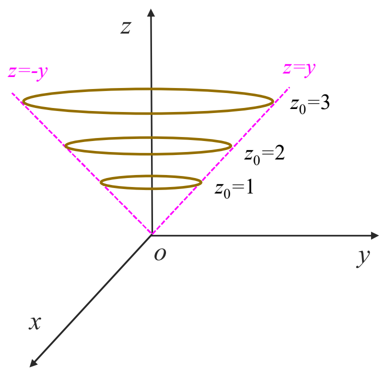
$$z = \sqrt{x^2 + y^2}.$$

Soln. Its intersection with any horizontal plane $z = z_0 \geq 0$ is the

$$x^2 + y^2 = z_0^2.$$

It is a circle with a radius z_0 and centred at the z-axis. It is thus a cone surface.



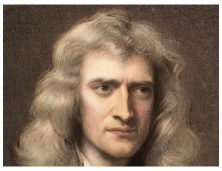


1.2 Partial derivatives **44354621**

Derivatives of single variable functions

Consider a furcation $y = f(x)$ at $x = a$. If the limit,

$$\lim_{\Delta x \to 0} \frac{f(a+\Delta x) - f(a)}{\Delta x},$$



Isaac Newton (1643-1727)

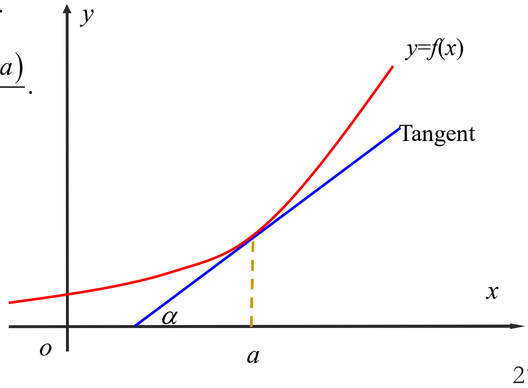
exists, we draw the 3 conclusions:

(1) $F(x)$ has a derivative at $x = a$.

(2) $$\frac{df(a)}{dx} \equiv \lim_{\Delta x \to 0} \frac{f(a+\Delta x) - f(a)}{\Delta x}.$$

(3) $f_x(a)$ is the slope of the tangent line of the curve at $x = a$,

$$\frac{df(a)}{dx} = \tan(\alpha)$$



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Partial derivatives **44354621**

For function $z = f(x, y)$, we fix $y = b =$ constant.

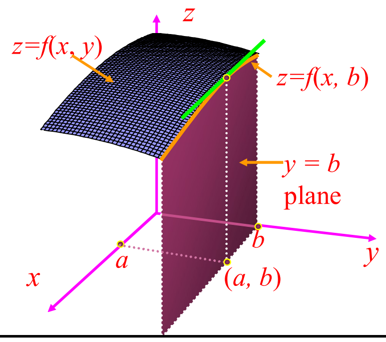
$z = f(x, b)$ reflects how $f(x, y)$ changes with x, while y is fixed.

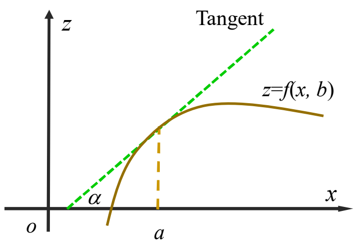
$z = f(x, b)$ is the intersection of the surface $z = f(x, y)$ and the plane $y = b$.

$$\left.\frac{df(x,b)}{dx}\right|_{x=a} = \lim_{\Delta x \to 0} \frac{\left.f(x,b)\right|_{x=a+\Delta x} - \left.f(x,b)\right|_{x=a}}{\Delta x} = \lim_{\Delta x \to 0} \frac{f(a+\Delta x, b) - f(a,b)}{\Delta x}$$

It is the slope of the tangent to the intersection curve $z = f(x, b)$ at (a, b).

$$\left.\frac{df(x,b)}{dx}\right|_{x=a} = \tan\alpha$$





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Partial derivatives

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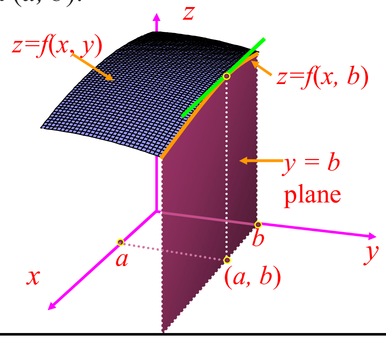
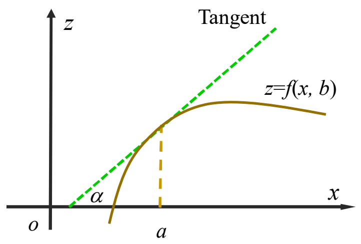
$$\left.\frac{df(x,b)}{dx}\right|_{x=a} = \lim_{\Delta x \to 0} \frac{f(a+\Delta x, b) - f(a,b)}{\Delta x}$$

This limit reflects how $f(x, y)$ changes with x at (a, b), while y is fixed. It is called the ***partial derivative*** of f w.r.t. x at the point (a, b), denoted as

$$\frac{\partial f(a,b)}{\partial x} \text{ or } f_x(a,b).$$

$f_x(a, b)$ is the slope of the tangent to the intersection curve $z = f(x, b)$ at (a, b).





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Partial derivatives

$$\frac{\partial f(a,b)}{\partial x} \equiv \left.\frac{df(x,b)}{dx}\right|_{x=a} \equiv \lim_{\Delta x \to 0} \frac{f(a+\Delta x, b) - f(a,b)}{\Delta x}.$$

In calculating the partial derivative of $f(x, y)$ to x:

(i) Treat y as a constant, $y = b$,

(ii) Calculate the "ordinary derivative" of $f(x, b)$ to x.

In calculating $f_x(x, y)$, y is treated as a constant.

The ***partial derivative*** of $f(x, y)$ w.r.t. y at the point (a, b) is defined as

$$\frac{\partial f(a,b)}{\partial y} \equiv \lim_{\Delta y \to 0} \frac{f(a, b+\Delta y) - f(a,b)}{\Delta y},$$

$f_y(a, b)$ reflects how $f(x, y)$ changes with y at (a, b), when x is fixed.

It is the slope of the tangent line to the curve $z = f(a, y)$ at $y = b$.

When calculating $f_y(x, y)$, x is treated as a constant.

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Examples 1. & 2.

Example 1. Calculate $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ for $f = y^2 + x^2 y$.

In calculating f_x, we treat $y = y_0$ as a constant and calculate the ordinary derivative of $f(x, y_0)$. As such, the partial derivative operation is linear for functions..

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}\left(y^2 + x^2 y\right) = \frac{\partial}{\partial x}\left(y_0^2 + x^2 y_0\right)$$
$$= 0 + 2xy_0 = 2xy$$
$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}\left(y^2 + x^2 y\right) = 2y + x^2$$

Example 2. Calculate $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ for $f = x^y$.

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x} x^y = \frac{\partial}{\partial x} x^{y_0} = y_0 x^{y_0 - 1} = y x^{y-1} \qquad \therefore \frac{d}{dx} x^n = n x^{n-1}$$
$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y} x^y = \frac{\partial}{\partial y} x_0^y = x_0^y \ln x_0 = x^y \ln x \qquad \therefore \frac{d}{dx} a^x = a^x \ln a$$

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Linearity, product and quotient rules

For $u = u(x, y)$ and $v = v(x, y)$, and constants a and b

$$\frac{\partial}{\partial x}\left(au(x,y) + bv(x,y)\right) = \frac{d}{dx}\left(au(x,y_0) + bv(x,y_0)\right)$$
$$= a\frac{du(x,y_0)}{dx} + b\frac{dv(x,y_0)}{dx} = a\frac{\partial u}{\partial x} + b\frac{\partial v}{\partial x},$$

$$\frac{\partial}{\partial x}\left(u(x,y)v(x,y)\right) = \frac{d}{dx}\left(u(x,y_0)v(x,y_0)\right)$$
$$= \frac{du(x,y_0)}{dx}v(x,y_0) + u(x,y_0)\frac{dv(x,y_0)}{dx} = \frac{\partial u}{\partial x}v + u\frac{\partial v}{\partial x},$$

i.e. $\frac{\partial(uv)}{\partial x} = \frac{\partial u}{\partial x}v + u\frac{\partial v}{\partial x}.$

Similarly one can show

$$\frac{\partial}{\partial x}\left(\frac{u}{v}\right) = \frac{1}{v^2}\left(\frac{\partial u}{\partial x}v - u\frac{\partial v}{\partial x}\right).$$

One can show the same results for partial derivatives w.r.t y.

The product and quotient rules stand for partial differentiation.

Partial derivatives

A function $f(x, y, z)$ is said to have a partial derivative to y at the point (x_0, y_0, z_0) if the following limit exists

$$\frac{\partial f(x_0, y_0, z_0)}{\partial y} \equiv \lim_{\Delta y \to 0} \frac{f(x_0, y_0 + \Delta y, z_0) - f(x_0, y_0, z_0)}{\Delta y} .$$

When calculating f_y, x and z are treated as constants.

Example 2. Calculate f_x, f_y and f_z for $f = x\sin(y)+yz$.

Solution.

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}(x\sin y + yz) = \sin y + 0 = \sin y,$$

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}(x\sin y + yz) = x\cos y + z,$$

$$\frac{\partial f}{\partial z} = \frac{\partial}{\partial z}(x\sin y + yz) = 0 + y = y,$$

High order partial derivatives

As the first order partial derivatives of $f(x, y)$ are functions of (x, y), one can perform partial derivative to them, which are termed as the second-order partial derivatives:

$$\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) \equiv \frac{\partial^2 f}{\partial x^2} \equiv f_{xx},$$ 2^{nd} order partial derivative to x

$$\frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) \equiv \frac{\partial^2 f}{\partial y^2} \equiv f_{yy},$$ 2^{nd} order partial derivative to y

$$\frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) \equiv \frac{\partial^2 f}{\partial y \partial x} \equiv f_{xy},$$ Mixed 2^{nd} order partial derivative firstly to x then y

$$\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) \equiv \frac{\partial^2 f}{\partial x \partial y} \equiv f_{yx},$$ Mixed 2^{nd} order partial derivative firstly to y then x

One can define higher order partial derivatives as follows:

$$\frac{\partial}{\partial x}\left(\frac{\partial^2 f}{\partial x^2}\right) \equiv \frac{\partial^3 f}{\partial x^3} \equiv f_{xxx}, \quad \frac{\partial}{\partial y}\left(\frac{\partial^2 f}{\partial x^2}\right) \equiv \frac{\partial^3 f}{\partial y \partial x^2} \equiv f_{xxy} .$$

Example 3

Calculate f_{xy} and f_{yx} for $f = ye^x + x\sin y$.

Solution.

$$f_x = \frac{\partial f}{\partial x} = \frac{\partial}{\partial x}\left(ye^x + x\sin y\right) = ye^x + \sin y$$

$$f_{xy} = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y}\frac{\partial f}{\partial x} = \frac{\partial}{\partial y}\left(ye^x + \sin y\right) = e^x + \cos y$$

$$f_y = \frac{\partial f}{\partial y} = \frac{\partial}{\partial y}\left(ye^x + x\sin y\right) = e^x + x\cos y$$

$$f_{yx} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}\frac{\partial f}{\partial y} = \frac{\partial}{\partial x}\left(e^x + x\cos y\right) = e^x + \cos y$$

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y}$$

Theorem. If the partial derivatives concerned are continuous, the mixed derivatives commute; that is, the order of differentiation does not matter.

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1.3 Differentiable functions

Little – *o*: If $\lim\limits_{x\to c} \frac{f(x)}{g(x)} = 0$

We say "f is little-oh of g as x approaches c" and write

$f(x) = o(g(x))$ as $x \to c$.

The little o notation indicates that the decay rate of a certain function is faster than that of another function.

Intuitively, this means the magnitude of $f(x)$ is much smaller than $g(x)$ for x near c.

Examples

Since $\lim\limits_{x\to 0} \frac{x^{4/3}}{x} = 0$

we have $x^{4/3} = o(x)$, as $x \to 0$.

Since $\lim\limits_{x\to 2} \frac{(x-2)^{3/2}}{x-2} = 0,$

we have $(x-2)^{3/2} = o(x-2)$, as $x \to 2$.

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Examples using small o notion

4551 8141

If $\lim_{x\to c}\frac{f(x)}{g(x)}=0$, we say that $f(x)$ is higher order compared to $g(x)$, denoted as

$$f(x)=o(g(x)) \quad \text{as } x\to c.$$

We know the error bounds for a number. For example, the height of somebody is h = 1.71 ± 0.005 m.

Litter o is a similar tool in the estimation of functions.

1. $\sin x = x-\frac{x^3}{3!}+\frac{x^5}{5!}-\cdots \;= x+f(x), \quad f(x)=-\frac{x^3}{3!}+\frac{x^5}{5!}-\cdots,$

since $\lim_{x\to 0}\frac{f(x)}{x}=\lim_{x\to 0}\frac{1}{x}\left(-\frac{x^3}{3!}+\frac{x^5}{5!}-\cdots\right)=0$

$\Rightarrow f(x)=o(x) \quad \Rightarrow \quad \sin x = x+o(x).$

$$\sin x = x+o(x) \quad \Rightarrow \quad \begin{array}{l}\text{As } x \text{ is near } 0:\\ |o(x)| << |x| << 1,\\ \sin x\approx x.\end{array}$$

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Examples using small o notion

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2. $\sin(x-2)=x-2-\frac{(x-2)^3}{3!}+\frac{(x-2)^5}{5!}-\cdots$

$\Leftrightarrow \quad \sin(x-2)=x-2+f(x), \quad f(x)=-\frac{(x-2)^3}{3!}+\frac{(x-2)^5}{5!}-\cdots$

$\lim_{x\to 2}\frac{f(x)}{x-2}=\lim_{x\to 2}\frac{1}{x-2}\left(-\frac{(x-2)^3}{3!}+\frac{(x-2)^5}{5!}-\cdots\right)=0 \;\Rightarrow\; f(x)=o(x-2)$

$\Rightarrow \sin(x-2)=x-2+o(x-2), \quad \text{as } x\to 2.$

As x approaches 2, sin(x -2) can be approximated by x-2, and the error of this estimation is a higher-order small quantity than x-2.

$$\sin(x-2)=x-2+o(x-2) \quad \Rightarrow \quad \begin{array}{l}\text{As } x \text{ is near } 2:\\ |o(x-2)| << |x-2| << 1,\\ \sin(x-2)\approx x-2.\end{array}$$

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Differentiable functions of single variable **4551 8141**

A function $y = f(x)$ is called differentiable at x_0 if

$$y - y_0 = f_x(x_0)(x - x_0) + o(x - x_0),$$

$$\text{or} \quad \Delta y = f_x(x_0)\Delta x + o(\Delta x), \qquad (1)$$

where $y_0 = f(x_0)$, $\Delta x = x\text{-}x_0$, $\Delta y = y\text{-}y_0$.

As x is near x_0, $|\Delta x| << 1$

$f_x(x_0)\Delta x$ is the same order as Δx.

$o(\Delta x)$ approaches 0 faster than Δx.

Eq. (1) becomes: $\Delta y \approx f_x(x_0)\Delta x$

or $y\text{-}y_0 \approx f_x(x_0)\,(x\text{-}x_0)$.

Being differentiable at a point:

The change of the function in its neighbourhood is approximately linear to the changes of the variable, with the coefficient of this linear approximation being the derivative.

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Geometrical interpretation of differentiable functions

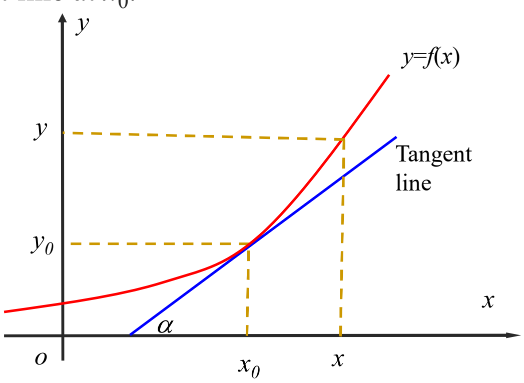
As $y = f(x)$ is differentiable at x_0, the curve near x_0 satisfies:

$y\text{-}y_0 \approx f_x(x_0)\,(x\text{-}x_0)$

The tangent line of the curve at x_0.

$y\text{-}y_0 = f_x(x_0)\,(x\text{-}x_0)$

When a curve is differentiable at x_0, the curve near x_0 can be approximated by its tangent line at x_0.



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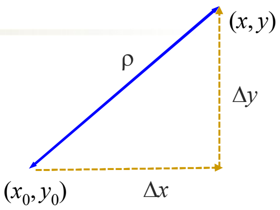
Differentiable functions with two variables



As function $f(x, y)$ is called differentiable at (x_0, y_0),
if $\Delta f = f(x, y)$- $f(x_0, y_0)$ satisfies:

$$\Delta f = f_x(x_0, y_0)\Delta x + f_y(x_0, y_0)\Delta y + o(\rho) \ ,$$

where $\Delta x = x$-x_0, $\Delta y = y$-y_0.

$\rho = \sqrt{(\Delta x)^2 + (\Delta y)^2}$ is the distance between (x, y) and (x_0, y_0).

As (x, y) is near (x_0, y_0), or (x, y) is in the neighbourhood of (x_0, y_0):

$f_x(x_0, y_0)\ \Delta x$ and $f_y\ (x_0, y_0)\Delta y$ are the same order as ρ.

o(ρ) approaches 0 faster than ρ.

Therefore $\quad \Delta f \approx f_x(x_0, y_0)\Delta x + f_y(x_0, y_0)\Delta y.$

Near a differentiable point, the change of a function is approximately linear to the changes of variables, with the coefficients being the corresponding partial derivatives.

As ρ is infinitely small, we have

$$df(x, y) = f_x(x, y)dx + f_y(x, y)dy.$$

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Geometrical interpretation of differentiable functions

As $z = f(x, y)$ is differentiable at (x_0, y_0), the surface near the point satisfies

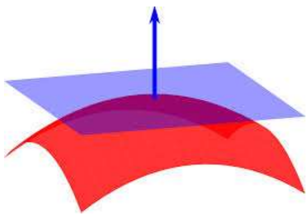
$$\Delta f \approx f_x(x_0, y_0)\Delta x + f_y(x_0, y_0)\Delta y,$$

or $\quad z - z_0 \approx f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$

The tangent plane for the surface $z = f(x, y)$ at (x_0, y_0) is (to be shown later in the chapter)

$$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

When a surface is differentiable at (x_0, y_0), the surface near (x_0, y_0) can be approximated by its tangent plane at (x_0, y_0).



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Being differentiable & having derivatives

For single variable functions, being differentiable is equivalent to having a derivative, i.e.

$$f(x) \in D \quad \Leftrightarrow \quad \exists f_x \,.$$

For multi-variable functions, a differentiable function has all partial derivatives

$$f(x,y) \in D \quad \Rightarrow \quad \exists f_x, f_y.$$

But the existence of all partial derivatives does not guarantee the function being differentiable

$$f(x,y) \in D \quad \not\Leftarrow \quad \exists f_x, f_y$$

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1.4 The chain rule

Informal definition. A function is said to be sufficiently smooth if the function and as many of its partial derivatives as required are continuous where they need to be.

Below, we will assume all our functions to be sufficiently smooth and differentiable.

The chain rule for functions of single variable

Suppose $y = y(x)$ and $x = x(u)$. Denoting $y = y(x(u)) = Y(u)$, we have

$$\frac{dY}{du} = \frac{dy}{dx}\frac{dx}{du}.$$

Conventionally we just write

$$\frac{dy}{du} = \frac{dy}{dx}\frac{dx}{du}.$$

We will use this convention for multivariable functions.

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Derivation of chain rule for single variable functions

Suppose $y = y(x)$ and $x = x(u)$ being differentiable:

$$dy = y_x dx,$$

$$dx = x_u du.$$

Thus $dy = y_x x_u du.$

Denoting $y = y(x(u)) = Y(u)$, assuming $Y(u)$ being differentiable, we have

$$dy = Y_u du.$$

From the above two eqs., we know that

$$Y_u du = y_x x_u du,$$

thus $Y_u = y_x x_u$,

or

$$\frac{dY}{du} = \frac{dy}{dx}\frac{dx}{du}.$$

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Chain rule for functions of two variables

Suppose $f = f(x, y)$ and $x = x(u, v)$, $y = y(u, v)$ and all are differentiable.

As $f(x, y)$ is differentiable: $df = f_x dx + f_y dy.$

As $x = x(u, v)$ and $y = y(u, v)$ are differentiable:

$$dx = x_u du + x_v dv, \qquad dy = y_u du + y_v dv.$$

Combining the above 3 equations

$$\begin{aligned} df &= f_x\left(x_u du + x_v dv\right) + f_y\left(y_u du + y_v dv\right) \\ &= \left(f_x x_u du + f_y y_u du\right) + \left(f_x x_v dv + f_y y_v dv\right) \\ &= \left(f_x x_u + f_y y_u\right) du + \left(f_x x_v + f_y y_v\right) dv. \end{aligned}$$

Since $f = f\left(x(u,v), y(u,v)\right) = f(u,v) \quad \Rightarrow \quad df = f_u du + f_v dv.$

du and dv are infinitely small but otherwise arbitrary. By setting $du \neq 0$, $dv=0$ and $du=0$, $dv \neq 0$, respectively, we obtain the *Chain Rule*

$$f_u = f_x x_u + f_y y_u, \quad f_v = f_x x_v + f_y y_v.$$

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A way to remember/write chain rule

For $f = f(x)$, $x = x(u)$,

$$\frac{df(x)}{du} = \frac{df}{\quad}\frac{\quad}{du}. \quad \Rightarrow \quad \frac{df(x)}{du} = \frac{df}{dx}\frac{dx}{du}.$$

For $f = f(x, y)$, $x = x(u, v)$, $y = y(u, v)$,

$$\frac{\partial f(x,y)}{\partial u} = \frac{\partial f}{\quad}\frac{\quad}{\partial u} + \frac{\partial f}{\quad}\frac{\quad}{\partial u} \quad \Rightarrow \quad \frac{\partial f(x,y)}{\partial u} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial u}$$

$$\frac{\partial f(x,y)}{\partial v} = \frac{\partial f}{\quad}\frac{\quad}{\partial v} + \frac{\partial f}{\quad}\frac{\quad}{\partial v} \quad \Rightarrow \quad \frac{\partial f(x,y)}{\partial v} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial v}$$

In Step 1, we put the function and the target variables in correct positions. For two intermediate variables, we will have two terms

In Step 2, we add an intermediate variable for each term.

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Example 1

If $f = e^x \sin y$, $x = uv^2$ and $y = uv$, find $\partial f/\partial u$ at $(u, v) = (1, \pi)$.

Solution. Using the chain rule we have

$$\frac{\partial f(x,y)}{\partial u} = \frac{\partial f}{\quad}\frac{\quad}{\partial u} + \frac{\partial f}{\quad}\frac{\quad}{\partial u} \quad \Rightarrow \quad \frac{\partial f(x,y)}{\partial u} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial u}. \qquad (1)$$

Calculate the partial derivatives needed

$$\frac{\partial f(x,y)}{\partial x} = \frac{\partial}{\partial x}\left(e^x \sin y\right) = e^x \sin y, \qquad \frac{\partial f(x,y)}{\partial y} = \frac{\partial}{\partial y}\left(e^x \sin y\right) = e^x \cos y,$$

$$\frac{\partial x(u,v)}{\partial u} = \frac{\partial}{\partial u}\left(uv^2\right) = v^2, \qquad \frac{\partial y(u,v)}{\partial u} = \frac{\partial}{\partial u}\left(uv\right) = v.$$

Substituting them to eq. (1)

$$f_u = e^x \sin y\ v^2 + e^x \cos y\ v = ve^x\left(v \sin y + \cos y\right)$$

$$= ve^{uv^2}\left(v \sin(uv) + \cos(uv)\right), \quad \text{(expressing target vars.)}$$

$$f_u\big|_{(u,v)=(1,\pi)} = \pi e^{\pi^2}\left(\pi \sin(\pi) + \cos(\pi)\right) = -\pi e^{\pi^2}.$$

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Chain rule for functions of multi-variables

In a general case, we have the following chain rule.

Fact: Let f be a function of the variables $x_1, x_2, \cdots, x_n$,

$f = f(x_1, x_2, \cdots, x_n)$,

where each x_j is a function of the variables $t_1, t_2, \cdots, t_m$

$x_j = x_j(t_1, t_2, \cdots, t_m),\ j = 1, 2, \cdots, n$.

If f and x_j are sufficiently smooth, then

$$\boxed{\frac{\partial f}{\partial t_i} = \frac{\partial f}{\partial x_1}\frac{\partial x_1}{\partial t_i} + \frac{\partial f}{\partial x_2}\frac{\partial x_2}{\partial t_i} + \cdots + \frac{\partial f}{\partial x_n}\frac{\partial x_n}{\partial t_i}, \quad i = 1, 2, \cdots, \text{m}.}$$

The summarization runs over all the intermediate variables.

One way to remember the chain rule is writing the following step firstly, then adding all intermediate variables.

$$\frac{\partial f(x_1, x_2, \cdots, x_n)}{\partial t_i} = \frac{\partial f}{\quad}\frac{\quad}{\partial t_i} + \frac{\partial f}{\quad}\frac{\quad}{\partial t_i} + \cdots + \frac{\partial f}{\quad}\frac{\quad}{\partial t_i}.$$

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Example 2

Assume $w = e^x yz,\ x = ts,\ y = s^2,\ z = t\text{-}s$. Find w_t.

Solution. Using the chain rule,

$$\frac{\partial w(x,y,z)}{\partial t} = \frac{\partial w}{\quad}\frac{\quad}{\partial t} + \frac{\partial w}{\quad}\frac{\quad}{\partial t} + \frac{\partial w}{\quad}\frac{\quad}{\partial t} \Rightarrow \frac{\partial w}{\partial t} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial t} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial t} \qquad (1)$$

Calculate the derivatives needed

$$\frac{\partial w}{\partial x} = \frac{\partial}{\partial x}\left(e^x yz\right) = e^x yz, \quad \frac{\partial w}{\partial y} = \frac{\partial}{\partial y}\left(e^x yz\right) = e^x z, \quad \frac{\partial w}{\partial z} = \frac{\partial}{\partial z}\left(e^x yz\right) = e^x y,$$

$$\frac{\partial x}{\partial t} = \frac{\partial}{\partial t}(ts) = s, \quad \frac{\partial y}{\partial t} = \frac{\partial}{\partial t}\left(s^2\right) = 0, \quad \frac{\partial z}{\partial t} = \frac{\partial}{\partial t}(t-s) = 1.$$

Substituting them into (1) yields

$$\frac{\partial w}{\partial t} = e^x yz \cdot s + e^x z \cdot 0 + e^x y \cdot 1 = e^x y(zs+1)$$

$$= e^{ts} s^2\left((t-s)s+1\right) \qquad \text{(express } x, y, z \text{ in terms of } s, t\text{)}$$

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Composite functions

Example 3. Calculate $\dfrac{\partial^2 f}{\partial y\partial x}$ for $f=\sin(x\sin y)$

Soln. Let $u = x\sin y$, we have $f = \sin u$

$$f_x=\frac{\partial f}{\partial x}=\frac{\partial \sin u}{\partial x}=\frac{\partial \sin u}{\partial u}\frac{\partial u}{\partial x}$$

$$=\cos u\frac{\partial}{\partial x}(x\sin y)=\cos u\sin y$$

$$\frac{\partial^2 f}{\partial y\partial x}=\frac{\partial}{\partial y}f_x=\frac{\partial}{\partial y}(\cos u\sin y)=\frac{\partial \cos u}{\partial y}\sin y+\cos u\cos y$$

$$\therefore\ \frac{\partial \cos u}{\partial y}=\frac{\partial \cos u}{\partial u}\frac{\partial u}{\partial y}=-\sin u\frac{\partial}{\partial y}(x\sin y)=-\sin u\ x\cos y$$

$$\frac{\partial^2 f}{\partial y\partial x}=-\sin u\ x\cos y\ \sin y+\cos u\cos y$$

$$=(-\sin u\ x\sin y\ +\cos u)\cos y$$

$$=(-\sin(x\sin y)\ x\sin y+\cos(x\sin y))\cos y$$

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Implicit partial differentiation

Example 4. Find $\dfrac{\partial z}{\partial x}$ and $\dfrac{\partial z}{\partial y}$, where $z(x,y)$ is defined by

$$yz-\ln z=x+y.$$

Solution. We differentiate both sides of the eq. to x:

$$\frac{\partial(yz)}{\partial x}-\frac{\partial \ln z}{\partial x}=\frac{\partial x}{\partial x}+\frac{\partial y}{\partial x}$$

As $\dfrac{\partial yz}{\partial x}=y\dfrac{\partial z}{\partial x}=yz_x$, $\quad \dfrac{\partial \ln z}{\partial x}=\dfrac{\partial \ln z}{\partial z}\dfrac{\partial z}{\partial x}=\dfrac{z_x}{z}$

$$\Rightarrow\quad yz_x-\frac{z_x}{z}=1+0\quad\Rightarrow\quad z_x=1/\left(y-\frac{1}{z}\right)=\frac{z}{yz-1}$$

Similarly $\dfrac{\partial(yz)}{\partial y}-\dfrac{\partial \ln z}{\partial y}=\dfrac{\partial x}{\partial y}+\dfrac{\partial y}{\partial y}\quad\Rightarrow\quad z+yz_y-\dfrac{z_y}{z}=0+1$

$$z_y\left(y-\frac{1}{z}\right)=1-z\quad\Rightarrow\quad z_y=(1-z)/\left(y-\frac{1}{z}\right)=\frac{z-z^2}{yz-1}$$

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Example 5 **6092 9713**

For the partial differential equation: $x\frac{\partial u}{\partial x} - y\frac{\partial u}{\partial y} = 2x^2$ (1)

change the variables (x, y) to $s = xy$, $t = x/y$.

Solution. $u = u(x, y) = u(x(s, t), y(s, t)) = U(s, t)$

$$\frac{\partial u}{\partial x} = \frac{\partial U(s,t)}{\partial x} = \frac{\partial U}{\partial s}\frac{\partial s}{\partial x} + \frac{\partial U}{\partial t}\frac{\partial t}{\partial x} = U_s y + U_t \frac{1}{y}, \qquad (2)$$

$$\frac{\partial u}{\partial y} = \frac{\partial U(s,t)}{\partial y} = \frac{\partial U}{\partial s}\frac{\partial s}{\partial y} + \frac{\partial U}{\partial t}\frac{\partial t}{\partial y} = U_s x + U_t \left(-\frac{x}{y^2}\right), \qquad (3)$$

where we used $s_x = y,\ \ s_y = x,\ \ t_x = \frac{1}{y},\ \ t_y = -\frac{x}{y^2}.$

Substituting eqs. (2) and (3) in PDE (1) yields:

$$x\left(U_s y + U_t \frac{1}{y}\right) - y\left(U_s x + U_t\left(-\frac{x}{y^2}\right)\right) = 2x^2$$

$$\Leftrightarrow \quad U_t \frac{x}{y} + U_t \frac{x}{y} = 2x^2 \quad \Leftrightarrow \quad U_t = xy \quad \Leftrightarrow \quad U_t = s$$

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Example 6

For the partial differential equation: $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 0$ (1)

change the variables (x, y) to. $r = \sqrt{x^2 + y^2}$, $\theta = \tan^{-1}\frac{y}{x}$

Solution. $u = u(x, y) = u(x(r, \theta), y(r, \theta)) = U(r, \theta)$

$$r_x = \frac{x}{r},\ \ r_y = \frac{y}{r},\ \ \theta_x = \frac{-y/x^2}{1+(y/x)^2} = -\frac{y}{r^2},\ \ \theta_y = \frac{1/x}{1+(y/x)^2} = \frac{x}{r^2}.$$

$$\frac{\partial u}{\partial x} = \frac{\partial U(r,\theta)}{\partial x} = \frac{\partial U}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial U}{\partial \theta}\frac{\partial \theta}{\partial x} = U_r \frac{x}{r} - U_\theta \frac{y}{r^2}, \qquad (2)$$

$$\frac{\partial u}{\partial y} = \frac{\partial U(r,\theta)}{\partial y} = \frac{\partial U}{\partial r}\frac{\partial r}{\partial y} + \frac{\partial U}{\partial \theta}\frac{\partial \theta}{\partial y} = U_r \frac{y}{r} + U_\theta \frac{x}{r^2}. \qquad (3)$$

Substituting eqs. (2) and (3) in PDE (1) yields

$$x\left(U_r \frac{x}{r} - U_\theta \frac{y}{r^2}\right) + y\left(U_r \frac{y}{r} + U_\theta \frac{x}{r^2}\right) = 2x^2$$

$$\Leftrightarrow \quad U_t \frac{x^2 + y^2}{r} = 0 \quad \Leftrightarrow \quad U_r r = 0 \quad \Leftrightarrow \quad U_r = 0$$

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1.5 Vectors and vector geometry

A scalar just has a magnitude.

A vector has both a magnitude and a direction.

A vector $\boldsymbol{v}$ is noted

$\vec{v}$,

v: using boldface letters in printing, or

$\underline{v}$ in writing

$|\boldsymbol{v}|$ or v denotes the magnitude of $\boldsymbol{v}$. If $\boldsymbol{v} = (v_1, v_2, v_3)$, then

$$v = |\boldsymbol{v}| = \sqrt{v_1^2 + v_2^2 + v_3^2}$$

A unit vector has magnitude of 1.

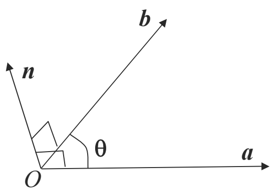
In Cartesian coordinate system $\boldsymbol{i}, \boldsymbol{j}$ and $\boldsymbol{k}$ denote the unit coordinate vectors along the x, y, z-axes respectively.

Dot product and cross-product of two vectors

Consider two vectors

$$\boldsymbol{a} = a_1\boldsymbol{i} + a_2\boldsymbol{j} + a_3\boldsymbol{k}, \quad \boldsymbol{b} = b_1\boldsymbol{i} + b_2\boldsymbol{j} + b_3\boldsymbol{k}\,.$$

The scalar or dot product of the two vectors is



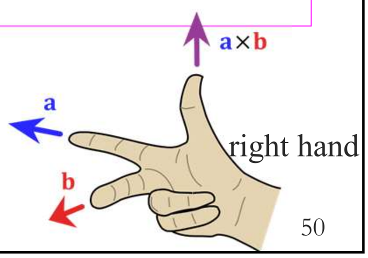
$$\boxed{\boldsymbol{a}\cdot\boldsymbol{b} = |\boldsymbol{a}||\boldsymbol{b}|\cos\theta = a_1b_1 + a_2b_2 + a_3b_3\,,}$$

where $\theta \in [0, \pi]$ is the angle between $\boldsymbol{a}$ and $\boldsymbol{b}$.

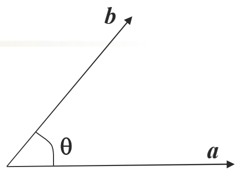
The vector or cross product of the two vectors is

$$\boxed{\mathbf{a}\times\mathbf{b} = |\mathbf{a}||\mathbf{b}|\sin\theta\mathbf{n} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = (a_2b_3 - a_3b_2)\mathbf{i} - (a_1b_3 - a_3b_1)\mathbf{j} + (a_1b_2 - a_2b_1)\mathbf{k}}$$

where $\boldsymbol{n}$ is the unit vector normal to both $\boldsymbol{a}$ and $\boldsymbol{b}$ such that $(\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{n})$ forms a right handed system.



Parallel and perpendicular vectors



If $\boldsymbol{a} \neq 0$, $\boldsymbol{b} \neq 0$, then

$\boldsymbol{a}$ and $\boldsymbol{b}$ are perpendicular if and only if $\boldsymbol{a}\cdot\boldsymbol{b} = 0$

$$\begin{aligned}\boldsymbol{a}\cdot\boldsymbol{b} = ab\cos\theta = 0 \quad &\Leftrightarrow \quad \cos\theta = 0 \quad \because\ a \neq 0,\ \mathrm{b} \neq 0\\ &\Leftrightarrow \quad \theta = \pi/2 \quad \because\ \theta \in [0,\pi]\\ &\Leftrightarrow \quad \boldsymbol{a} \perp \boldsymbol{b}\end{aligned}$$

$\boldsymbol{a}$ and $\boldsymbol{b}$ are parallel if and only if $\boldsymbol{a}\times\boldsymbol{b} = 0$

$$\begin{aligned}\boldsymbol{a}\times\boldsymbol{b} = ab\sin\theta\boldsymbol{n} = 0 \quad &\Leftrightarrow \quad \sin\theta = 0 \quad \because a \neq 0,\ \mathrm{b} \neq 0\\ &\Leftrightarrow \quad \theta = 0 \text{ or } \pi \quad \because \theta \in [0,\pi]\\ &\Leftrightarrow \quad \boldsymbol{a} \,||\, \boldsymbol{b}\end{aligned}$$

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Equation of a line

The equation of a line is

$$\boxed{\boldsymbol{r} = \boldsymbol{a}+t\boldsymbol{d} \text{ for } t \in \mathbb{R}}$$

where $\boldsymbol{a}$ is a point at the line, $\boldsymbol{d}$ is a vector along the line and $\boldsymbol{r}$ is a general point at the line.

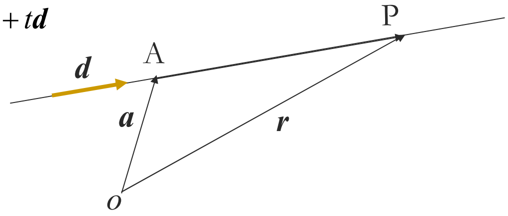
Proof. As shown in the figure:

$$\boldsymbol{r} = \boldsymbol{a} + \overrightarrow{AP} \quad \Leftrightarrow \quad \overrightarrow{AP} = \boldsymbol{r} - \boldsymbol{a}$$

Since $\overrightarrow{AP} \,||\, \boldsymbol{d}$

We know $\boldsymbol{r} - \boldsymbol{a} \,||\, \boldsymbol{d}$

Thus $\boldsymbol{r} - \boldsymbol{a} = t\boldsymbol{d} \quad \Leftrightarrow \quad \boldsymbol{r} = \boldsymbol{a} + t\boldsymbol{d}$



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Equation of a plane

The vector equation of a *plane* is

$\boxed{\boldsymbol{r} \cdot \boldsymbol{n} = \boldsymbol{a} \cdot \boldsymbol{n},}$

where $\boldsymbol{a}$ is a point at the plane, $\boldsymbol{n}$ is the normal vector of the plane and $\boldsymbol{r}$ a general point at the plane.

Proof. As shown in the figure:

$$\boldsymbol{r} = \boldsymbol{a} + \overrightarrow{AP}$$

$$\Leftrightarrow \quad \overrightarrow{AP} = \boldsymbol{r} - \boldsymbol{a}$$

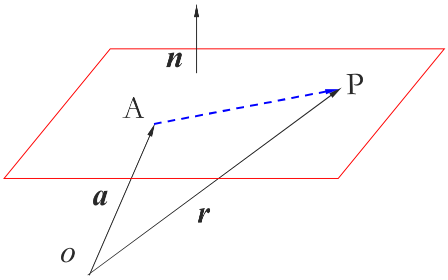
Since AP is on the plane

$$\overrightarrow{AP} \perp \boldsymbol{n}$$

We know $\quad \boldsymbol{r} - \boldsymbol{a} \perp \boldsymbol{n}$

Thus $\quad (\boldsymbol{r} - \boldsymbol{a}) \cdot \boldsymbol{n} = 0$

$$\Leftrightarrow \quad \boldsymbol{r} \cdot \boldsymbol{n} = \boldsymbol{a} \cdot \boldsymbol{n}$$



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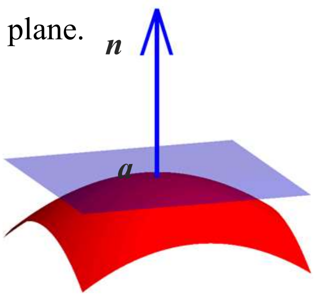
Equation of a tangent plane

Fact: If $\boldsymbol{n}$ is the normal vector to a surface S at a point $\boldsymbol{a}$, then the equation of the tangent plane to the surface S at the point $\boldsymbol{a}$ is given by

$$\boldsymbol{r} \cdot \boldsymbol{n} = \boldsymbol{a} \cdot \boldsymbol{n} \,,$$

where $\boldsymbol{r}$ is a general point on the tangent plane.

- The tangent plane to the surface at point **a** passes through the point $\boldsymbol{a}$.
- The normal of the tangent plane is the same as the normal **n** of the surface at $\boldsymbol{a}$.



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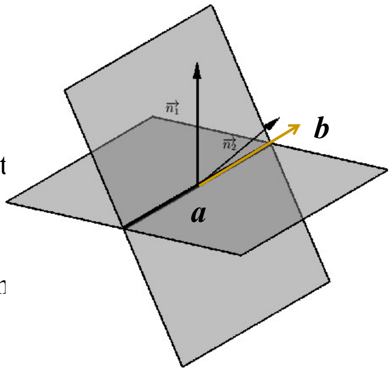
Example 1

Show that a line can be represented as an intersection of two planes.

Proof. A line can be expressed as:

$$\boldsymbol{r} = \boldsymbol{a}+t\boldsymbol{b}.$$

where $\boldsymbol{a}$ is a point at the line, $\boldsymbol{b}$ along the line direction. Any point $\boldsymbol{r}$ at the line satisfies this equation.



Choose two different vectors $\boldsymbol{n}_1$ and $\boldsymbol{n}_2$ that are perpendicular to $\boldsymbol{b}$, thus

$$\boldsymbol{b}\cdot\boldsymbol{n}_1 = 0, \quad \boldsymbol{b}\cdot\boldsymbol{n}_2 = 0.$$

Performing dot product to the line equation by $\boldsymbol{n}_1$ yields

$$\boldsymbol{r}\cdot\boldsymbol{n}_1 = (\boldsymbol{a}+t\boldsymbol{b})\cdot\boldsymbol{n}_1 = \boldsymbol{a}\cdot\boldsymbol{n}_1 + t\boldsymbol{b}\cdot\boldsymbol{n}_1 = \boldsymbol{a}\cdot\boldsymbol{n}_1$$

i.e. $\quad \boldsymbol{r}\cdot\boldsymbol{n}_1 = \boldsymbol{a}\cdot\boldsymbol{n}_1$.

Similarly we can obtain: $\boldsymbol{r}\cdot\boldsymbol{n}_2 = \boldsymbol{a}\cdot\boldsymbol{n}_2$.

As any point $\boldsymbol{r}$ at the line satisfies the two plane equations, the line must be at their intersection.

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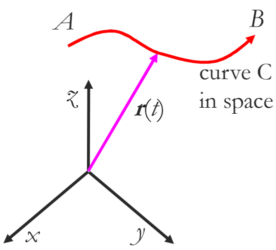
Representation of curves

A curve C can be expressed in the parametric form

$$\boldsymbol{r} = \boldsymbol{r}(t) = x(t)\boldsymbol{i}+y(t)\boldsymbol{j}+z(t)\boldsymbol{k},\ t\in[a, b].$$



Thus, the starting point A at $\boldsymbol{r}(a)$

and the terminal point B at $\boldsymbol{r}(b)$.

Example: $y = f(x)$, $x\in[a, b]$ is a curve. It can be expressed as

$$x = t,\quad y = f(t) \quad \Leftrightarrow \quad \mathbf{r} = \mathbf{r}(t) = (t, f(t)),\ t\in[a,b].$$

Example: Consider a tiny fly flying around. Its position at time t is at the point

$$\boldsymbol{r} = \boldsymbol{r}(\mathrm{t}) = (x(t), y(t), z(t)).$$

The trajectory of the fly from time $t = a$ to $t = b$ is a curve, which can be described by

$$\boldsymbol{r} = \boldsymbol{r}(t),\ t\in[a, b].$$

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Tangent vector of a curve

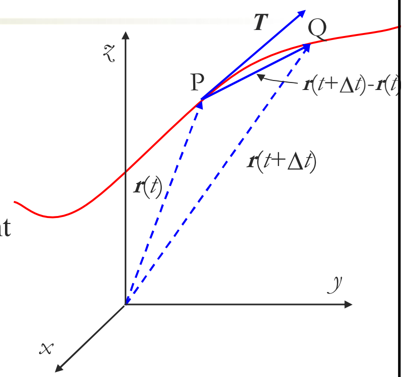
Consider a curve $\boldsymbol{r} = \boldsymbol{r}\,(t)$

and two points P and Q on the curve:

P: $\boldsymbol{r}(t) = x(t)\boldsymbol{i}+y(t)\boldsymbol{j}+z(t)\boldsymbol{k}$

Q: $\boldsymbol{r}(t+\Delta t) = x(t+\Delta t)\boldsymbol{i}+y(t+\Delta t)\boldsymbol{j}+z(t+\Delta t)\boldsymbol{k}$.

As Q approaches P, $\overrightarrow{PQ}$ is along the tangent of the curve at P.



The tangent vector $\boldsymbol{T}$ of the curve at P:

$$\boldsymbol{T} = \lim_{\Delta t\to 0}\frac{\overrightarrow{PQ}}{\Delta t} = \lim_{\Delta t\to 0}\frac{\boldsymbol{r}(t+\Delta t)-\boldsymbol{r}(t)}{\Delta t}$$

$$= \lim_{\Delta t\to 0}\frac{x(t+\Delta t)-x(t)}{\Delta t}\boldsymbol{i} + \lim_{\Delta t\to 0}\frac{y(t+\Delta t)-y(t)}{\Delta t}\boldsymbol{j} + \lim_{\Delta t\to 0}\frac{z(t+\Delta t)-z(t)}{\Delta t}\boldsymbol{k}$$

$$= \frac{dx}{dt}\boldsymbol{i}+\frac{dy}{dt}\boldsymbol{j}+\frac{dz}{dt}\boldsymbol{k} \equiv \frac{d\boldsymbol{r}}{dt}$$

$$\boxed{\boldsymbol{T} = \frac{d\boldsymbol{r}}{dt} = \boldsymbol{r}' = \boldsymbol{r}_t}$$

In general, for a sufficiently smooth curve one can always have parameterization for which $\boldsymbol{r}' \neq 0$.

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1.6 Gradient vector and directional derivative

If $f(x, y, z)$ has all partial derivatives at a point $\boldsymbol{a}$, then the gradient vector of f at $\boldsymbol{a}$ is defined as

$$\boxed{\nabla f(\boldsymbol{a}) = \mathrm{grad} f(\boldsymbol{a}) \equiv \frac{\partial f(\boldsymbol{a})}{\partial x}\boldsymbol{i} + \frac{\partial f(\boldsymbol{a})}{\partial y}\boldsymbol{j} + \frac{\partial f(\boldsymbol{a})}{\partial z}\boldsymbol{k}}$$

∇f is read as delta f, nabla f, or gradient f.

The gradient of a n-variable function is a vector function in n dimensions.

For example if $f: \mathbb{R}^2 \to \mathbb{R}$, then $\nabla f = f_x\boldsymbol{i} + f_y\boldsymbol{j}$.

Example 1. Find the gradient of $f(x, y, z) = x^2yz$ at (1, 1, 2).

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right) = \left(\frac{\partial}{\partial x}x^2yz, \frac{\partial}{\partial y}x^2yz, \frac{\partial}{\partial z}x^2yz\right)$$

$$= \left(2xyz, x^2z, x^2y\right)$$

$$\nabla f(1,1,2) = \left(2xyz, x^2z, x^2y\right)\Big|_{(1,1,2)} = (4,2,1)$$

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Linearity of ∇

It is often useful to consider ∇ as an operator, i.e. 'something that you do to functions':

$$\nabla f = \left(\boldsymbol{i}\frac{\partial}{\partial x} + \boldsymbol{j}\frac{\partial}{\partial y} + \boldsymbol{k}\frac{\partial}{\partial z} \right) f = \frac{\partial f}{\partial x}\boldsymbol{i} + \frac{\partial f}{\partial y}\boldsymbol{j} + \frac{\partial f}{\partial z}\boldsymbol{k}$$

Like usual differentiation, ∇ is a linear operator, i.e. for any functions f and g and any constants λ and μ, we have

$$\nabla (\lambda f + \mu g) = \lambda \nabla f + \mu \nabla g .$$

In fact:

$$\begin{aligned}\nabla (\lambda f + \mu g) &= \left(\frac{\partial}{\partial x}(\lambda f + \mu g), \frac{\partial}{\partial y}(\lambda f + \mu g), \frac{\partial}{\partial z}(\lambda f + \mu g) \right) \\ &= (\lambda f_x + \mu g_x, \lambda f_y + \mu g_y, \lambda f_z + \mu g_z) \\ &= \lambda (f_x, f_y, f_z) + \mu (g_x, g_y, g_z) \\ &= \lambda \nabla f + \mu \nabla g\end{aligned}$$

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Grad as normal for a level curve

Theorem: If $\boldsymbol{a} = (a, b)$ is a point on a level curve $f(x, y) = k$ and $\nabla f(a, b) \neq 0$, then the vector $\nabla f(a, b)$ is normal to the curve at the point.

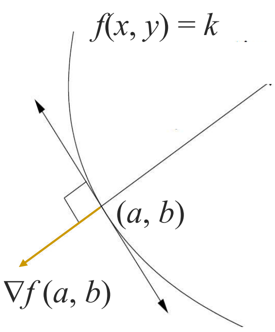
Proof: Express $f(x, y) = k$ in the parametric form:

$\boldsymbol{r}(t) = x(t)\boldsymbol{i} + y(t)\boldsymbol{j} = (x(t), y(t))$.

Its tangent is given by: $\boldsymbol{r}' = (x'(t), y'(t))$.

Every point $(x(t), y(t))$ at the curve satisfies:

$f(x(t), y(t)) = k$.



$$\frac{d}{dt} f\big(x(t), y(t)\big) = \frac{dk}{dt}$$

$$\Leftrightarrow \quad f_x x' + f_y y' = 0$$

$$\Leftrightarrow \quad (f_x, f_y) \cdot (x', y') = 0$$

$$\Leftrightarrow \quad \nabla f \cdot \boldsymbol{r}' = 0 \quad \Rightarrow \quad \nabla f(\boldsymbol{a}) \cdot \boldsymbol{r}'|_{\boldsymbol{a}} = 0$$

$\because \nabla f(\boldsymbol{a}) \neq 0, \quad \boldsymbol{r}'|_{\boldsymbol{a}} \neq 0$ with suitable parameterization

$\Rightarrow \quad \nabla f(\boldsymbol{a}) \perp \boldsymbol{r}'|_{\boldsymbol{a}}$

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Grad as normal for level surface

2654 5863

Theorem: Let $f(x, y, z)$ be sufficiently smooth and $\boldsymbol{a}$ be a point on the level surface $f(x, y, z) = k$ for some constant k. If $\nabla f(\boldsymbol{a}) \neq 0$, then the vector $\nabla f(\boldsymbol{a})$ is normal to the level surface at the point $\boldsymbol{a}$.

Proof: Draw an arbitrary curve C at the surface passing through $\boldsymbol{a}$.

Express the curve in the parametric form

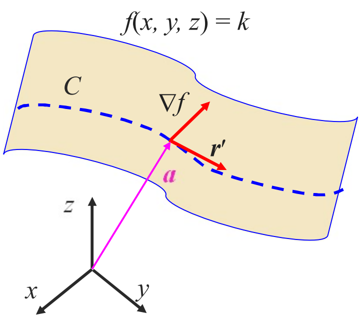
$$\boldsymbol{r}(t) = x(t)\boldsymbol{i} + y(t)\boldsymbol{j} + z(t)\boldsymbol{k}.$$

Its tangent is given by

$$\boldsymbol{r}' = (x'(t), y'(t), z'(t)).$$

Every point $(x(t), y(t), z(t))$ on the curve lies on the surface and thus satisfies the surface equation:

$$f(x(t), y(t), z(t)) = k.$$



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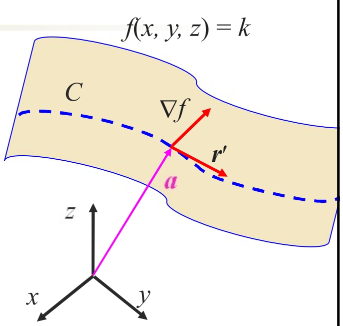
Grad as normal for level surface (cont.)

If $\boldsymbol{a}$ is a point on surface $f(x, y, z) = k$, then $\nabla f(\boldsymbol{a})$ is normal to the surface at the point $\boldsymbol{a}$.



Proof: $f(x(t), y(t), z(t)) = k$

$$\frac{d}{dt} f\left(x(t), y(t), z(t)\right) = \frac{dk}{dt}$$

$$\Leftrightarrow \quad f_x x' + f_y y' + f_z z' = 0$$

$$\Leftrightarrow \quad \left(f_x, f_y, f_z\right)\cdot\left(x', y', z'\right) = 0$$

$$\Leftrightarrow \quad \nabla f \cdot \boldsymbol{r}' = 0 \quad \Rightarrow \quad \nabla f(\boldsymbol{a})\cdot \boldsymbol{r}'\big|_{\boldsymbol{a}} = 0$$

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$\because \nabla f(\boldsymbol{a}) \neq 0, \quad \boldsymbol{r}'\big|_{\boldsymbol{a}} \neq 0$ with suitable parameterization

$$\Rightarrow \quad \nabla f(\boldsymbol{a}) \perp \boldsymbol{r}'\big|_{\boldsymbol{a}}$$

For an arbitrary curve on the surface passing $\boldsymbol{a}$, its tangent at $\boldsymbol{a}$ is perpendicular to $\nabla f(\boldsymbol{a})$.

$\nabla f(\boldsymbol{a})$ is therefore the normal of the surface at $\boldsymbol{a}$.

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Example 2 $\boldsymbol{r}\cdot\boldsymbol{n} = \boldsymbol{a}\cdot\boldsymbol{n}$

Find the tangent plane for the cone $z^2 = 4(x^2+y^2)$ at point P: (1, 0, 2).

Solution. Express a surface as a level surface $f(x, y, z) = 0$, then its normal at a point is given by grad f. The surface considered can be expressed as

$$f(x, y, z) = 4(x^2+y^2)- z^2 = 0 .$$

Calculate its gradient

$$\text{grad} f = (f_x, f_y, f_z) = (8x, 8y, -2z)$$

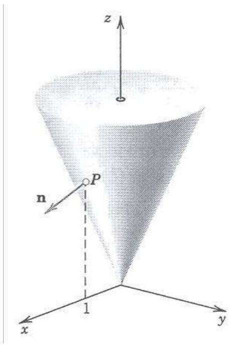
The normal to the surface & tangent plane at P, $\boldsymbol{n}$ = grad $f(1, 0, 2) = (8, 0, -4)$.

The tangent plane

$$\boldsymbol{r}\cdot\boldsymbol{n} = \boldsymbol{a}\cdot\boldsymbol{n}$$

$\Rightarrow$ $(x, y, z) \cdot (8, 0, -4) = (1, 0, 2) \cdot (8, 0, -4)$

i.e. $8x-4z=0$ $\Rightarrow 2x-z=0$.



Cone and unit normal vector **n**

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Example 3

Find a tangent vector to the intersection of the two surfaces:

Surface 1: $z = x^2-y^2$ and Surface 2: $xyz = -30$

at the point $\boldsymbol{a}$ = (-3, 2, 5).

Solution. Denote the tangent of the intersection at the point as **T**. Denote the normal vectors of the two surfaces at the point $\boldsymbol{a}$ as $\boldsymbol{n}_1$ and $\boldsymbol{n}_2$.

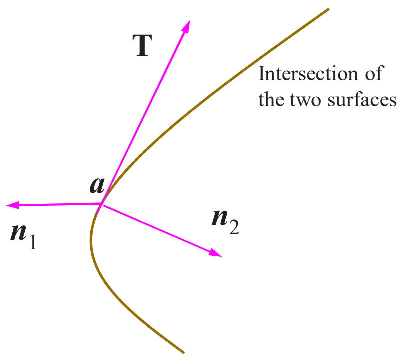
As the intersection is on surface 1,

$$\boldsymbol{T} \perp \boldsymbol{n}_1.$$

As the intersection is on surface 2,

$$\boldsymbol{T} \perp \boldsymbol{n}_2.$$

As such $\boldsymbol{T} = \boldsymbol{n}_1 \times \boldsymbol{n}_2$.



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Example 3 (cont.)

Surface 1: $z = x^2 - y^2$ $\Leftrightarrow$ Level surface $f = x^2 - y^2 - z = 0$.
Surface 2: $xyz = -30$ $\Leftrightarrow$ Level surface $g = xyz = -30$.
At the point (-3, 2, 5).
Solution. The normal $\boldsymbol{n}_1$ and $\boldsymbol{n}_2$ of the two surfaces at the point are

$$\mathbf{n}_1 = \nabla f\big|_{(-3,2,5)} = \nabla\left(x^2 - y^2 - z\right)\big|_{(-3,2,5)}$$

$$= \left(2x\mathbf{i} - 2y\mathbf{j} - \mathbf{k}\right)\big|_{(-3,2,5)} = -6\mathbf{i} - 4\mathbf{j} - \mathbf{k}$$

$$\mathbf{n}_2 = \nabla g\big|_{(-3,2,5)} = \nabla\left(xyz\right)\big|_{(-3,2,5)}$$

$$= \left(yz\mathbf{i} + xz\mathbf{j} + xy\mathbf{k}\right)\big|_{(-3,2,5)} = 10\mathbf{i} - 15\mathbf{j} - 6\mathbf{k}$$

$$\boldsymbol{T} = \boldsymbol{n}_1 \times \boldsymbol{n}_2 = \begin{vmatrix} \boldsymbol{i} & \boldsymbol{j} & \boldsymbol{k} \\ -6 & -4 & -1 \\ 10 & -15 & -6 \end{vmatrix} = 9\boldsymbol{i} - 46\boldsymbol{j} + 130\boldsymbol{k}$$

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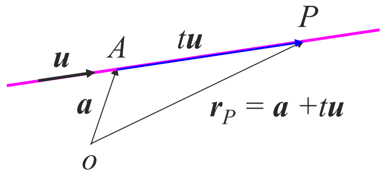
Directional derivatives

Consider a function $f(x, y, z)$ at point A: $\boldsymbol{a} = (a, b, c)$.

We want to find the rate of change of f in the direction of $\boldsymbol{u} = (u_1, u_2, u_3)$, where $\boldsymbol{u}$ is a unit vector.

Draw a line AP along $\boldsymbol{u}$. Denote the distance of AP as t

$$\left|\overrightarrow{AP}\right| = t, \quad \overrightarrow{AP} = t\boldsymbol{u}.$$



Point P is at $\boldsymbol{r}_P = \boldsymbol{a} + \overrightarrow{AP} = \boldsymbol{a} + t\boldsymbol{u}$

$$= (a,b,c) + t(u_1,u_2,u_3)$$

$$= (a + tu_1, b + tu_2, c + tu_3)$$

The directional derivative of f at $\boldsymbol{a}$ in the direction of $\boldsymbol{u}$ is defined as

$$\lim_{P\to A}\frac{f(P) - f(A)}{\left|\overrightarrow{AP}\right|} = \lim_{t\to 0}\frac{f(\boldsymbol{r}_P) - f(\boldsymbol{a})}{t}$$

$$= \lim_{t\to 0}\frac{f(a + tu_1, b + tu_2, c + tu_3) - f(a,b,c)}{t} \equiv D_{\boldsymbol{u}}f(a,b,c)$$

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Directional derivatives (cont.)

$$D_{\boldsymbol{u}} f(a,b,c) = \lim_{t\to 0}\frac{f(a+tu_1, b+tu_2, c+tu_3) - f(a,b,c)}{t}$$

Denote $F(t) = f(a+tu_1, b+tu_2, c+tu_3)$, we have $F(0) = f(a,b,c)$

$$D_u f(a,b,c) = \lim_{t\to 0}\frac{F(t)-F(0)}{t} = \left.\frac{dF(t)}{dt}\right|_{t=0}$$

$$= \left.\frac{d}{dt} f(a+tu_1, b+tu_2, c+tu_3)\right|_{t=0}$$

$$= \left\{ \begin{array}{l} \dfrac{\partial}{\partial x} f(a+tu_1, b+tu_2, c+tu_3)\dfrac{d(a+tu_1)}{dt} \\ +\dfrac{\partial f(\cdots)}{\partial y}\dfrac{d(b+tu_2)}{dt} + \dfrac{\partial f(\cdots)}{\partial z}\dfrac{d(c+tu_3)}{dt} \end{array} \right\}_{t=0}$$

$$= u_1\frac{\partial}{\partial x} f(a,b,c) + u_2\frac{\partial}{\partial y} f(a,b,c) + u_3\frac{\partial}{\partial x} f(a,b,c)$$

$$= (u_1, u_2, u_3)\cdot\left(\frac{\partial f(a,b,c)}{\partial x}, \frac{\partial f(a,b,c)}{\partial y}, \frac{\partial f(a,b,c)}{\partial z}\right) = \boldsymbol{u}\cdot\nabla f(a,b,c)$$

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Relation between gradient and directional derivative

The directional derivative of $f(x, y, z)$ at a point $\boldsymbol{a} = (a, b, c)$ in the direction of a unit vector $\boldsymbol{u} = u_1\boldsymbol{i} + u_2\boldsymbol{j} + u_3\boldsymbol{k}$ is given by

$$\boxed{D_{\boldsymbol{u}} f(\boldsymbol{a}) = \boldsymbol{u}\cdot\nabla f(\boldsymbol{a})}$$

The directional derivative equals the dot product of the unit directional vector and the gradient.

The directional derivative is a scalar, not a vector.

$D_{\boldsymbol{u}}$ is linear, i.e. for any functions f and g and any constants λ and μ,

$$D_{\boldsymbol{u}}(\lambda f + \mu g) = \lambda D_{\boldsymbol{u}} f + \mu D_{\boldsymbol{u}} g .$$

In fact

$$D_{\boldsymbol{u}}(\lambda f + \mu g) = \boldsymbol{u}\cdot\nabla(\lambda f + \mu g) \quad (\text{as } \nabla \text{ is an liner operator})$$

$$= \boldsymbol{u}\cdot(\lambda\nabla f + \mu\nabla g)$$

$$= \lambda\boldsymbol{u}\cdot\nabla f + \mu\boldsymbol{u}\cdot\nabla g = \lambda D_{\boldsymbol{u}} f + \mu D_{\boldsymbol{u}} g$$

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Example 4 $D_{\boldsymbol{u}} f(\boldsymbol{a}) = \boldsymbol{u} \cdot \nabla f(\boldsymbol{a})$

Find the directional derivative for $f = xy+yz+zx$ at the point (1, 1, 1) in the direction from this point to (2, 2, 3).

Solution: The directional derivative is: $D_{\mathbf{u}}f(1, 1, 1) = \mathbf{u}\cdot\nabla f(1, 1, 1)$,

where **u** is a unit vector.

The directional vector from (1, 1, 1) to (2, 2, 3) is

$\mathbf{U} = (2, 2, 3) - (1, 1, 1) = (1, 1, 2)$.

The corresponding unit vector is $\boldsymbol{u} = \dfrac{(1,1,2)}{\sqrt{1^2+1^2+2^2}} = \dfrac{1}{\sqrt{6}}(1,1,2)$

The gradient of f is $\nabla f = \left(f_x, f_y, f_z\right) = \left(y+z, z+x, x+y\right)$

$$\nabla f(1,\ 1,\ 1) = (2,2,2)$$

Finally we have the directional derivative

$$D_{\mathbf{u}} f(1,1,1) = \mathbf{u}\cdot\nabla f(1,1,1) = \frac{1}{\sqrt{6}}(1,1,2)\cdot(2,2,2) = \frac{2+2+4}{\sqrt{6}} = \frac{8}{\sqrt{6}}$$

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Example 5 **4034 7593**

Consider $f(x, y, z) = xyz$ at point $\boldsymbol{a} = (2, 1, 1)$, Find

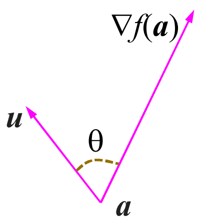
(a) The direction and rate of steepest ascent for f at $\boldsymbol{a}$.

(b) The direction and rate of steepest descent for f at $\boldsymbol{a}$.

(c) The equation for a level surface passing through the point with the normal $\nabla f(\boldsymbol{a})$.

Soln. Consider the directional derivative of the function along direction $\boldsymbol{u}$ ($u = 1$) at the point $\boldsymbol{a}$

$$D_{\boldsymbol{u}} f(\boldsymbol{a}) = \boldsymbol{u}\cdot\nabla f(\boldsymbol{a}) = u\left|\nabla f(\boldsymbol{a})\right|\cos\theta = \left|\nabla f(\boldsymbol{a})\right|\cos\theta,$$



where θ is the angle between $\boldsymbol{u}$ and $\nabla f(\boldsymbol{a})$.

$$\nabla f = \frac{\partial(xyz)}{\partial x}\boldsymbol{i} + \frac{\partial(xyz)}{\partial y}\boldsymbol{j} + \frac{\partial(xyz)}{\partial z}\boldsymbol{k}$$

$$= yz\boldsymbol{i} + zx\boldsymbol{j} + xy\boldsymbol{k}$$

$$\nabla f(\boldsymbol{a}) = \nabla f(2,1,1) = \boldsymbol{i} + 2\boldsymbol{j} + 2\boldsymbol{k}, \qquad \left|\nabla f(\boldsymbol{a})\right| = 3.$$

$$D_{\boldsymbol{u}} f(\boldsymbol{a}) = \left|\nabla f(\boldsymbol{a})\right|\cos\theta = 3\cos\theta.$$

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Example 5 (cont.)

4034 7593

(a) The direction and rate of steepest ascent for *f* at $\boldsymbol{a}$.
(b) The direction and rate of steepest descent for *f* at $\boldsymbol{a}$.
(c) Write the equation for a level surface at the point normal to $\nabla f(\boldsymbol{a})$.

Soln. $D_u f(\boldsymbol{a}) = 3\cos\theta.$

(a) As $\theta = 0$, $\boldsymbol{u}$ in the direction $\nabla f(\boldsymbol{a}) = \boldsymbol{i}+2\boldsymbol{j}+2\boldsymbol{k}$, *f* has its steepest ascent. The maximum rate is 3.

(b) As $\theta = \pi$, $\boldsymbol{u}$ is in the direction of $-\nabla f(\boldsymbol{a}) = -\boldsymbol{i}-2\boldsymbol{j}-2\boldsymbol{k}$, *f* has its steepest descent. The minimum rate is -3.

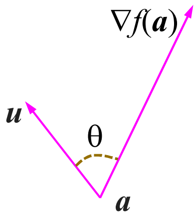
(c) A level surface normal to $\nabla f(\boldsymbol{a})$ is given as

$f(x, y, z) = xyz = k.$

As point $\boldsymbol{a} = (2, 1, 1)$ is on the surface, thus $k = 2$.

The level surface is:

$xyz = 2.$



Steepest ascent theorem

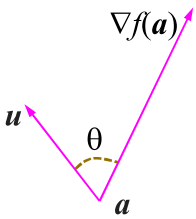
Consider a function $f(x, y, z)$ at a point $\boldsymbol{a} = (a, b, c)$. We want to find the direction along which the function increases or decreases most rapidly most rapidly at the point. Let $\boldsymbol{u}$ be a unit vector,

$$D_u f(\boldsymbol{a}) = \boldsymbol{u}\cdot\nabla f(\boldsymbol{a}) = |\boldsymbol{u}||\nabla f(\boldsymbol{a})|\cos\theta = |\nabla f(\boldsymbol{a})|\cos\theta, \quad \text{as } u = 1,$$

where θ is the angle between $\boldsymbol{u}$ and $\nabla f(\boldsymbol{a})$.

(1) As $\theta = 0$, i.e. $\boldsymbol{u}$ and $\nabla f(\boldsymbol{a})$ in the same direction:
$\nabla f(\boldsymbol{a})$ is in the direction of steepest ascent of *f* at $\boldsymbol{a}$.
$|\nabla f(\boldsymbol{a})|$ is the rate of steepest ascent.

(2) As $\theta = \pi$, i.e. $\boldsymbol{u}$ is opposite to $\nabla f(\boldsymbol{a})$:
$-\nabla f(\boldsymbol{a})$ is in the direction of steepest descent of *f* at $\boldsymbol{a}$.
$-|\nabla f(\boldsymbol{a})|$ is the rate of steepest decent.



Example 5

Consider $f(x, y, z) = xyz$ at point $\boldsymbol{a} = (2, 1, 1)$, Find

(a) The direction and rate of steepest ascent for f at $\boldsymbol{a}$.

(b) The direction and rate of steepest descent for f at $\boldsymbol{a}$.

Soln.
$$\nabla f = \frac{\partial(xyz)}{\partial x}\boldsymbol{i} + \frac{\partial(xyz)}{\partial y}\boldsymbol{j} + \frac{\partial(xyz)}{\partial z}\boldsymbol{k}$$
$$= yz\boldsymbol{i} + zx\boldsymbol{j} + xy\boldsymbol{k}$$
$$\nabla f(\boldsymbol{a}) = \nabla f(2,1,1) = \boldsymbol{i} + 2\boldsymbol{j} + 2\boldsymbol{k}, \qquad |\nabla f(\boldsymbol{a})| = 3.$$

(a)The direction and rate of steepest ascent for f at $\boldsymbol{a}$ are
$$\nabla f(\boldsymbol{a}) = \nabla f(2,1,1) = \boldsymbol{i} + 2\boldsymbol{j} + 2\boldsymbol{k}, \qquad |\nabla f(\boldsymbol{a})| = 3.$$

(b) The direction and rate of steepest descent for f at $\boldsymbol{a}$ are
$$-\nabla f(\boldsymbol{a}) = -(\boldsymbol{i} + 2\boldsymbol{j} + 2\boldsymbol{k}), \qquad -|\nabla f(\boldsymbol{a})| = -3.$$

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Example. A1

Show product and quotient rules for partial derivatives using the chain rule.

For $u = u(x, y)$ and $v = v(x, y)$, using the chain rule, we have
$$\frac{\partial f(u,v)}{\partial x} = \frac{\partial f}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial f}{\partial v}\frac{\partial v}{\partial x},$$
Let $f = uv$:
$$\frac{\partial(uv)}{\partial x} = \frac{\partial(uv)}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial(uv)}{\partial v}\frac{\partial v}{\partial x} = \frac{\partial u}{\partial x}v + u\frac{\partial v}{\partial x},$$
Let $f = u/v$:
$$\frac{\partial}{\partial x}\left(\frac{u}{v}\right) = \frac{\partial}{\partial u}\left(\frac{u}{v}\right)\frac{\partial u}{\partial x} + \frac{\partial}{\partial v}\left(\frac{u}{v}\right)\frac{\partial v}{\partial x} = \frac{1}{v}\frac{\partial u}{\partial x} - \frac{u}{v^2}\frac{\partial v}{\partial x} = \frac{1}{v^2}\left(\frac{\partial u}{\partial x}v - u\frac{\partial v}{\partial x}\right).$$
i.e. $$\frac{\partial(uv)}{\partial x} = \frac{\partial u}{\partial x}v + u\frac{\partial v}{\partial x}, \qquad \frac{\partial}{\partial x}\left(\frac{u}{v}\right) = \frac{1}{v^2}\left(\frac{\partial u}{\partial x}v - u\frac{\partial v}{\partial x}\right).$$

One can show the same results for partial derivatives w.r.t y.

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II. Integration of Functions of Several Real Variables

2.1 Double integrals (Adams, 14.1, 14.2)

2.2 Triple integrals (Adams, 14.5)

This chapter is for 4 lectures in Week 3.

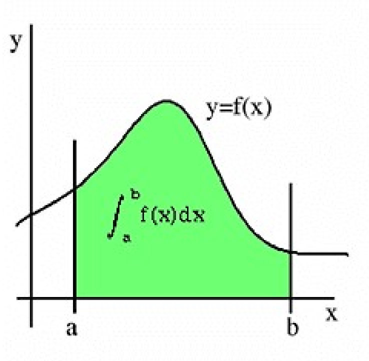
2.1 Double integrals

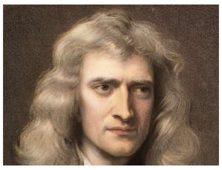
Integrals of single variable functions.

The integral of a function $f(x)$ over an interval $[a, b]$,

$$\int_a^b f(x)dx,$$

is the area below the graph of $y = f(x)$ and above the interval $[a, b]$.





Isaac Newton (1643-1727)

Gottfried Wilhelm Leibnitz (1646-1716)

Single integrals

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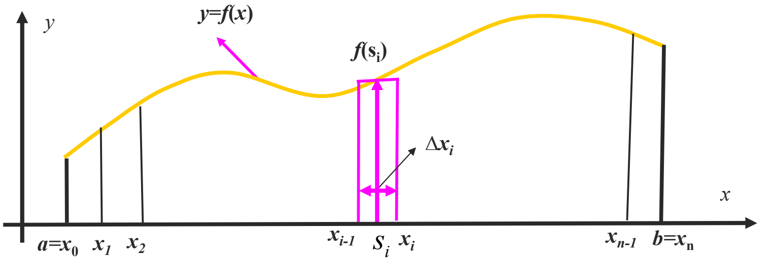
1. Divide $[a, b]$ into n segments using: $a = x_0 < x_1 < \cdots < x_n = b$.
2. Calculate the subdomain area above $[x_{i-1}, x_i]$:

   Choose $s_i \in [x_{i-1}, x_i]$ & use $f(s_i)$ to approximate $f(x)$ in the segment,

   Approximating the subdomain as a rectangle with area $A_i = f(s_i)\,\Delta x_i$.
3. Form the Riemann sum by adding the subdomain areas: $\sum_{i=1}^{n} f(s_i)\Delta x_i$.
4. As max $(\Delta x_i) \to 0$, the Riemann sum $\to$ the domain area:

$$\lim_{\max \Delta x_i \to 0} \sum_{i=1}^{n} f(s_i)\Delta x_i \equiv \int_a^b f(x)dx$$



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Definition of single integrals

Consider a function $f(x)$ in an interval $[a, b]$. After the partition

$$a = x_0 < x_1 < \cdots < x_n = b,$$

we form the Riemann sum $\sum_{i=1}^{n} f(s_i)\Delta x_i$,

where

$$\Delta x_i = x_i - x_{i-1} \text{ and } s_i \in [x_{i-1}, x_i].$$

If $\lim_{\max \Delta x_i \to 0} \sum_{i=1}^{n} f(s_i)\Delta x_i$ exists, $f(x)$ is integrable in [a, b] and the limit is the integral of $f(x)$ over $[a, b]$, i.e.

$$\int_a^b f(x)dx \equiv \lim_{\max \Delta x_i \to 0} \sum_{i=1}^{n} f(s_i)\Delta x_i.$$

Georg Friedrich Bernhard Riemann (1826-1866)

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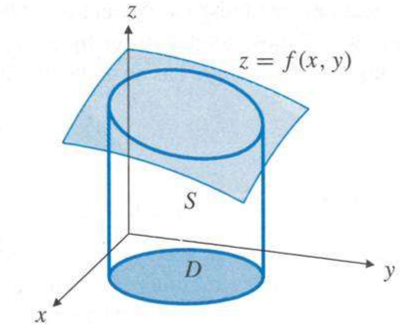
Geometrical interpretation of double integrals

The double integral of $f(x, y)$ in $D \in \mathbb{R}^2$, $\iint_D f(x,y)\,dxdy$,

is the volume of a solid region S lying above domain D in the xy-plane and below the surface $z = f(x, y)$.



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Partition of domain

Consider a rectangular domain D defined by:

$a \le x \le b,\ c \le y \le d.$

Draw a mesh as follows for the domain D:

$a = x_0 < x_1 < \cdots < x_n = b,$

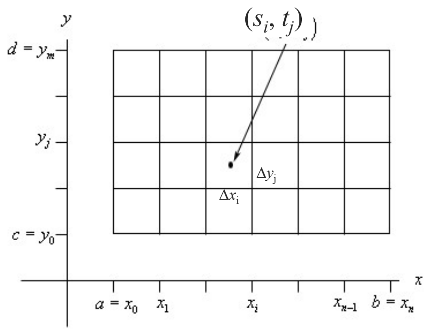
$c = y_0 < y_1 < \cdots < y_m = d.$

For each element of the mesh with sides

$$\Delta x_i = x_i - x_{i-1},\ \Delta y_j = y_j - y_{j-1}$$

choose a point (s_i, t_j)

$$s_i \in [x_{i-1}, x_i],\quad t_j \in [y_{j-1}, y_j]$$



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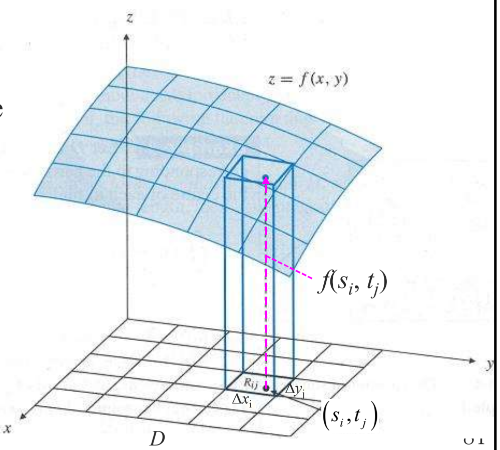
Volume above each element below the surface

Within the element with the sides Δx_i and Δy_j:

- $f(x, y) \approx f(s_i, t_j)$, which is the function value at the representing point of the element
- The curved surface is approximated by a horizontal plane at the height $f(s_i, t_j)$.

The volume below the surface and above the element is thus approximated by a cuboid, with the volume given by

$$f(s_i, t_j)\, \Delta x_i\, \Delta y_j.$$



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Riemann sum and accurate volume

Summarizing the volumes of cuboids above all elements, we form the Riemann sum,

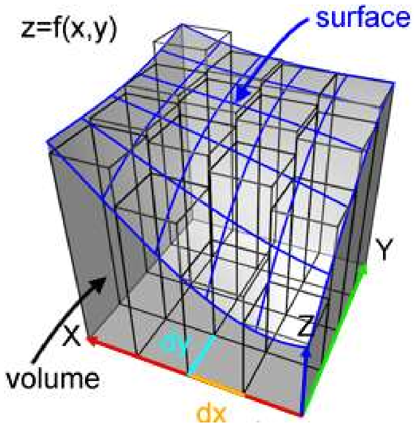
$$\sum_{j=1}^{m}\sum_{i=1}^{n} f\left(s_i, t_j\right) \Delta x_i \Delta y_j .$$

The sum approximates the volume of the solid S lying above the domain D in the xy-plane below the surface $z = f(x, y)$.

The limit of the sum,

$$\lim_{\max\{\Delta x_i, \Delta y_j\}\to 0} \sum_{j=1}^{m}\sum_{i=1}^{n} f\left(s_i, t_j\right) \Delta x_i \Delta y_j$$

is the volume of a solid region S.



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Definition of double integrals

For a function $f(x, y)$ in a rectangular domain D,

$D=\{(x, y): a \le x \le b,\ c \le y \le d\}$,

we mesh the domain using

$x = x_i,\ i = 0, 1, \cdots, n.$

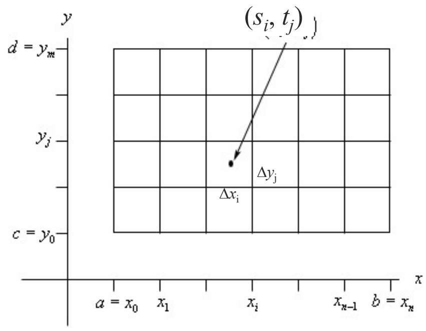
$y = y_j,\ j = 0, 1, \cdots, m.$

We then form the Riemann sum

$$\sum_{j=1}^{m}\sum_{i=1}^{n} f\left(s_i, t_j\right) \Delta x_i \Delta y_j,$$

where $\Delta x_i = x_i - x_{i-1},\ \ \Delta y_j = y_j - y_{j-1}$,

$s_i \in [x_{i-1}, x_i], \quad t_j \in \left[y_{j-1}, y_j\right]$



As $\max\left\{\Delta x_i, \Delta y_j\right\} \to 0$, if the Riemann sum has a limit, we say $f(x, y)$ is integrable in D, and the integral $f(x, y)$ over D is the limit, i.e.

$$\iint_D f(x, y) dxdy \equiv \lim_{\max\left\{\Delta x_i, \Delta y_j\right\} \to 0} \sum_{j=1}^{m}\sum_{i=1}^{n} f\left(s_i, t_j\right) \Delta x_i \Delta y_j \ .$$

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General domains

For function $f(x, y)$ over a non-rectangular and confined D:

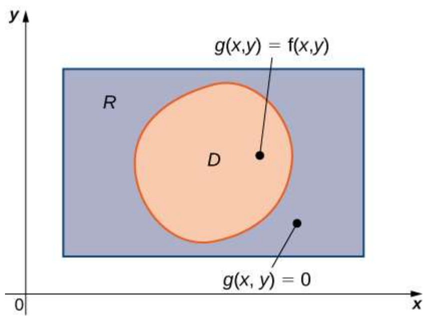
(i) Choose a rectangular domain R containing D, with sides parallel to the coordinate axes.

(ii) Extend $f(x, y)$ to $g(x, y)$ defined on R defined as

$$g(x, y) = \begin{cases} f(x, y), & \text{if } (x, y) \in D, \\ 0, & \text{if } (x, y) \notin D. \end{cases}$$

(iii) If $g(x, y)$ is integrable over R, then the integral of $f(x, y)$ over D is defined as follows:

$$\iint_D f(x, y)\ dxdy \equiv \iint_R g(x, y)\ dxdy$$



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Remarks

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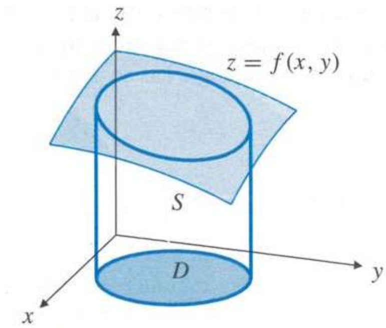
1. If D is a region of $\mathbb{R}^2$, then $\iint_D f(x,y)\,dxdy$

is the volume of a solid region S lying above domain D in the xy-plane and below the surface $z = f(x, y)$.

2. $\iint_D 1\,dxdy = \text{Area of } D$

3. In the figures shown before, $f(x, y)$ is larger than 0. However, the same definition applies to non-positive functions.



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Facts of double integrals

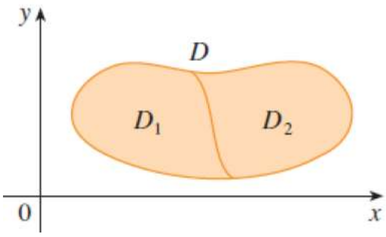
1. Multiple-integrals are linear to functions. Suppose f and g are continuous functions, c and d are constants, then

$$\iint_D [cf(x,y)+dg(x,y)]dxdy = c\iint_D f(x,y)dxdy + d\iint_D g(x,y)dxdy$$

2. Let $D = D_1 \cup D_2$, where D_1 and D_2 be non-overlapping regions, and $f(x, y)$ be continuous, then

$$\iint_D f(x,y)dxdy = \iint_{D_1} f(x,y)dxdy + \iint_{D_2} f(x,y)dxdy \; .$$

3. A continuous function is always integrable over a bounded domain.



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Basic idea for calculating double integrals

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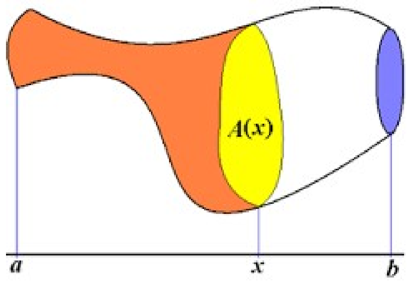
We will illustrate how to calculate a double integral by calculating the volume of a solid.

To calculate the volume of a solid:

Step 1. Consider its intersection with the plane

$$x = x,$$

we calculate the area of the cross-section $A(x)$. **In this step, x is treated as a constant.**



Step 2. $A(x)dx$ is the volume of a thin slice of area $A(x)$ and thickness dx. The volume of the solid S is obtained by summing the volumes of all such thin slices.

$$V = \int_a^b A(x)\,dx$$

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Geometrical interpretation of Fubini's theorem

9261 4514

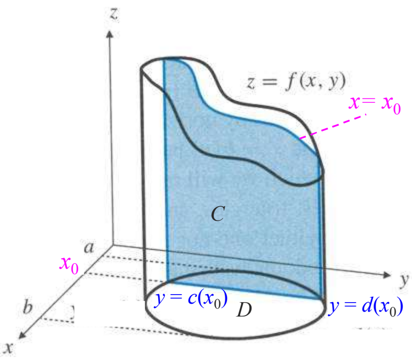
Consider the solid below $z = f(x, y)$ above domain D in the xy plane.

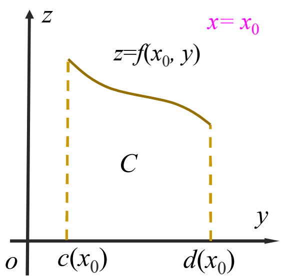
Step 1: Consider the cross-section C (shaded in the figure) of the solid S with a vertical plane $x = x_0$:

$$C = \{(y,z) : c(x_0) \le y \le d(x_0),\ 0 \le z \le f(x_0, y)\}$$

where $c(x_0)$ and $d(x_0)$ are the intersections of the boundary of D with the plane $x = x_0$. The area of the cross-section C is

$$A(x_0) = \int_{c(x_0)}^{d(x_0)} f(x_0, y)\,dy$$





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Geometrical interpretation of Fubini's theorem

Consider the volume below $z = f(x, y)$ above domain D in the xy plane.

Step 1: In the last slide we considered the cross-section of the solid with a vertical plane $x = x_0$ and found its area

$$A(x_0) = \int_{c(x_0)}^{d(x_0)} f(x_0, y)\, dy$$

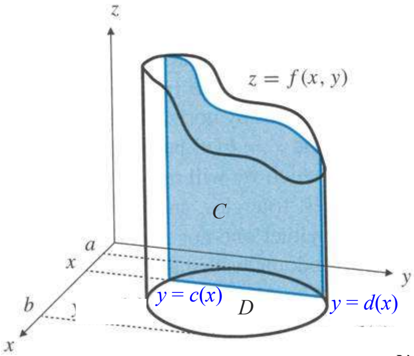
The area of the cross-section C with a vertical plane $x = x$

$$A(x) = \int_{c(x)}^{d(x)} f(x, y)\, dy$$

In this step, x is treated as constant.

Step 2: The volume of the solid S

$$V = \int_a^b A(x) dx = \int_a^b \left(\int_{c(x)}^{d(x)} f(x, y)\, dy \right) dx$$



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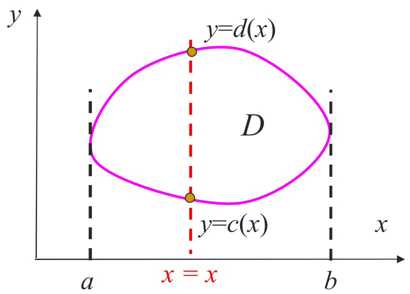
Fubini's theorem

$$\iint_D f(x, y) dxdy = \int_a^b \left(\int_{c(x)}^{d(x)} f(x, y)\, dy \right) dx \quad D = \{(x, y):\ a \le x \le b, c(x) \le y \le d(x)\}$$

- a and b are the minimum and maximum values of x on the boundary of the domain D.
- $y = c(x)$ and $y = d(x)$ are the points where the domain boundary intersects $x = x$.
- In the inner integral we treat x as constant, because we are calculating the cross-section area of the solid with the plane $x = x$.



Guido Fubini
(1879 – 1943), Italian

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Fubini's theorem (cont.)

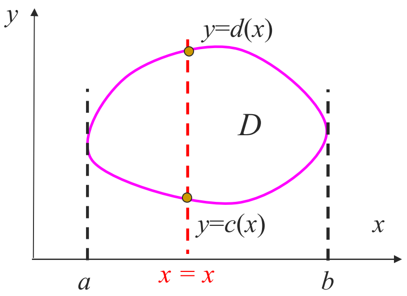
Step 1, we first describe the domain D as

$$D=\{(x,y):\ a\le x\le b, c(x)\le y\le d(x)\}$$

- a and b are the minimum and maximum values of x on the boundary of the domain D.
- $y = c(x)$ and $y = d(x)$ are the lower and upper boundary curves of the domain D.

Step 2: Express double integral as repeated integrals.

$$\iint_D f(x,y)dxdy=\int_a^b\left(\int_{c(x)}^{d(x)} f(x,y)\,dy\right)dx$$



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Example 1.

$$D=\{(x,y):a\le x\le b,\ c(x)\le y\le d(x)\},$$

$$\iint_D f(x,y)dxdy=\int_a^b\left(\int_{c(x)}^{d(x)} f(x,y)\,dy\right)dx$$

Calculate $I=\iint_Q(4-x-y)\,dxdy$

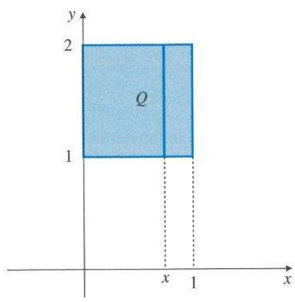
where $Q=\{(x,y):\ 0\le x\le 1,\ 1\le y\le 2\}$.

Solution. The domain is a rectangle.

$$I=\int_{x=0}^{x=1}\left(\int_{y=1}^{y=2}(4-x-y)\,dy\right)dx=\int_{x=0}^{x=1}A(x)\,dx$$

$$A(x)=\int_{y=1}^{y=2}(4-x-y)\,dy=\left(4y-xy-\frac{y^2}{2}\right)_{y=1}^{y=2}$$

$$=(8\text{-}2x\text{-}2)-(4-x-1/2)=\frac{5}{2}-x.$$



Treat x as a constant while integrate to y.

$$I=\int_{x=0}^{x=1}A(x)\,dx=\int_0^1\left(\frac{5}{2}-x\right)dx=\left(\frac{5x}{2}-\frac{x^2}{2}\right)_{x=0}^{x=1}=2$$

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Example 2

$$D=\{(x,y):\ x_{\min}\le x\le x_{\max},\, c(x)\le y\le d(x)\},$$

$$\iint_D f(x,y)\,dxdy=\int_{x_{\min}}^{x_{\max}}\left(\int_{c(x)}^{d(x)} f(x,y)\,dy\right)dx$$

Find $I=\iint_T xy\,dxdy$, where the triangle T with vertices (0,0), (1,0), (1,1).

Solution. Draw the domain. Determine the range of x in the domain T:

$$0=x_{\min}\le x\le x_{\max}=1.$$

where $x_{\min}$ and $x_{\max}$ are the minimum and maximum values of x achieved on the domain boundary.

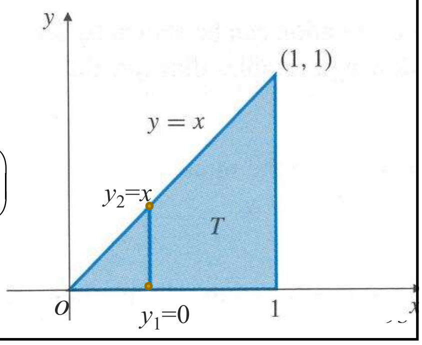
To determine the range for y for a given value of x, draw the line $x = x$ parallel to the y-axis crosses the boundary at y_1 and y_2:

$$0=y_1\le y\le y_2=\text{x}.$$

$$I=\int_0^1\left(\int_0^x xy\,dy\right)dx\quad (\text{Using Fubini's thm.})$$

$$=\int_0^1 x\left(\frac{y^2}{2}\right)_{y=0}^{y=x}dx\quad \left(\begin{array}{l}\text{Treat } x \text{ as a contant while}\\ \text{doing integral to } y.\end{array}\right)$$

$$=\int_0^1 x\left(\frac{x^2}{2}-0\right)dx=\frac{1}{8}$$



Convention for writing repeated integrals

The following convention is often used in multivariable integrals:

$$\int_a^b A(x)\,dx\equiv\int_a^b dx\,A(x)\qquad A(x)=\int_{c(x)}^{d(x)} f(x,y)\,dy$$

$$\int_a^b\left(\int_{c(x)}^{d(x)} f(x,y)\,dy\right)dx\equiv\int_a^b dx\int_{c(x)}^{d(x)} f(x,y)\,dy$$

Using this convention, the solution to example 1 is given as following

$$I=\int_0^1\left(\int_0^x xy\,dy\right)dx\equiv\int_0^1 dx\int_0^x xy\,dy$$

$$=\int_0^1 dx\,x\left(\frac{y^2}{2}\right)_{y=0}^{y=x}=\int_0^1 dx\,\frac{x^3}{2}$$

$$\equiv\int_0^1\frac{x^3}{2}dx=\frac{1}{8}$$

$$I=\int_0^1\left(\int_0^x xy\,dy\right)dx\equiv\int_0^1 dx\int_0^x xy\,dy$$

$$\equiv\int_0^1 x\left(\frac{y^2}{2}\right)_{y=0}^{y=x}dx=\int_0^1\frac{x^3}{2}dx=\frac{1}{8}$$

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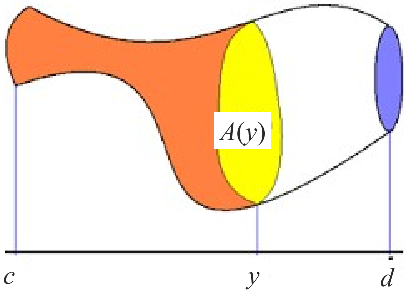
Fubini's theorem (cont.)

Alternatively, we can consider the cross-section of a solid with a vertical plane $y = y$ and find the cross-section area $A(y)$.
We then calculate the volume of the solid by integrating the cross-sectional area $A(y)$ to y from c to d.



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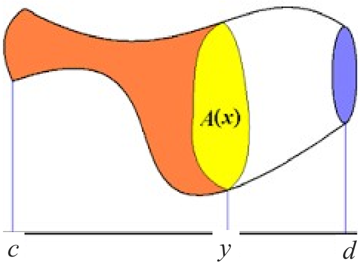
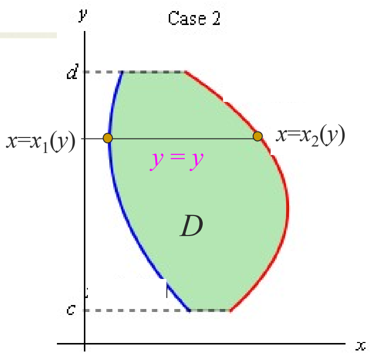
Fubini's theorem (cont.)





When D is given as

$$D = \{(x, y) : c \le y \le d, x_1(y) \le x \le x_2(y)\},$$

where $x_1(y)$ and $x_2(y)$ are smooth, then

$$\boxed{\iint_D f(x, y)\, dxdy = \int_c^d \left(\int_{x_1(y)}^{x_2(y)} f(x, y)\, dx \right) dy}$$

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Example 1: method 2

$$D=\{(x,y):\ c\le y\le d,\, x_1(y)\le x\le x_2(y)\},$$

$$\iint_D f(x,y)\,dxdy=\int_c^d\left(\int_{x_1(y)}^{x_2(y)} f(x,y)\,dx\right)dy$$

Find $I=\iint_Q (4-x-y)\,dxdy$,

$Q=\{(x,y):0\le x\le 1,\ 1\le y\le 2\}.$

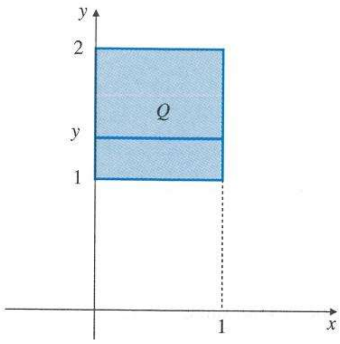
Solution. Integrating in x firstly.

$$I=\int_{y=1}^{y=2}\left(\int_{x=0}^{x=1}(4-x-y)\,dx\right)dy=\int_{y=1}^{y=2}A(y)\,dy$$

$$A(y)=\int_{x=0}^{x=1}(4-x-y)\,dx=\left(4x-\frac{x^2}{2}-xy\right)_{x=0}^{x=1}$$

$$=\left(4-\frac{1}{2}-y\right)-0=\frac{7}{2}-y$$

$$I=\int_{y=1}^{y=2}\left(\frac{7}{2}-y\right)dy=\left(\frac{7y}{2}-\frac{y^2}{2}\right)_{y=1}^{y=2}=2$$



The order of integration does not matter.
Choose the order that leads to an easier integral.

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Example 2. (cont.)

$$D=\{(x,y):\ c\le y\le d,\, x_1(y)\le x\le x_2(y)\},$$

$$\iint_D f(x,y)\,dxdy=\int_c^d\left(\int_{x_1(y)}^{x_2(y)} f(x,y)\,dx\right)dy$$

Soln 2. Determine the range of y:

$$0=y_{\min}\le y\le y_{\max}=1.$$

where $y_{\min}$ and $y_{\max}$ are the minimum and maximum values of y achieved on the domain boundary.

To determine the range for x for a given value of y, draw the line $y=y$ parallel to the x-axis, which crosses the boundary at x_1 and x_2:

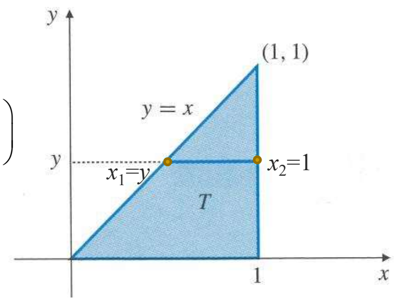
$$y=x_1\le x\le x_2=1.$$

$$I=\int_0^1\left(\int_y^1 xy\,dx\right)dy\quad\text{(Using Fubini's thm.)}$$

$$\equiv\int_0^1 dy\int_y^1 xy\,dx\quad\left(\begin{array}{l}\text{Write the inner integral behind}\\ dy\text{ using the convention.}\end{array}\right)$$

$$=\int_0^1 y\left(\frac{x^2}{2}\right)_{x=y}^{x=1}dy=\int_0^1 y\left(\frac{1}{2}-\frac{y^2}{2}\right)dy$$

$$=\left(\frac{y^2}{4}-\frac{y^4}{8}\right)_0^1=\frac{1}{8}$$



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Example 3

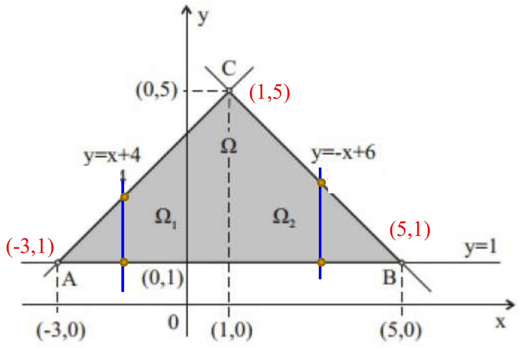
Determine integration limits for $I = \iint\limits_{\Omega} f(x,y)dxdy$

where Ω is bounded by equations $y = 1$, $y = x + 4$, $y = 6 - x$.

Solution 1. Divide Ω into

$\Omega_1 : -3 \le x \le 1,\ 1 \le y \le x + 4$

$\Omega_2 : 1 \le x \le 5,\ 1 \le y \le 6 - x.$



$$I = \iint\limits_{\Omega} f(x,y)dxdy$$

$$= \iint\limits_{\Omega_1} f(x,y)dxdy$$

$$+ \iint\limits_{\Omega_2} f(x,y)dxdy$$

$$= \int_{-3}^{1} dx \int_{1}^{x+4} f(x,y)\,dy + \int_{1}^{5} dx \int_{1}^{6-x} f(x,y)\,dy$$

In this approach, we have to divide Ω into two subdomains, since their upper curves are expressed by different functions.

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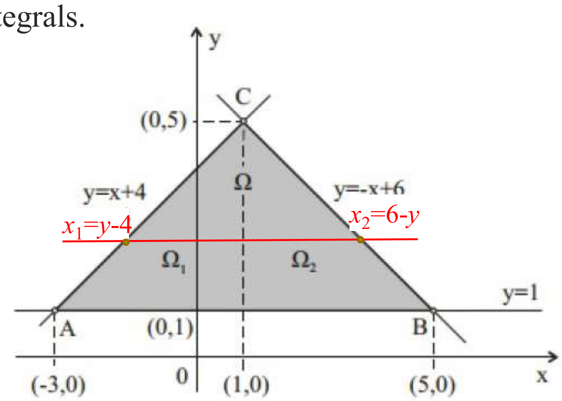
Example 3 (cont.)

Solution 2. The domain can be expressed as

$$1 \le y \le 5, \quad y - 4 = x_1 \le x \le x_2 = 6 - y,$$

$$I = \iint\limits_{\Omega} f(x,y)dxdy = \int_{1}^{5} dy \int_{y-4}^{6-y} f(x,y)\,dx$$

In this integral order, the integral can be expressed by repeated integrals.



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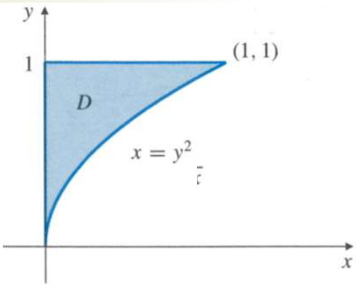
Example 4

Evaluate the integral $I = \int_0^1 dx \int_{\sqrt{x}}^1 e^{y^3}\, dy$



Solution. There is no simple antiderivative for e^{y^3}, so we should try the other order of integration.

From the integral limits, we know that the domain D can be described by

$$0 = x_{\min} \le x \le x_{\max} = 1,$$
$$\sqrt{x} = y_1(x) \le y \le y_2(x) = 1.$$

The lower curve: $y = \sqrt{x}$

The upper curve: $y = 1$

We thus can draw D as shown in the figure.

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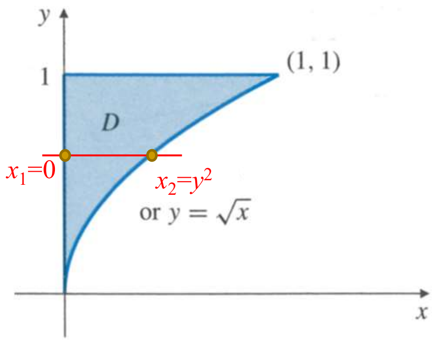
Example 4 (cont.)

To integrate with x firstly, we describe D as

$$0 = y_{\min} \le y \le y_{\max} = 1, \qquad 0 = x_1 \le x \le x_2 = y^2.$$

Then the double integral becomes

$$I = \int_0^1 dy \int_0^{y^2} e^{y^3} dx = \int_0^1 e^{y^3} dy \int_0^{y^2} 1\, dx$$
$$= \int_0^1 e^{y^3}\, x\Big|_{x=0}^{x=y^2} dy = \int_0^1 e^{y^3}\left(y^2 - 0\right) dy$$
$$= \int_0^1 y^2 e^{y^3} dy = \frac{e^{y^3}}{3}\bigg|_0^1 = \frac{e-1}{3}.$$



The above integral is provided as:

Let $u = y^3$, $du = 3y^2 dy$, we have

$$\int y^2 e^{y^3} dy = \frac{1}{3}\int e^u du = \frac{e^u}{3} + c = \frac{e^{y^3}}{3} + c$$

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Example 5 (added)

Show that as a, b, c, d are constants, we have

$$\int_a^b dx \int_c^d f(x)g(y)dy = \left(\int_a^b f(x)dx\right)\left(\int_c^d g(y)dy\right)$$

Proof. $$LHS = \int_a^b \left(\int_c^d f(x)g(y)dy\right)dx \ = \int_a^b f(x)\left(\int_c^d g(y)dy\right)dx$$

Because $\int_c^d g(y)dy$ is constant denote it as $h = \int_c^d g(y)dy$

$$LHS = \int_a^b hf(x)\,dx = \left(\int_a^b f(x)\,dx\right)h = \left(\int_a^b f(x)dx\right)\left(\int_c^d g(y)dy\right) = RHS$$

If the integral function is separable into variables and the integral limits are constants, a double integral can be expressed as the product of two single integrals. For example:

$$\int_0^1 dx\int_0^2 x^2 y\,dy = \left(\int_0^1 x^2dx\right)\left(\int_0^2 ydy\right) = \frac{1}{3}\times\frac{4}{2} = \frac{2}{3}$$

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2.2 Triple integrals

Consider a function $f(x, y, z)$ in a domain S.

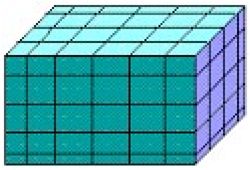
Perform a partition of S using

$x = x_i$ for $i = 0, 1, \ldots, n$,

$y = y_j$ for $j = 0, 1, \ldots, m$,

$z = z_k$ for $k = 0, 1, \ldots, l$.

S



This meshes the domain S into cuboids.

Each cuboid has three sides:

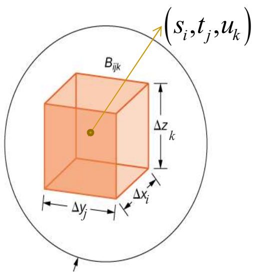
$$\Delta x_i = x_i - x_{i-1},\ \Delta y_j = y_j - y_{j-1}\ ,\ \Delta z_k = z_k - z_{k-1}.$$

The volume of a cuboid $\Delta V_{ijk} = \Delta x_i \Delta y_j \Delta z_k$.

Choose a representing point (s_i, t_j, u_k)

for each cuboid

$$s_i \in [x_{i-1}, x_i],\quad t_j \in [y_{j-1}, y_j],\quad u_k \in [z_{k-1}, z_k].$$



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Triple integrals

Definition: The triple integral of a continuous function $f(x, y, z)$ defined on a closed domain S in the real space is the following number

$$\iiint_S f(x,y,z)\,dxdydz \equiv \lim_{\max\{\Delta x_i,\Delta y_j,\Delta z_k\}\to 0} \sum_{k=1}^{l}\sum_{j=1}^{m}\sum_{i=1}^{n} f(s_i,t_j,u_k)\,\Delta x_i \Delta y_j \Delta z_k,$$

where the sum taken over all $(x_i, y_j, z_k) \in S$.

If $f(x, y, z) = 1$

$$\text{Volume of } S = \iiint_S 1\,dxdydz\ .$$

If $f(x, y, z)$ is the density distribution of a material in S, then the mass of the material in the region S is given by the triple integral

$$\text{Mass} = \iiint_S f(x,y,z)\,dxdydz\ .$$

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Description of 3D domains

Consider a 3D domain T as shown in the figure.

The projection of T on the xy plane is denoted as P. We know

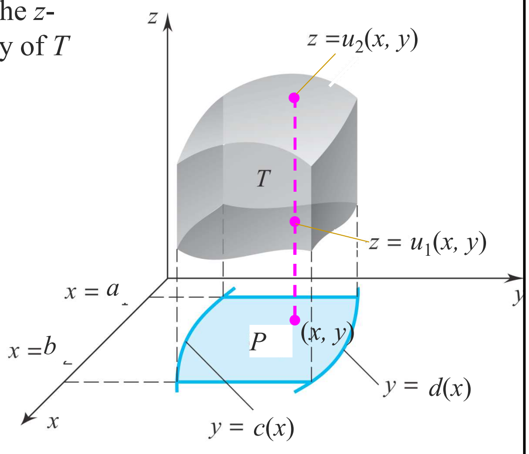
$$P:\ \{(x,y):\ a \le x \le b,\quad c(x) \le y \le d(x)\}.$$

Draw a vertical line (parallel to the z-axis), which crosses the boundary of T at $z = u_1(x, y)$ and $z = u_2(x, y)$.

$$u_1(x,y) \le z \le u_2(x,y)$$

T can thus be described as

$$T = \left\{ \begin{array}{l} (x,y,z):\ a \le x \le b, \\ c(x) \le y \le d(x), \\ u_1(x,y) \le z \le u_2(x,y) \end{array} \right\}$$



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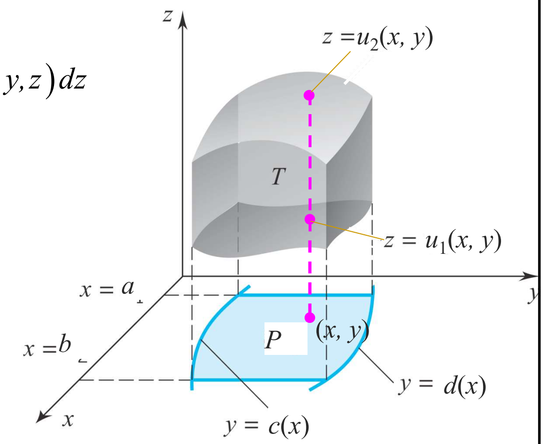
Fubini's theorem for triple integrals

$$I = \iiint_T f(x,y,z)\,dxdydz$$

$$= \iint_P dxdy \int_{z=u_1(x,y)}^{z=u_2(x,y)} f(x,y,z)\,dz$$

$$= \int_{x=a}^{x=b} dx \int_{y=c(x)}^{y=d(x)} dy \int_{z=u_1(x,y)}^{z=u_2(x,y)} f(x,y,z)\,dz$$

$$T = \left\{ \begin{array}{l} (x,y,z):\ a \le x \le b, \\ c(x) \le y \le d(x), \\ u_1(x,y) \le z \le u_2(x,y) \end{array} \right\}$$

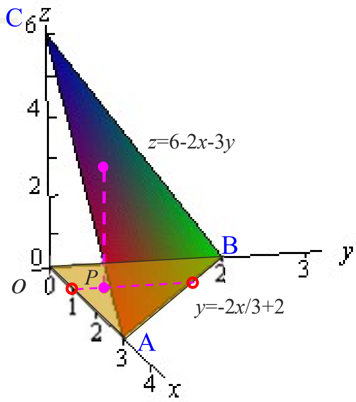


Example 1

Evaluate the integral $I = \iiint_S 2x\,dxdydz$ where S is bounded by the planes $x = 0$, $y = 0$, $z = 0$ and $2x+3y+z = 6$.

Solution. Draw the solid S, , which is a tetrahedron. The intersection of the inclined plane, $2x+3y+z = 6$, with the x-axis is

$$\begin{cases} 2x+3y+z=6, \\ y=z=0. \end{cases} \qquad \text{A: } (3, 0, 0)$$



The intersections of the inclined plane with the y- and z-axes are
B: (0, 2, 0), C: (0, 0, 6)

It is a tetrahedron:

The back face: $x = 0$;

The bottom face: $z = 0$,

The left face: $y = 0$,

The front face: $2x+3y+z = 6$.

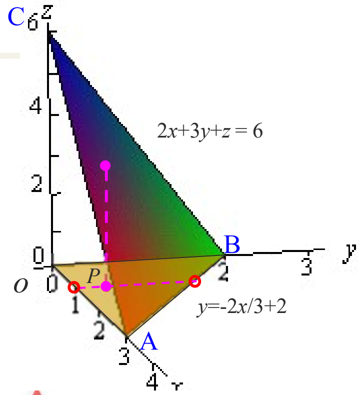
Example 1 (cont.)

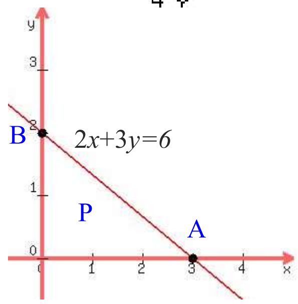
With A (3, 0, 0) and B (0, 2, 0), we can draw the projection P of the domain onto the xy plane. P can be thus described as

$$P = \{0 \le x \le 3,\ 0 \le y \le -2x/3+2\}.$$

To determine the range of z for given x & y, we draw a vertical line parallel to the z-axis. It crosses the boundary of the domain at z_1 and z_2:

$$0 = z_1 < z < z_2 = 6\text{-}2x\text{-}3y$$





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Example 1 (cont.)

$$S = \{0 \le x \le 3,\ 0 \le y \le -2x/3+2,\ 0 \le z \le 6-2x-3y\}.$$

$$\iiint_S 2x\,dxdydz = \int_0^3 dx \int_0^{-2x/3+2} dy \int_0^{6-2x-3y} 2x\,dz$$

$$= \int_0^3 dx \int_0^{-2x/3+2} 2xz\Big|_{z=0}^{z=6-2x-3y} dy = \int_0^3 dx \int_0^{-2x/3+2} 2x\left(6-2x-3y-0\right)dy$$

$$= \int_0^3 2x\,dx \int_0^{-2x/3+2} \left(6-2x-3y\right)dy = \int_0^3 2x\left(6y-2xy-1.5y^2\right)\Big|_{y=0}^{y=-2x/3+2} dx$$

$$= \int_0^3 2x\left(6\left(-2x/3+2\right)-2x\left(-2x/3+2\right)-1.5\left(-2x/3+2\right)^2-0\right)dx$$

$$= \int_0^3 2x\left(-4x+12+\frac{4}{3}x^2-4x-\frac{2}{3}x^2+4x-6\right)dx$$

$$= \int_0^3 \left(\frac{4}{3}x^3-8x^2+12x\right)dx \ = 9$$

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Fubini's theorem for triple integrals

A triple integral is integrated with z firstly:

$$\iiint_E f(x,y,z)\,dxdydz = \iint_D dxdy \int_{z=u_1(x,y)}^{z=u_2(x,y)} f(x,y,z)\,dz$$

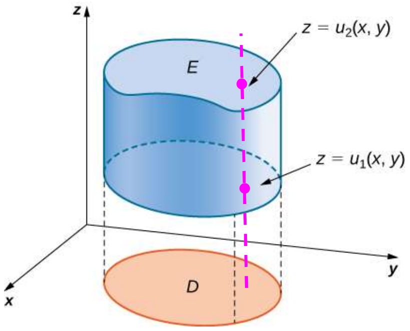
Denote $F(x,y) = \int_{z=u_1(x,y)}^{z=u_2(x,y)} f(x,y,z)\,dz$

$$\iiint_E f(x,y,z)\,dxdydz = \iint_D F(x,y)\,dxdy$$

where D is the projection of the 3D domain E onto the xy plane,

$z = u_1(x, y)$ is the lower boundary surface, and

$z = u_2(x, y)$ is the upper boundary surface.



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Fubini's theorem for triple integrals

A triple integral is integrated with x firstly:

$$\iiint_E f(x,y,z)\,dxdydz = \iint_D dydz \int_{x=u_1(y,z)}^{x=u_2(y,z)} f(x,y,z)\,dx$$

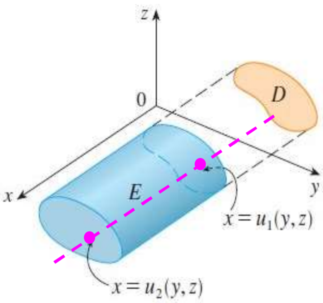
Denote $F(y,z) = \int_{x=u_1(y,z)}^{x=u_2(y,z)} f(x,y,z)\,dx$

$$\iiint_E f(x,y,z)\,dxdydz = \iint_D F(y,z)\,dydz$$

where D is the projection of the 3D domain E onto the yz plane,

$x = u_1(y, z)$ is the right boundary surface, and

$x = u_2(y, z)$ is the left boundary surface, as shown in the figure.



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Fubini's theorem for triple integrals

A triple integral is integrated with y firstly:

$$\iiint_E f(x,y,z)\,dxdydz = \iint_D dxdz \int_{y=u_1(x,z)}^{y=u_2(x,z)} f(x,y,z)\,dy$$

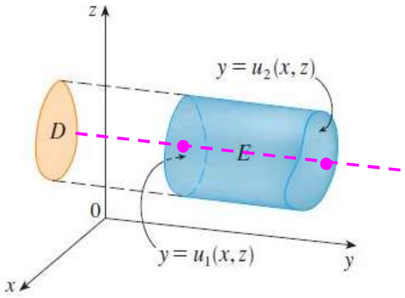
Denote $F(x,z) = \displaystyle\int_{y=u_1(x,z)}^{y=u_2(x,z)} f(x,y,z)\,dy$

$$\iiint_E f(x,y,z)\,dxdydz = \iint_D F(x,z)\,dxdz$$

where D is the projection of the 3D domain E onto the zx plane,

$y = u_1(x, z)$ is the left boundary surface, and

$y = u_2(x, z)$ is the right boundary surface, as shown in the figure.



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Example 2.

Evaluate the integral $I = \iiint_E z^2\,dxdydz$

where E is bounded by the planes $x = 0$, $y = 0$, $z = 0$ and $x+y+z = 1$.

Solution. Draw the integral domain

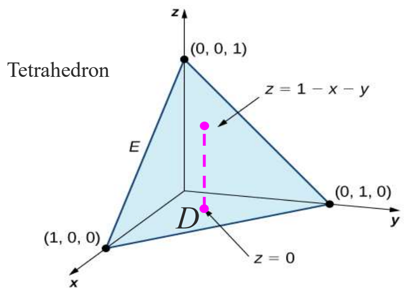
The plane $x+y+z = 1$ has intersections with the x-, y and z-axes at

(1, 0, 0), (0, 1, 0) and (0, 0, 1).

Back face: $x = 0$

Left side face: $y = 0$

Bottom face: $z = 0$



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Example 2. (cont.)

As the function is in z, we should integrate in z lastly. We may either integrate with x or y first. Let us integrate with y first.

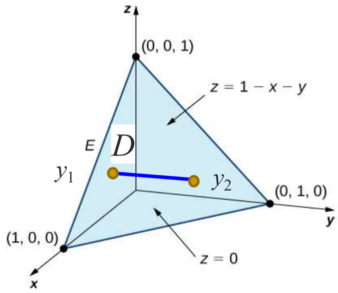
Draw a horizontal line along the y-axis. It crosses the domain boundary at two points:

$$0 = y_1 \le y \le y_2 = 1 - x - z$$

$$I = \iint_D dxdz \int_0^{1-x-z} z^2 dy$$

$$= \iint_D z^2 \left. y \right|_{y=0}^{y=1-x-z} dxdz$$

$$= \iint_D z^2 \left[(1-x-z) - 0 \right] dxdz$$



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Example 2. (cont.)

After integrating in y, we have a double integral in x and z.

$$I = \iint_D z^2 (1-x-z) dxdz$$

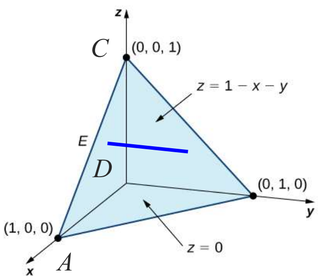
We then describe the projection D of the solid S onto the zx plane.

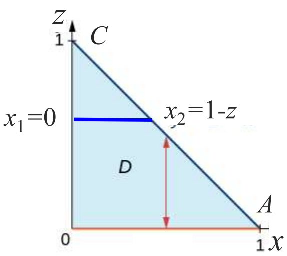
To integrate to z last, we need to define its range first

$$0 = z_{\min} \le z \le z_{\max} = 1.$$

To determine the range of x, we draw a line parallel to the x-axis:

$$0 = x_1 \le x \le x_2 = 1\text{-z}.$$





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Example 3. (cont.)

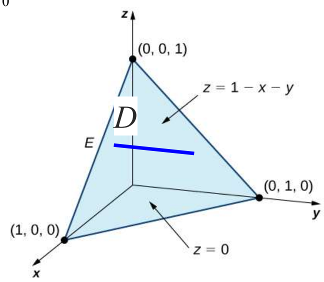
The projection D of the solid on the zx plane: $D=\{0\le z\le 1,\ 0\le x\le 1-z\}$.

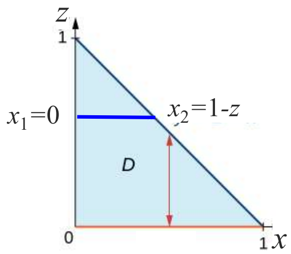
$$I=\iint_D z^2(1-x-z)\,dxdz=\int_0^1 dz\int_0^{1-z} z^2(1-x-z)\,dx$$

$$=\int_0^1 z^2dz\int_0^{1-z}(1-x-z)\,dx=\int_0^1 z^2\left(x-\frac{1}{2}x^2-xz\right)\Big|_{x=0}^{x=1-z}dz$$

$$=\int_0^1 z^2\left((1-z)-\frac{1}{2}(1-z)^2-(1-z)z-0\right)dz$$

$$=\int_0^1 z^2\frac{1}{2}(1-z)\big(2-(1-z)-2z\big)dz=\frac{1}{2}\int_0^1 z^2(1-z)^2dz=\frac{1}{60}$$





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Example 3

Evaluate the integral $I=\iiint_S xy\,dxdydz$

where S is enclosed by the surfaces $z=0$, $z=x+y$, $y=x^2$ and $x=y^2$.

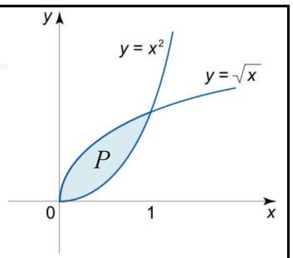
Solution: Geometry analysis.

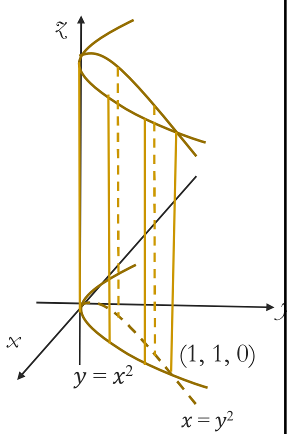
$y=x^2$ and $x=y^2$ are cylindrical surfaces parallel to the z-axis, as their equations do not depend on z.

The cross-sections of the two surfaces in the xy plane are the two curves $y=x^2$ and $x=y^2$, enclosing a domain P.

The two surfaces form a vertical cylinder parallel to the z-axis.

The domain S is a part of the vertical cylinder cut below by the plane $z=0$ and above by the plane $z=x+y$.

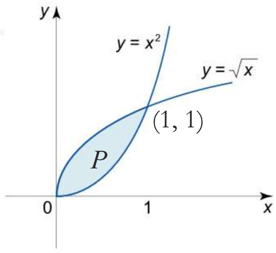




Example 3 (cont.)

Evaluate the integral $I = \iiint\limits_S xydxdydz$

where S is enclosed by the surfaces $z=0$, $z=x+y$, $y=x^2$ and $x=y^2$.

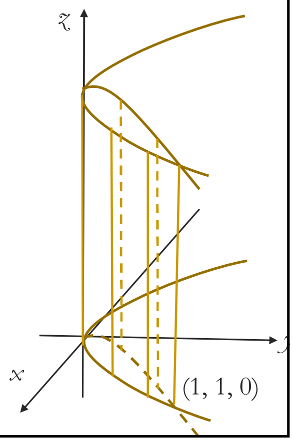


Solution: The domain S is the vertical cylinder cut below by the plane $z = 0$ and above by the plane $z = x+y$.

$$0 \le z \le x + y$$

$$I = \iint\limits_P dxdy \int_0^{x+y} xydz = \iint\limits_P xy\, z\Big|_{z=0}^{z=x+y} dxdy$$

$$= \iint\limits_P xy(x+y)dxdy$$



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Example 3 (cont.)

$$I = \iint\limits_P xy(x+y)dxdy$$

The projection P of the domain onto the xy plane can be described as

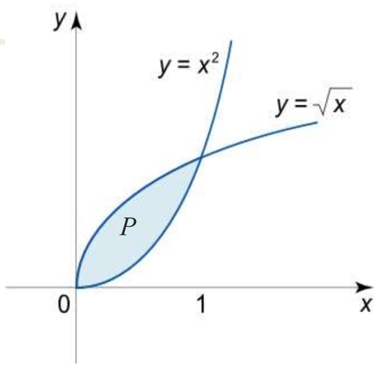
$$P = \left\{0 \le x \le 1,\ x^2 \le y \le \sqrt{x}\right\}.$$

$$I = \int_0^1 dx \int_{x^2}^{\sqrt{x}} xy(x+y)dy$$

$$= \int_0^1 dx \int_{x^2}^{\sqrt{x}} \left(x^2y + xy^2\right)dy$$

$$= \int_0^1 \left(\frac{x^2y^2}{2} + \frac{xy^3}{3}\right)\Bigg|_{y=x^2}^{y=\sqrt{x}} dx$$

$$= \int_0^1 \left(\left(\frac{x^3}{2} + \frac{x^{5/2}}{3}\right) - \left(\frac{x^6}{2} + \frac{x^7}{3}\right)\right)dx = \frac{3}{28}$$



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Example A1

Evaluate $I = \iiint_S (2 + x - \sin z)dxdydz$,

where S is a sphere with radius a and centred at the origin.

Solution. The integration of x in the sphere is zero because:

The integral domain is symmetric to the plane $x = 0$.

For every dV in the half of the sphere where $x > 0$, there is a corresponding element in the other half of the sphere where x has the same magnitude but the opposite sign.

Similarly, the integration of sin$z$ in the sphere is zero for the same reason.

$$I = \iiint_S 2dxdydz = 2\iiint_S 1dxdydz = 2 \cdot \frac{4\pi}{3}a^3 = \frac{8\pi}{3}a^3$$

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Even and odd functions in symmetrical domains

If S is symmetric to the plane $x = 0$
and $f(x, y, z)$ is odd in x, i.e. $f(-x, y, z) = -f(x, y, z)$, then

$$\iiint_S f(x,y,z)dv = 0$$

If S is symmetric to the plane $x = 0$
and $f(x, y, z)$ is even in x, i.e. $f(-x, y, z) = f(x, y, z)$, then

$$\iiint_S f(x,y,z)dv = 2\iiint_{S_{half}} f(x,y,z)dv,$$

where $S_{half} = \{(x, y, z): (x, y, z) \in S \ \& \ x > 0 \}$.

One can write the same symmetry properties for S being symmetric with respect to the planes $y = 0$ and $z = 0$, respectively.

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Even and odd functions in symmetric domains

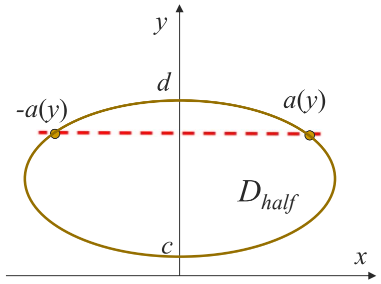
If D is symmetric to $x = 0$ and $f(x, y)$ is odd in x, then

$$\iint_D f(x,y)\,dxdy = 0.$$

If D is symmetric to $x = 0$ and $f(x, y)$ is even in x, then

$$\iint_D f(x,y)\,dxdy = 2\iint_{D_{half}} f(x,y)\,dxdy,$$

where D_{half} is the half of D with $x \geq 0$.



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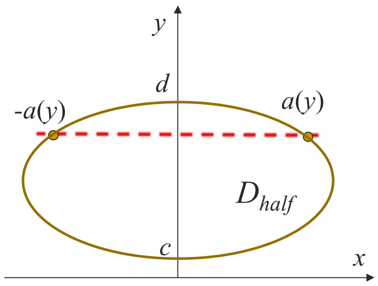
Even and odd functions in symmetric domains



Proof. As D is symmetric in x, we have

$$I = \iint_D f(x,y)\,dxdy = \int_c^d dy \int_{-a(y)}^{a(y)} f(x,y)\,dx$$

If $f(x, y)$ is odd in x, $I = 0$. If $f(x, y)$ is even in x

$$I = 2\int_c^d dy \int_0^{a(x)} f(x,y)\,dy = 2\iint_{D_{half}} f(x,y)\,dxdy$$

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Hits for Qu. 10 in Prb. Sheet 2

Evaluate $I = \iiint\limits_R \left(xy + z^2\right) dxdydz$, where $R = \left\{(x, y, z):\ 0 \le z \le 1 - |x| - |y|\right\}$

R is with the lower surface $\quad z = f_1(x, y) = 0$

R is with the upper surface $\quad z = f_2(x, y) = 1 - |x| - |y|$.

Both of the two surfaces are symmetric to

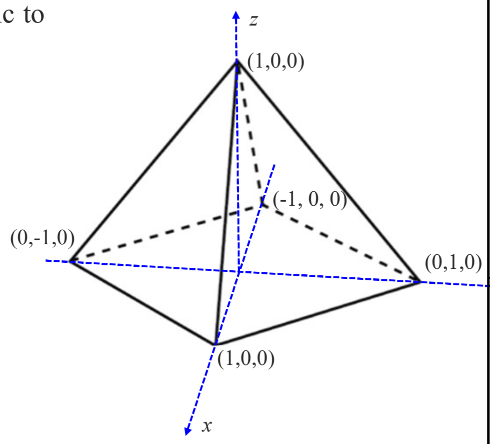
$x = 0$ (the yz plane) and

$y = 0$ (the zx plane).

The integration of xy in R is zero because

(i) The integral domain is symmetric to the plane $x = 0$.

(ii) For every dV in the half of R where $x > 0$, there is a corresponding element in the other half of R where xy has the same magnitude but the opposite sign.



Hits for Qu. 10 in Prb. Sheet 2 (cont.)

Evaluate $I = \iiint\limits_R \left(xy + z^2\right) dxdydz$, where $R = \left\{(x, y, z):\ 0 \le z \le 1 - |x| - |y|\right\}$

R is symmetric to $x = 0$. For every dV in the half of R where $x > 0$, there is a corresponding element in the other half of R where z^2 has the same value. Therefore

$$I = \iiint\limits_R z^2 dV = 2\iiint\limits_{R_H} z^2 dV,\ R_H = \left\{(x, y, z) \in R:\ x \ge 0\right\}$$

R_H is symmetric to $y = 0$. For every dV in the half of R_H where $y > 0$, there is a corresponding element in the other half of R_H where z^2 has the same value. As such,

$$\iiint\limits_{R_H} z^2 dV = 2\iiint\limits_{R_Q} z^2 dV,\ R_Q = \left\{(x, y, z) \in R:\ x \ge 0, y \ge 0\right\}$$

As the result, $I = 4\iiint\limits_{R_Q} z^2 dV$.

R_Q is the same as the domain E in Example 2.

III. Change of variables for multiple integrals 4525 3594

3.1 Double integrals in polar coordinates (Adams, 14.4)
3.2 Triple integrals in cylindrical coordinates (Adams, 14.6)
3.3 Triple integrals in spherical coordinates (Adams, 14.6)
This chapter is for 4 lectures for Week 4.

Change of variables for single integrals. Consider the integral $\int_a^b f(x)dx$

Introduce change of variable $x = x(u)$ with $a = x(\alpha)$ & $b = x(\beta)$. We have

$$\int_a^b f(x)dx = \int_\alpha^\beta f(x(u))x_u du$$

There are three changes here:
1. Change of function: $f(x)$ to $f(x(u))$
2. Change of integration domain: $x \in [a, b]$ to $u \in [\alpha, \beta]$
3. Change of integral element: $dx = x_u du$.

We are expecting similar changes for multiple integrals.

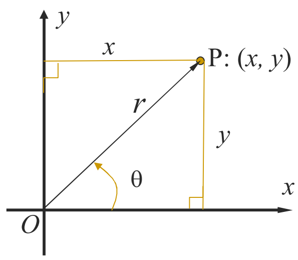
3.1 Double integrals in polar coordinates

4525 3594

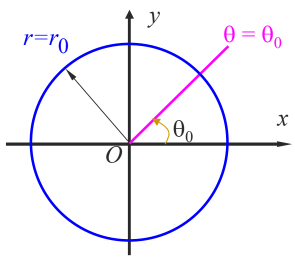
Polar coordinates (r, θ):
Consider a point P (x, y).
r is the distance from O to P,
θ is the angle OP makes with the x-axis, i.e.

$$\begin{cases} r = \sqrt{x^2 + y^2}, & r \ge 0 \\ \tan\theta = y/x, & \theta \in [0, 2\pi) \text{ or } \theta \in (-\pi, \pi] \end{cases}$$

$$\begin{cases} x = r\cos\theta \\ y = r\sin\theta \end{cases}$$



The coordinate curves of polar coordinates:
- $r = r_0$: a circle centred at the origin with radius r_0.
- $\theta = \theta_0$: a ray starting from the origin making an angle θ_0 with the x-axis.



Partition of 2D domain in polar coordinates

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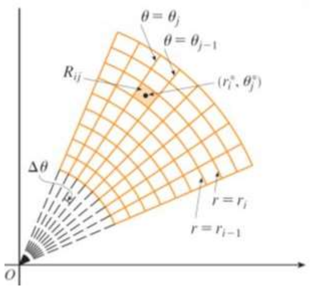
A 2D domain R is partitioned by two sets of the coordinate curves:

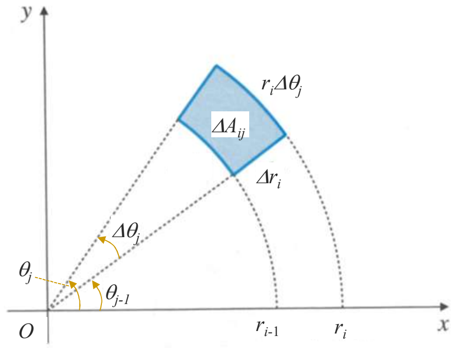
$$r = r_i\,,\ i = 0, 1, \ldots, n; \qquad \theta = \theta_j, j = 0, 1, \ldots, m.$$

A mesh element is bounded by two circles $r = r_{i-1}, r_i$ and two rays $\theta = \theta_{j-1}, \theta_j$.

As $\Delta r_i \ll 1$ and $\Delta\theta_j \ll 1$, the element is approximately a rectangle, with area

$$\Delta A_{ij} = r_i\, \Delta\theta_j\, \Delta r_i.$$





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Fubini's theorem for double integral in polar coordinates

For double integrals in polar coordinates

$$dA = r\,dr\,d\theta, \qquad f(x,y) = f(r\cos\theta, r\sin\theta).$$

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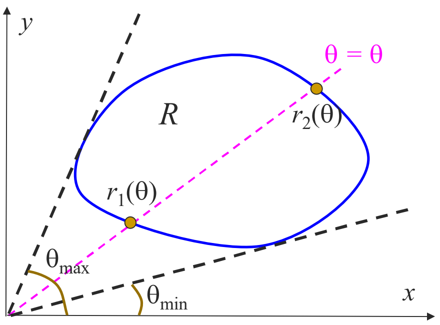
A domain R can be expressed as follows in polar coordinates:

$$R = \{(r,\theta) : \theta_{\min} \le \theta \le \theta_{\max}, r_1(\theta) \le r \le r_2(\theta)\}.$$

θ_{min} & θ_{max} are the minimum & maximum of θ on the domain boundary.

To find the range of r for a given value of θ, draw a ray $\theta = \theta$. It crosses the domain boundary at $r = r_1(\theta)$ and $r = r_2(\theta)$.

$$\iint_R f(x,y)\,dxdy$$
$$= \iint_R f(r\cos\theta, r\sin\theta)\; r\,dr\,d\theta$$
$$= \int_{\theta_{\min}}^{\theta_{\max}} d\theta \int_{r_1(\theta)}^{r_2(\theta)} f(r\cos\theta, r\sin\theta)\, r\,dr$$



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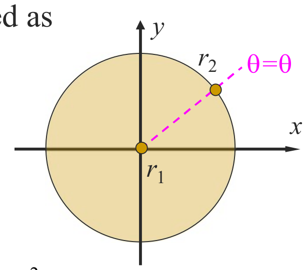
Example 1

Derive the formula for the area of a circle with radius a.

Solution. The domain can be described as

$$0 \le \theta \le 2\pi,$$
$$0 = r_1 \le r \le r_2 = a$$



$$A = \iint_R 1 dxdy = \iint_R r dr d\theta$$
$$= \int_0^{2\pi} d\theta \int_0^a r dr = \left(\int_0^{2\pi} d\theta \right) \left(\int_0^a r dr \right) = 2\pi \times \frac{a^2}{2} = \pi a^2$$

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Example 2

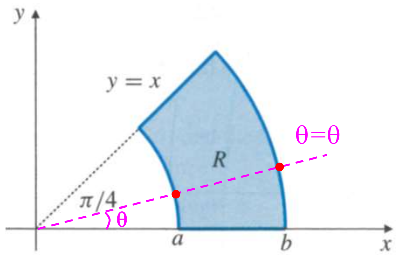
Evaluate $I = \iint_R \frac{y^2}{x^2} dA.$

R is a part of the annulus $0 < a^2 \le x^2 + y^2 \le b^2$ $(a, b \ge 0)$ lying in first quarter and below the line $y = x$.

Using the polar coordinates: $R = \{0 \le \theta \le \pi/4, a \le r \le b\}$.

$$\frac{y^2}{x^2} = \frac{r^2 \sin^2\theta}{r^2\cos^2\theta} = \tan^2\theta, \qquad dA = r dr d\theta,$$

$$I = \int_0^{\pi/4} \tan^2\theta \, d\theta \int_a^b r \, dr$$
$$= \frac{1}{2}(b^2 - a^2) \int_0^{\pi/4} (\sec^2\theta - 1) \, d\theta$$
$$= \frac{1}{2}(b^2 - a^2)(\tan\theta - \theta)\Big|_0^{\pi/4}$$
$$= \frac{1}{2}(b^2 - a^2)\left(1 - \frac{\pi}{4}\right) = \frac{4-\pi}{8}(b^2 - a^2).$$



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Example 3 — 7402 3572

Let D be the interior of the circle: $x^2+y^2=2x$. Find $I=\iint_D \sqrt{x^2+y^2}\,dA$.

Solution. (1) Draw the domain. The circle can be expressed as

$$x^2-2x+1+y^2=1, \qquad (x-1)^2+y^2=1.$$

(2) Find the range of θ:

$$-\pi/2=\theta_{min}\le\theta\le\theta_{max}=\pi/2$$

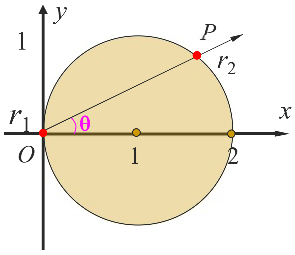
θ_{min} & θ_{max} are the minimum & maximum of θ on the domain boundary.

(**3**) To find the range of r for a given value of θ, we draw a ray θ = θ. It crosses the circle at two points $r=r_1$, $r=r_2$

$$0=r_1\le r\le r_2=2\cos\theta$$

In polar coordinates, the circle is

$$r^2=2r\cos\theta \quad\Rightarrow r_1=0 \text{ and } r_2=2\cos\theta.$$



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Example 3 (cont.) — 7402 3572

Let D be the interior of the circle: $x^2+y^2=2x$. Find $I=\iint_D \sqrt{x^2+y^2}\,dA$.

(4) Calculating the integral

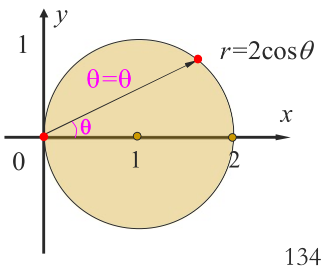
$$D=\{-\pi/2\le\theta\le\pi/2,\ 0\le r\le 2\cos\theta\}.$$

$$I=\iint_D\sqrt{r^2}\,r\,dr\,d\theta=\int_{-\pi/2}^{\pi/2}d\theta\int_0^{2\cos\theta}r^2dr$$

$$=\int_{-\pi/2}^{\pi/2}\left.\frac{r^3}{3}\right|_0^{2\cos\theta}d\theta=\int_{-\pi/2}^{\pi/2}\frac{8}{3}\cos^3\theta\,d\theta$$

Change the varaible $u = \sin\theta$, $du=\cos\theta\,d\theta$

$$I=\frac{8}{3}\int_{-1}^{1}\left(1-u^2\right)du=\frac{8}{3}\left.\left(u-\frac{u^3}{3}\right)\right|_{-1}^{1}=\frac{32}{9}$$



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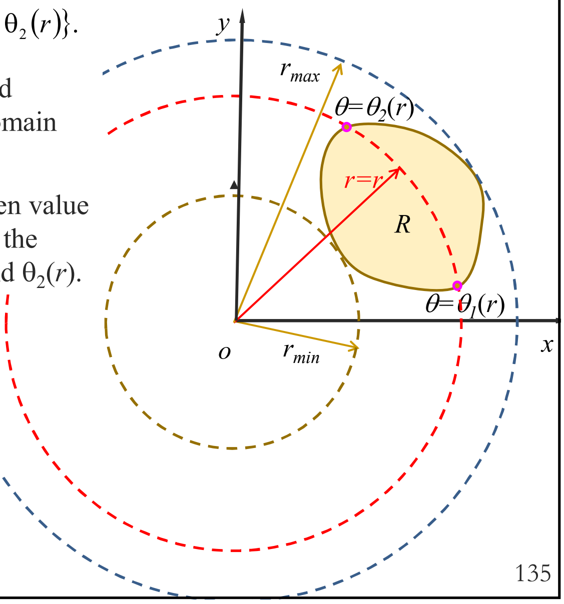
Double integral in polar coordinates (cont.)

A domain R can be expressed as follows in polar coordinates

$$R = \{r_{\min} \le r \le r_{\max},\ \theta_1(r) \le \theta \le \theta_2(r)\}.$$

r_{min} to r_{max} are the minimum and maximum values of r on the domain boundary.

To find the range of θ for a given value r, draw a circle $r = r$. It crosses the boundary at two points $\theta_1(r)$ and $\theta_2(r)$.



$$\iint_R f(x,y)\,dxdy$$

$$= \iint_R f(r\cos\theta, r\sin\theta)\ rdrd\theta$$

$$= \int_{r_{\min}}^{r_{\max}} dr \int_{\theta_1(r)}^{\theta_2(r)} f(r\cos\theta, r\sin\theta)\,rd\theta$$

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Example 3 (integrate to θ firstly)

Let D be the interior of the circle of $x^2+y^2=2x$. Find $I = \iint_D \sqrt{x^2+y^2}\,dA$.

Solution. The range of r in the domain: $0 = r_{\min} \le r \le r_{\max} = 2$.

$r_{\max}$ is determined as follows:

$r^2 = 2r\cos\theta \ \Rightarrow r = 0$ or $r = 2\cos\theta \ \Rightarrow\ r_{\max} = 2$.

To find the range of θ, we draw a circle $r = r$, it crosses the domain boundary at $\theta = \theta_1$ and $\theta = \theta_2$:

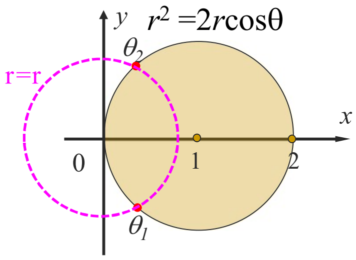
$r = 2\cos\theta \ \Rightarrow\ \theta_1 = -\cos^{-1}(r/2)$, $\theta_2 = \cos^{-1}(r/2)$,

The range of θ: $-\cos^{-1}(r/2) = \theta_1 \le \theta \le \theta_2 = \cos^{-1}(r/2)$,

$$I = \int_0^2 dr \int_{-\cos^{-1}(r/2)}^{\cos^{-1}(r/2)} r^2 d\theta =$$

$$= \int_0^2 r^2 dr \int_{-\cos^{-1}(r/2)}^{\cos^{-1}(r/2)} d\theta$$

$$= \int_0^2 r^2\, 2\cos^{-1}(r/2)\,dr \quad \left(= \frac{32}{9}\right)$$



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Example 4 Show that $I = \int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$.

Soln. Consider the double integral

$$J = \int_{-\infty}^{\infty} dx \int_{-\infty}^{\infty} e^{-(x^2+y^2)} dy = \int_{-\infty}^{\infty} dx \int_{-\infty}^{\infty} e^{-x^2} e^{-y^2} dy$$

$$= \int_{-\infty}^{\infty} e^{-x^2} dx \int_{-\infty}^{\infty} e^{-y^2} dy = \int_{-\infty}^{\infty} e^{-x^2} I dx \quad \left(\because \int_{-\infty}^{\infty} e^{-y^2} dy = \int_{-\infty}^{\infty} e^{-x^2} dx = I = \text{cont.} \right)$$

$$= I \int_{-\infty}^{\infty} e^{-x^2} dx = I^2 \qquad \text{i.e.} \quad J = I^2$$

The integral domain D for J is: $-\infty < x < \infty$, $-\infty < y < \infty$,

In the polar coordinates D is: $0 \le \theta \le 2\pi$, $0 \le r < \infty$.

$$J = \int_0^{2\pi} d\theta \int_0^{\infty} e^{-r^2} r dr = \int_0^{2\pi} \frac{1}{2} d\theta = \pi$$

$$\therefore \int_0^{\infty} e^{-r^2} r dr = \frac{1}{2} \int_0^{\infty} e^{-u} du = \frac{1}{2}, \quad u = r^2, \, du = 2rdr$$

Hence, $I = \int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$. It is the Gaussian integral.

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General cases

With the change of variables

$x = x(u, v)$ and $y = y(u, v)$

The Jacobian J for the change of variables is defined by the determinant

$$J \equiv \frac{\partial(x,y)}{\partial(u,v)} \equiv \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix}.$$

A double integral is changed as

$$\iint_D f(x,y)\, dxdy = \iint_{D'} f\big(x(u,v), y(u,v)\big) |J|\, dudv.$$

Here D' is the region of the (u, v)-plane corresponding to the region of D of the (x, y)-plane.

The area element in u and v $dA = |J|\, dudv$.

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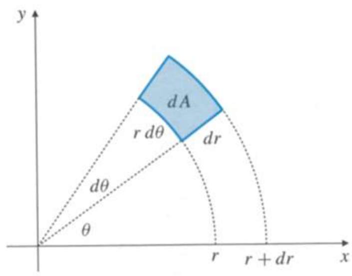
Jacobin for polar coordinates

For the polar coordinates, we have $x = r\cos\theta$, $y = r\sin\theta$

$$J = \frac{\partial(x,y)}{\partial(r,\theta)} = \begin{vmatrix} x_r & x_\theta \\ y_r & y_\theta \end{vmatrix}$$

$$= \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r\begin{vmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{vmatrix}$$

$$= r\left(\cos^2\theta + \sin^2\theta\right) = r$$



For double integrals in the polar coordinates

$$dA = |J|\,dr d\theta = r dr d\theta,$$

This is the same result as we derived geometrically before.

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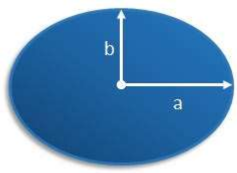
Example 5 (added)



Calculate the area of an ellipse with radii a and b.

Solution.

$$A = \iint_D 1\,dxdy, \quad D = \left\{(x,y) : \frac{x^2}{a^2} + \frac{y^2}{b^2} \le 1\right\}$$

Set $x = au$, $y = bv$

$$J = \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix} = \begin{vmatrix} a & 0 \\ 0 & b \end{vmatrix} = ab$$

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We thus have

$$A = \iint_{D'} 1\,dxdy, \quad D' = \left\{(u,v) : \frac{x^2}{a^2} + \frac{y^2}{b^2} \le 1\right\}$$

$$= \iint_{D'} |J|\,dudv$$

$$= \iint_{D'} ab dudv = ab\iint_{D'} 1 dudv = \pi ab.$$

$\because \iint_{D'} 1 dudv$ is the area of a unit circle.

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3.2 Triple integrals in cylindrical coordinates

Cylindrical coordinates (r, θ, z)

Consider a point P at (x, y, z).

Draw PR ⊥ the z-axis with R at the z-axis.

Draw PQ ⊥ the xy-plane with Q on the xy plane.

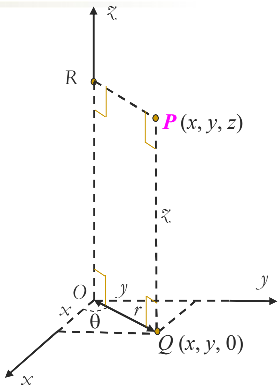
Thus ORPQ is a rectangle.

The cylindrical coordinates (r, θ, z) are defined as:

r is the distance from O to Q,

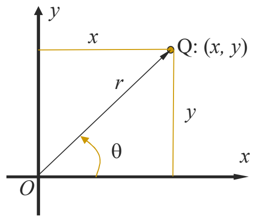
θ is the angle that OQ makes with the x-axis,

z is the same as in Cartesian coordinates.



$$\begin{cases} r = OQ = \sqrt{x^2+y^2} \\ \theta = \arctan(y/x) \\ z = z \end{cases} \Leftrightarrow \begin{cases} x = r\cos\theta \\ y = r\sin\theta \\ z = z \end{cases}$$

where $r \geq 0$, $0 \leq \theta < 2\pi$, $-\infty < z < \infty$.



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Coordinate surfaces of cylindrical coordinates (r, θ, z)

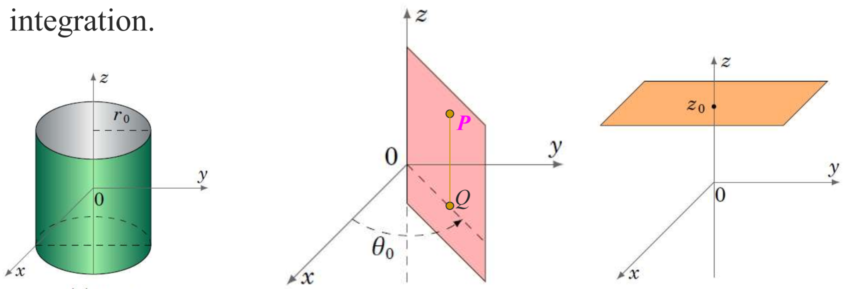
The **3 coordinate surfaces** of cylindrical coordinates are:

$r = r_0$: a cylindrical surface of radius r_0 centred along the z-axis, (the reason for the name "cylindrical coordinates").

$\theta = \theta_0$: a half-plane emanating from the z-axis, making an angle of θ_0 with the x-axis.

$z = z_0$: a plane parallel to the xy-plane.

Coordinate surfaces are essential in setting the limits of integration.



(a) $r = r_0$ **(b)** $\theta = \theta_0$ **(c)** $z = z_0$

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Examples

Find the cylindrical coordinates of the point with the Cartesian coordinates (-3, 3, -7).

Solution. $r = \sqrt{x^2 + y^2} = \sqrt{(-3)^2 + 3^2} = 3\sqrt{2}$

$$\theta = \tan^{-1}(y/x) = \tan^{-1}\left(\frac{3}{-3}\right) = 3\pi/4$$

$$z = -7$$

For a point with cylindrical coordinates (2, 2π/3, 1), find its Cartesian coordinates.

Solution:

$$x = r\cos\theta = 2\cos(2\pi/3) = -1$$

$$y = r\sin\theta = 2\sin(2\pi/3) = \sqrt{3}$$

$$z = 1$$

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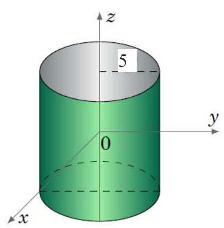
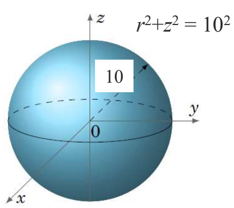
Examples 7837 0187

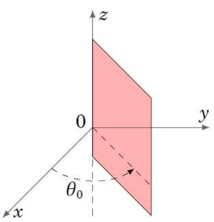
Identify the surface for each of the following equations:

$r = 5$, $r^2+z^2 = 100$, $z = r$

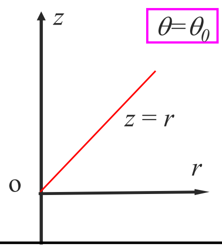


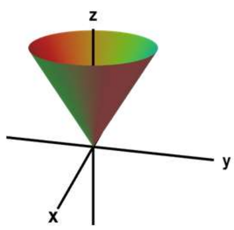


Draw their cross-sections in *orz*-plane.



(b) $\theta = \theta_0$





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The Jacobian and change of variables in triple integrals

7837 0187

The Jacobian of the change of variables

$$x = x(u, v, w), \quad y = y(u, v, w), \quad z = z(u, v, w)$$

defined as the determinant

$$J = \frac{\partial(x, y, z)}{\partial(u, v, w)} = \begin{vmatrix} x_u & x_v & x_w \\ y_u & y_v & y_w \\ z_u & z_v & z_w \end{vmatrix}$$

For triple integrals, we have $\boxed{dV = |J|\, du dv dw},$

$$\iiint_S f(x, y, z)\, dx dy dz$$

$$= \iiint_{S'} f(x(u, v, w), y(u, v, w), z(u, v, w))\, |J|\, du dv dw,$$

where S' is the region of the (u, v, w)-space corresponding to the region S of the (x, y, z)-space.

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Triple integrals in cylindrical coordinates

7837 0187

For the cylindrical coordinates, we have $x = r\cos\theta$, $y = r\sin\theta$, $z = z$

$$J = \frac{\partial(x, y, z)}{\partial(r, \theta, z)} = \begin{vmatrix} x_r & x_\theta & x_z \\ y_r & y_\theta & y_z \\ z_r & z_\theta & z_z \end{vmatrix} = \begin{vmatrix} x_r & x_\theta & x_z \\ y_r & y_\theta & y_z \\ 0 & 0 & 1 \end{vmatrix}$$

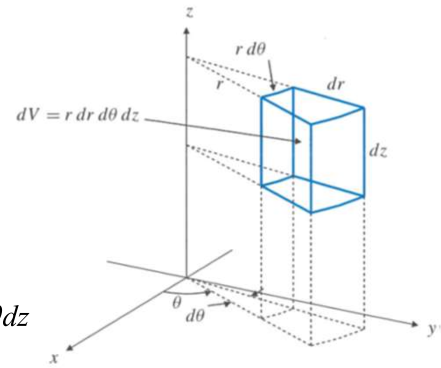
$$= \begin{vmatrix} x_r & x_\theta \\ y_r & y_\theta \end{vmatrix} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r$$

For triple integrals in the cylindrical coordinates

$$\boxed{dV = r dr d\theta dz},$$

$$\iiint_S f(x, y, z)\, dx dy dz$$

$$= \iiint_{S'} f(r\cos\theta, r\sin\theta, z)\; r\; dr d\theta dz$$



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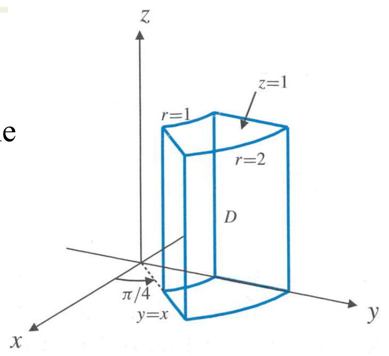
Example 1

Evaluate $I = \iiint\limits_D (x^2 + y^2)\,dV$

D is in the first octant bounded by the cylinders $x^2+y^2 = 1$ and $x^2+y^2 = 4$ and the planes, $x = y$, $x = 0$, $z = 0$, and $z = 1$.



Solution. In terms of cylindrical coordinates:

$x^2+y^2 = 1 \;\Rightarrow\; r = 1,$

$x^2+y^2 = 4 \;\Rightarrow\; r = 2,$

$x = y \;\Rightarrow\; \theta = \pi/4,$

$x = 0 \;\Rightarrow\; \theta = \pi/2,$

$z = 0$ and $z = 1$.

The domain is thus defined as

$$D = \{\pi/4 \le \theta \le \pi/2,\ 1 \le r \le 2, 0 \le z \le 1\}$$

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Example 1 (cont.) $I = \iiint\limits_D (x^2 + y^2)\,dV$

$$D = \{\pi/4 \le \theta \le \pi/2,\ 1 \le r \le 2, 0 \le z \le 1\}$$

The integrand is: $x^2+y^2 = r^2$,

$\boxed{dV = rdrd\theta dz\ ,}$

$$I = \iiint\limits_D r^2 rdrd\theta dz = \int_{\pi/4}^{\pi/2} d\theta \int_1^2 dr \int_0^1 r^3 dz = \int_{\pi/4}^{\pi/2} d\theta \int_1^2 \left(r^3 z\right)\Big|_{z=0}^{z=1} dr$$

$$= \int_{\pi/4}^{\pi/2} d\theta \int_1^2 r^3 (1-0)\,dr = \int_{\pi/4}^{\pi/2} \left(\frac{2^4}{4} - \frac{1^4}{4}\right) d\theta \;= \frac{15}{16}\pi$$

The above calculation can be simplified as

$$I = \int_{\pi/4}^{\pi/2} d\theta \int_1^2 dr \int_0^1 r^3 dz = \int_{\pi/4}^{\pi/2} d\theta \int_1^2 r^3 dr \int_0^1 dz$$

$$= \int_{\pi/4}^{\pi/2} d\theta \;\times\; \int_1^2 r^3 dr \;\times\; \int_0^1 dz = \left(\frac{\pi}{2} - \frac{\pi}{4}\right)\left(\frac{2^4}{4} - \frac{1^4}{4}\right)(1-0) = \frac{15}{16}\pi$$

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Example 2

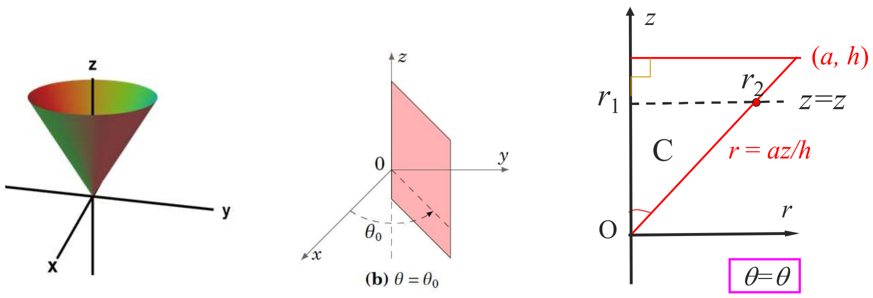
Find the volume of a cone of height h and radius a.

Solution. Build coordinates as shown in the left figure, with the origin at the cone vertex and the z-axis along the axis of symmetry. As it is axisymmetric about the z-axis: $0 \le \theta \le 2\pi$.

We draw the intersection C of the cone with $\theta = \theta$.

The range of z in C: $0 \le z \le h$.

The range of r for a given value of z: $0 = r_1 \le r \le r_2 = az/h$.



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Example 2 (cont.)

$$D = \{(r, \theta, z) : 0 \le \theta \le 2\pi,\ 0 \le z \le h,\ 0 \le r \le az/h\}$$

The volume of a domain is equal to the integral of unit over the domain.

$$V = \iiint_D 1 dV = \iiint_D 1 r dr dz d\theta = \int_0^{2\pi} d\theta \int_0^h dz \int_0^{az/h} r\, dr$$

$$= \int_0^{2\pi} d\theta \int_0^h \left[\frac{r^2}{2}\right]_{r=0}^{r=az/h} dz = \int_0^{2\pi} d\theta \int_0^h \frac{1}{2}\left(\left(\frac{az}{h}\right)^2 - 0\right) dz$$

$$= \frac{1}{2}\left(\frac{a}{h}\right)^2 \int_0^{2\pi} d\theta \int_0^h z^2\, dz$$

$$= \frac{1}{2}\left(\frac{a}{h}\right)^2 \int_0^{2\pi} \frac{h^3}{3} d\theta = \frac{1}{2}\left(\frac{a}{h}\right)^2 \cdot 2\pi \cdot \frac{h^3}{3}$$

$$V = \frac{\pi a^2 h}{3}$$

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Example 3 **7471 0007**

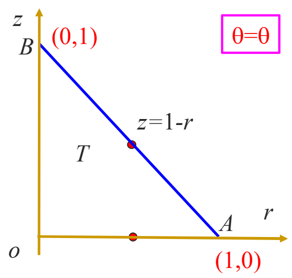
Let C be a solid cone with the boundary $x^2+y^2=(1-z)^2$, $0 \le z \le 1$. If the density of C at point (x, y, z) is z, find the mass of C.

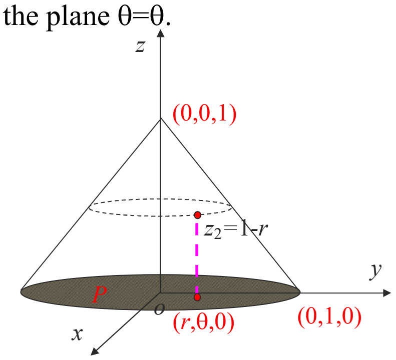
Solution. In cylindrical coordinates: $x^2+y^2=(1-z)^2$ for $0 \le z \le 1$

$$\Rightarrow \quad r=1-z \quad \text{for } 0 \le z \le 1$$

It is axisymmetric about the z-axis. We can draw the cross-section of the cone surface with the plane θ=θ, the line AB. Rotating AB by 2π around the z-axis forms the cone.

T is the cross-section of the cone with the plane θ=θ.





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Example 3 (cont.)

It is axisymmetric about the z-axis, thus $\quad 0 \le \theta \le 2\pi$

We then can determine the variation ranges for r and z as following:

$$0 \le r \le 1, \quad 0 \le z \le 1-r$$

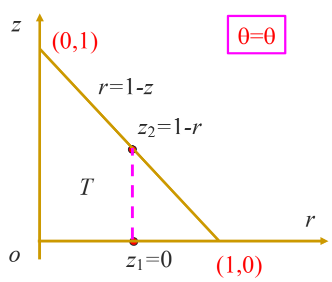
$$M = \iiint_C d(x,y,z)\,dV = \iiint_C z\,dV = \iiint_C z\; r\,dz\,dr\,d\theta$$

$$= \int_0^{2\pi} d\theta \int_0^1 dr \int_0^{1-r} zr\,dz$$

$$= \int_0^{2\pi} d\theta \int_0^1 \left[\frac{rz^2}{2}\right]_{z=0}^{z=1-r} dr = \int_0^{2\pi} d\theta \int_0^1 \frac{r(1-r)^2}{2}\,dr$$

$$= \left(\int_0^{2\pi} d\theta\right)\cdot\left(\int_0^1 \frac{r(1-r)^2}{2}\right) = 2\pi\cdot\frac{1}{24} = \frac{\pi}{12}$$

$$\therefore \int_0^1 \frac{r(1-r)^2}{2}\,dr = \frac{1}{2}\int_0^1 \left(r^3-2r^2+r\right)dr = \frac{1}{24}$$



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3.3 Triple integrals in spherical coordinates

Spherical coordinates (ρ, ϕ, θ)

Draw PR $\perp$ the z-axis with R at the z-axis.

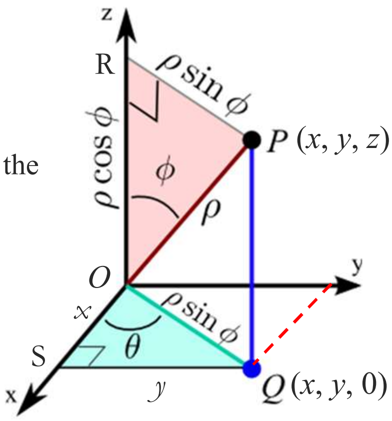
Draw PQ $\perp$ the xy-plane with Q on the xy plane. **Thus ORPQ is a rectangle**.

ρ is the distance from the point P(x, y, z) to the origin

$$\rho = \sqrt{x^2 + y^2 + z^2}, \quad \rho \geq 0.$$

ϕ is the angle OP makes with the z-axis. z = OR = $\rho\cos\phi$, thus

$$\phi = \arccos\left(\frac{z}{\rho}\right), \quad \phi \in [0, \pi].$$



θ is the angle OQ makes with the x-axis. θ in spherical coordinates are the same as θ in cylindrical coordinates.

$$\theta = \arctan\left(\frac{y}{x}\right), \quad \theta \in [0, 2\pi).$$

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Spherical coordinates (ρ, ϕ, θ)

ORPQ is a rectangle, as such

r = OQ = RP = $\rho\sin\phi$, (here r is the r-coordinate used in cylindrical coordinates)

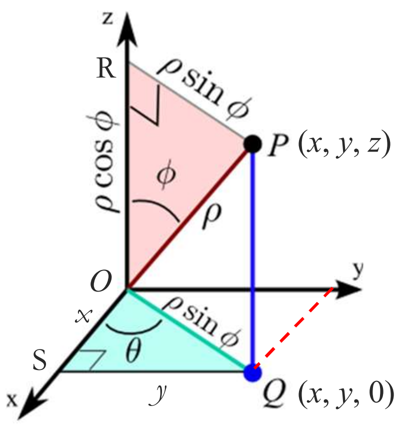
z = OR = $\rho\cos\phi$.

From (ρ, ϕ, θ) to (r, θ, z)

$$r = \rho \sin\phi$$
$$\theta = \theta$$
$$z = \rho\cos\phi$$

From (ρ, ϕ, θ) to (x, y, z)

$x = r\cos\theta = \rho\sin\phi\cos\theta$
$y = r\sin\theta = \rho\sin\phi\sin\theta$
$z = \rho\cos\phi$



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Examples

For the point with spherical coordinates (2, $\pi/4$, $\pi/3$), find its Cartesian coordinates and cylindrical coordinates.

Solution: In cylindrical coordinates, we have

$$r = \rho \sin\phi = 2\sin(\pi/4) = \sqrt{2}$$
$$\theta = \pi/3$$
$$z = \rho\cos\phi = 2\cos(\pi/4) = \sqrt{2}$$

$$r = \rho\sin\phi$$
$$z = \rho\cos\phi$$

In Cartesian coordinates, we have

$$x = r\cos\theta = \sqrt{2}\cos(\pi/3) = \sqrt{2}/2$$
$$y = r\sin\theta = \sqrt{2}\sin(\pi/3) = \sqrt{6}/2$$
$$z = \sqrt{2}$$

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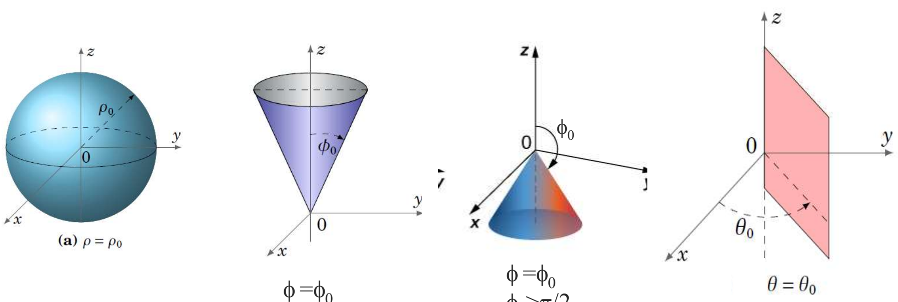
Coordinate surfaces of spherical coordinates **3769 4243**

The three coordinate surfaces of the spherical coordinates are

$\rho = \rho_0$: a spherical surface with radius ρ_0 centred at the origin. (The reason for the name "spherical coordinates")

$\phi = \phi_0$: a circular cone with the vertex at the origin being axisymmetric about the z-axis.

$\theta = \theta_0$: a half-plane emanating from the z-axis making angle θ_0 with the x-axis.



$\phi = \phi_0$, $\phi_0 < \pi/2$

$\phi = \phi_0$, $\phi_0 > \pi/2$

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Example

3769 4243

Identify the surface for each of the following equations

$\rho = 0$: The origin.

$\rho = 5$: Spherical surface cantered at the origin having radius 5.

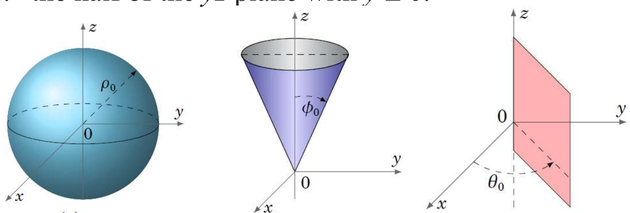
$\phi = 0$: The positive part of the z-axis.

$\phi = \pi/2$: The xy plane.

$\phi = \pi$: The negative part of the z-axis.

$\theta = \pi/4$: A vertical plane starting from the z-axis making angle $\pi/4$ with the x-axis.

$\theta = \pi/2$: the half of the yz plane with $y \geq 0$.



(a) $\rho = \rho_0$ **(c)** $\phi = \phi_0$ **(b)** $\theta = \theta_0$

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Jacobian of spherical coordinates

For the spherical coordinates, we have

$$x = \rho \sin\phi \cos\theta, \quad y = \rho \sin\phi \sin\theta, \quad z = \rho\cos\phi$$

$$J = \frac{\partial(x, y, z)}{\partial(\rho, \phi, \theta)} = \begin{vmatrix} x_\rho & x_\phi & x_\theta \\ y_\rho & y_\phi & y_\theta \\ z_\rho & z_\phi & z_\theta \end{vmatrix}$$

$$= \begin{vmatrix} \sin\phi\cos\theta & \rho\cos\phi\cos\theta & -\rho\sin\phi\sin\theta \\ \sin\phi\sin\theta & \rho\cos\phi\sin\theta & \rho\sin\phi\cos\theta \\ \cos\phi & -\rho\sin\phi & 0 \end{vmatrix}$$

$$= \cos\phi \begin{vmatrix} \rho\cos\phi\cos\theta & -\rho\sin\phi\sin\theta \\ \rho\cos\phi\sin\theta & \rho\sin\phi\cos\theta \end{vmatrix} + \rho\sin\phi \begin{vmatrix} \sin\phi\cos\theta & -\rho\sin\phi\sin\theta \\ \sin\phi\sin\theta & \rho\sin\phi\cos\theta \end{vmatrix}$$

$$= \rho^2 \sin\phi \ \cos^2\phi \begin{vmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{vmatrix} + \rho^2 \sin^3\phi \begin{vmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{vmatrix}$$

$$= \rho^2 \sin\phi \ \cos^2\phi + \rho^2 \sin^3\phi$$

$$= \rho^2 \sin\phi\left(\cos^2\phi + \sin^2\phi\right) = \rho^2 \sin\phi$$

$$\boxed{dV = |J|\, d\rho d\phi d\theta = \rho^2 \sin\phi d\rho d\phi d\theta.}$$

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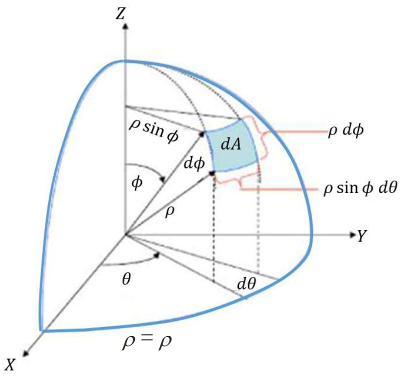
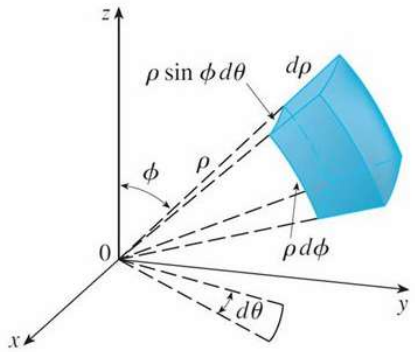
Triple integrals in spherical coordinates

$$J = \frac{\partial(x,y,z)}{\partial(\rho,\phi,\theta)} = \rho^2 \sin\phi$$

$$\boxed{dV = |J|\, d\rho\, d\phi\, d\theta = \rho^2 \sin\phi\, d\rho\, d\phi\, d\theta,}$$





Example 1

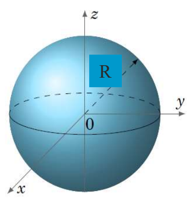
Find the volume of a sphere of radius R.

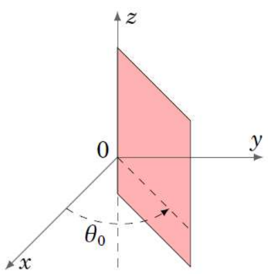
Solution: Sphere: $S = \{\theta \in [0, 2\pi],\ \phi \in [0,\pi],\ \rho \in [0, R]\}$.

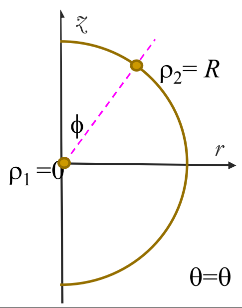
$$V = \iiint_S 1 dV = \iiint_S \rho^2 \sin\phi\, d\rho\, d\phi\, d\theta$$

$$= \int_0^{2\pi} d\theta \int_0^{\pi} d\phi \int_0^{R} \rho^2 \sin\phi\, d\rho = \int_0^{2\pi} d\theta \int_0^{\pi} \sin\phi\, d\phi \int_0^{R} \rho^2 d\rho$$

$$= \left(\int_0^{2\pi} d\theta\right) \cdot \left(\int_0^{\pi} \sin\phi\, d\phi\right) \cdot \left(\int_0^{R} \rho^2 d\rho\right) = 2\pi \cdot 2 \cdot \left.\left(\frac{\rho^3}{3}\right)\right|_0^R = \frac{4}{3}\pi R^3$$







Example 2

Find the volume of a cone of height h and radius a using spherical coordinates.

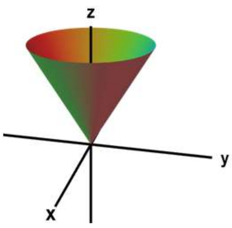
Solution. Being axisymmetric: $0 \le \theta \le 2\pi$.

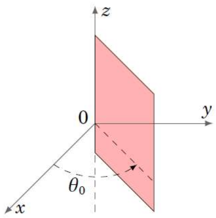
Draw the intersection of the domain in the Orz plane.

The range of ϕ: $0 \le \phi \le \alpha = \tan^{-1}(a/h)$.

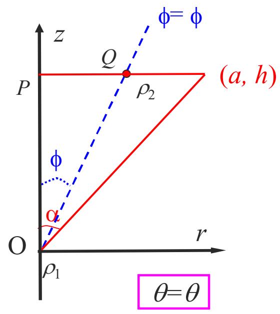
To find the range for ρ, we draw a ray $\phi = \phi$. It cross the boundaries at ρ_1 and ρ_2:

$$0 = \rho_1 \le \rho \le \rho_2 = h \sec \phi$$





(b) $\theta = \theta_0$



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Example 2 (cont.) $C = \{(r, \phi, \theta) : 0 \le \theta \le 2\pi,\ 0 \le \phi \le \alpha,\ 0 \le \rho \le h \sec \phi\}$

$$V = \iiint_C 1 dV = \iiint_C 1\rho^2 \sin\phi d\rho d\phi d\theta = \int_0^{2\pi} d\theta \int_0^{\alpha} d\phi \int_0^{h\sec\phi} \rho^2 \sin\phi \, d\rho$$

$$= \int_0^{2\pi} d\theta \int_0^{\alpha} \sin\phi \left[\frac{\rho^3}{3}\right]_{\rho=0}^{\rho=h\sec\phi} d\phi \quad = \int_0^{2\pi} d\theta \int_0^{\alpha} \frac{h^3}{3} \frac{\sin\phi}{\cos^3\phi} d\phi$$

$$= \int_0^{2\pi} d\theta \times \int_0^{\alpha} \frac{h^3}{3} \frac{\sin\phi}{\cos^3\phi} d\phi = 2\pi \frac{h^3}{3} \int_0^{\alpha} \frac{\sin\phi}{\cos^3\phi} d\phi$$

Let $u = \cos\phi,\ du = -\sin\phi d\phi,$ we have

$$\int \frac{\sin\phi}{\cos^3\phi} d\phi = -\int \frac{du}{u^3} = \frac{1}{2u^2} + c = \frac{1}{2} \frac{1}{\cos^2\phi} + c = \frac{1}{2}\sec^2\phi + c$$

$$V = \frac{2\pi h^3}{3} \frac{1}{2} \sec^2\phi \Big|_0^{\alpha} = \frac{\pi h^3}{3}\left(\sec^2\alpha - 1\right)$$

$$= \frac{\pi h^3}{3} \tan^2\alpha = \frac{\pi a^2 h}{3} \quad (\because\ h\tan\alpha = a)$$

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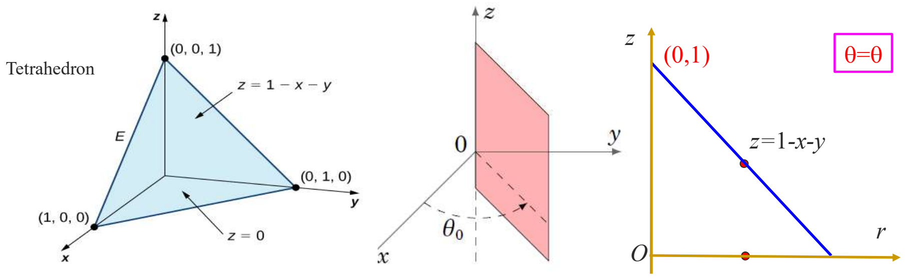
Example 3.

Consider the integral $I = \iiint_E z^2 dV$

where E is bounded by the planes $x = 0$, $y = 0$, $z = 0$ and $x+y+z = 1$.

Express the integral as repeated integral in cylindrical coordinates and spherical coordinates.

Soln. The range of θ in E: $0 \le \theta \le \pi/2$.



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Example 3. (cont.)

$I = \iiint_E z^2 dV$

Cylindrical coordinates

The range of θ in E: $0 \le \theta \le \pi/2$.

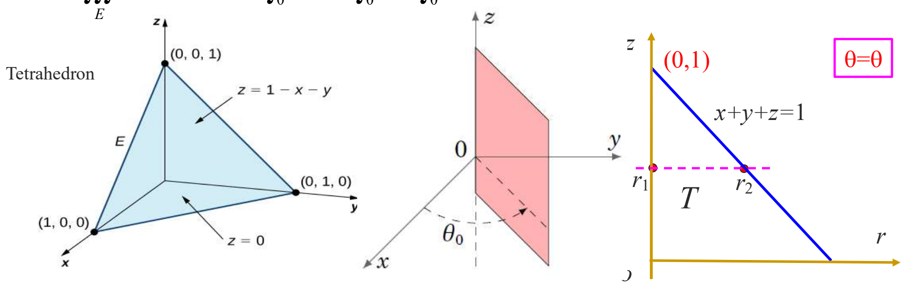
Denote the cross section of E with θ = θ as T:

The range of z in T is: $0 \le z \le 1$.

The range of r in T for a given value of z is:

$$0 = r_1 \le r \le r_2 \qquad r_2\cos\theta + r_2\sin\theta + z = 1, \quad r_2 = \frac{1-z}{\cos\theta+\sin\theta}$$

$$I = \iiint_E z^2\ r dz dr d\theta = \int_0^{\pi/2} d\theta \int_0^1 dz \int_0^{\frac{1-z}{\cos\theta+\sin\theta}} z^2 r\ dr$$



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Example 3. (cont.)

$$I = \iiint_E z^2 dV$$

Spherical coordinates

The range of θ in E: $0 \le \theta \le \pi/2$.

The range of ϕ in T is: $0 \le \phi \le \pi/2$.

The range of ρ in T for a given value of ϕ is:

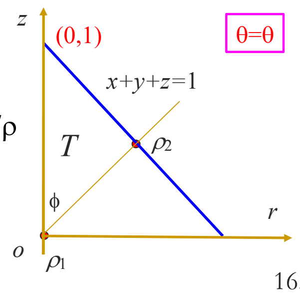
$$0 = \rho_1 \le \rho \le \rho_2$$

$$x + y + z = 1 \quad \Leftrightarrow \quad \rho_2 \sin\phi\cos\theta + \rho_2 \sin\phi\sin\theta + \rho_2 \cos\phi = 1$$

$$\Leftrightarrow \quad \rho_2 = \frac{1}{\sin\phi\cos\theta + \sin\phi\sin\theta + \cos\phi}$$

$$I = \iiint_E (\rho\cos\phi)^2 \ \rho^2 \sin\phi dz dr d\theta$$

$$= \int_0^{\pi/2} d\theta \int_0^{\pi/2} \cos^2\phi \sin\phi \ d\phi \int_0^{\frac{1}{\sin\phi\cos\theta+\sin\phi\sin\theta+\cos\phi}} \rho^4 \ d\rho$$



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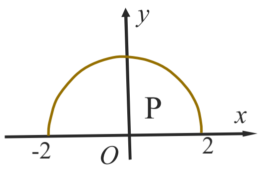
Example 4

Change to spherical coordinates and compute the integral

$$I = \int_{-2}^{2} dx \int_0^{\sqrt{4-x^2}} dy \int_0^{\sqrt{4-x^2-y^2}} y\sqrt{x^2+y^2+z^2} dz$$

Soln. Projection P of the domain on the xy plane



$$-2 \le x \le 2$$

$$0 = y_1(x) \le y \le y_2(x) = \sqrt{4-x^2}$$

$\Rightarrow$ The lower curve: $y_1 = 0$

The upper curve: $y_2 = \sqrt{4-x^2} \quad \Rightarrow \quad x^2 + y_2^2 = 2^2.$

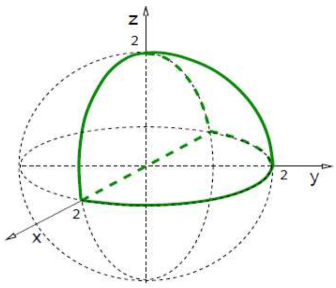
Upper and lower surfaces

$$0 = z_1(x,y) \le z \le z_2(x,y) = \sqrt{4-x^2-y^2}$$

$\Rightarrow$ The lower surface: $z_1 = 0$

The upper surface: $z_2 = \sqrt{4-x^2-y^2}$

$$\Rightarrow \ x^2 + y^2 + z_2^2 = 2^2.$$



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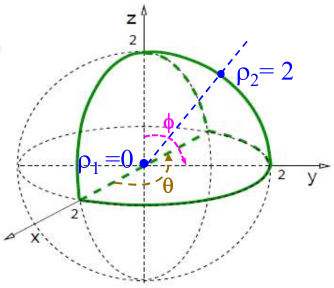
Example 4 (cont.)



The range of θ: $0 \le \theta \le \pi$

The range of ϕ in P: $0 \le \phi \le \pi/2$

The range for ρ: $0 \le \rho \le 2$

$$dV = \rho^2 \sin\phi d\rho d\phi d\theta,$$

$$f = y\sqrt{x^2+y^2+z^2} = \rho\sin\phi\sin\theta\ \rho, \quad \therefore y = \rho\sin\phi\sin\theta,$$

$$fdV = \rho^4\sin^2\phi\sin\theta d\rho d\phi d\theta$$

$$I = \int_0^{\pi} d\theta \int_0^{\pi/2} d\phi \int_0^2 \rho^4 \sin^2\phi \sin\theta d\rho$$

$$= \left(\int_0^{\pi} \sin\theta d\theta\right)\times\left(\int_0^{\pi/2} \sin^2\phi d\phi\right)\times\left(\int_0^2 \rho^4 d\rho\right) = 2\times\frac{\pi}{4}\times\frac{2^5}{5} = \frac{16\pi}{5}$$

$$\therefore \int_0^{\pi/2}\sin^2\phi d\phi = \int_0^{\pi/2}\frac{1-\cos 2\phi}{2}d\phi = \frac{1}{2}\left(\phi - \frac{1}{2}\sin 2\phi\right)_0^{\pi/2} = \frac{\pi}{4}$$

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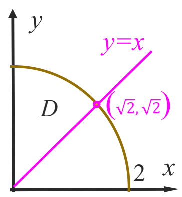
Example A1

Change to polar coordinates and compute the integral

$$I = \int_0^{\sqrt{2}} dx \int_x^{\sqrt{4-x^2}} \sqrt{x^2+y^2}dy$$



Soln. Denote the domain as D.

$$0 = z_1(x,y) \le z \le z_2(x,y) = \sqrt{4-x}$$

$$\Rightarrow \text{The lower surface: } z_1 = 0$$

$$\text{The upper surface: } z_2 = \sqrt{4-x^2}$$

$$\Rightarrow x^2 + y^2 + z_2^2$$

In polar coordinates, D can be described as:

$\pi/4 \le \theta \le \pi/2$, $0 \le r \le 2$,

$$I = \int_{-2}^{2} dx \int_0^{\sqrt{4-x^2}} dy \int_0^{\sqrt{4-x^2-y^2}} y\sqrt{x^2+y^2+z^2}dz$$

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Example A2

Change to polar coordinates and compute the integrals

$$I=\int_0^2 dx\int_0^{\sqrt{4-x^2}}\sqrt{x^2+y^2}\,dy+\int_{-\sqrt{2}}^{0}dx\int_x^{\sqrt{4-x^2}}\sqrt{x^2+y^2}\,dy$$

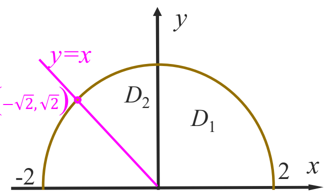
Soln. Denote the domains for the first and second integrals as D_1 and D_2.

$D_1: 0\le x\le 2$

$0=y_1(x)\le y\le y_2(x)=\sqrt{4-x^2}$

$\Rightarrow$ The lower boundary: $y_1=0$

The upper boundary: $y_2=\sqrt{4-x^2}\quad\Rightarrow\quad x^2+y_2^2=2^2.$



$D_2: -\sqrt{2}\le x\le 0$

$x=y_1(x)\le y\le y_2(x)=\sqrt{4-x^2}$

$\Rightarrow$ The lower boundary: $y_1=x$

The boundary curve: $y_2=\sqrt{4-x^2}\quad\Rightarrow\quad x^2+y_2^2=2^2.$

$$I=\iint_D r\,dA=\iint_D r\,r\,dr\,d\theta=\int_0^{3\pi/4}d\theta\int_0^2 r\,r\,dr=\left(\int_0^{3\pi/4}d\theta\right)\left(\int_0^2 r^2dr\right)=\frac{3\pi}{4}\frac{2^3}{3}=2\pi$$

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Example A3.

Find the volume of a sphere $x^2+y^2+z^2\le 1$ contained between planes $z=1/2$ and $z=0$ using spherical coordinates.

Soln: The domain is axisymmetric: $0\le\theta\le 2\pi$.

Its intersection with $\theta=\theta$ is shown in the figure.

The domain is divided into D_1 and D_2 by $\phi\le\pi/3$.

D_1: $0\le\phi\le\pi/3$,

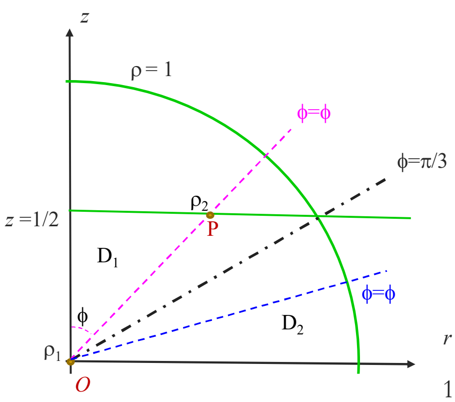
$0=\rho_1\le\rho\le\rho_2=\sec\phi/2$,

as $OP\cos\phi=1/2$

or $\rho_2\cos\phi=1/2$.

D_2: $\pi/3\le\phi\le\pi/2$,

$0\le\rho\le 1$.



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Example A3.

Soln: $0 \le \theta \le 2\pi$.

D_1: $0 \le \phi \le \pi/3$, $0 \le \rho \le \sec\phi/2$,

D_2: $\pi/3 \le \phi \le \pi/2$, $0 \le \rho \le 1$.

$$V_1 = \iiint_{D_1} dV = \iiint_{D_1} \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta = \int_0^{2\pi} d\theta \int_0^{\pi/3} d\phi \int_0^{\sec\phi/2} \rho^2 \sin\phi \, d\rho$$

$$= \int_0^{2\pi} d\theta \int_0^{\pi/3} \frac{1}{3} \sin\phi \, \rho^3 \Big|_0^{\sec\phi/2} d\phi = \int_0^{2\pi} d\theta \int_0^{\pi/3} \frac{1}{24} \sin\phi \sec^3\phi \, d\phi$$

$$= \frac{1}{24}\left(\int_0^{2\pi} d\theta\right)\left(\int_0^{\pi/3} \sin\phi \sec^3\phi \, d\phi\right) = \frac{1}{24} \times 2\pi \times \frac{3}{2} = \frac{\pi}{8}$$

$$\therefore \text{ Let } u = \cos\phi, \ du = -\sin\phi \, d\phi, \quad \int_0^{\pi/3} \sin\phi \sec^3\phi \, d\phi = -\int_1^{1/2} \frac{du}{u^3} = 3/2$$

$$V_2 = \int_0^{2\pi} d\theta \int_{\pi/3}^{\pi/2} d\phi \int_0^1 \rho^2 \sin\phi \, d\rho = \left(\int_0^{2\pi} d\theta\right)\left(\int_{\pi/3}^{\pi/2} \sin\phi \, d\phi\right)\left(\int_0^1 \rho^2 \, d\rho\right)$$

$$= 2\pi \times \frac{1}{2} \times \frac{1}{3} = \frac{\pi}{3}$$

$$\therefore \int_{\pi/3}^{\pi/2} \sin\phi \, d\phi = \frac{1}{2}\int_{\pi/3}^{\pi/2} (1 - \cos 2\phi) \, d\phi = (-\cos\phi)_{\pi/3}^{\pi/2} = \frac{1}{2}$$

$$V = V_1 + V_2 = \frac{11\pi}{24}$$

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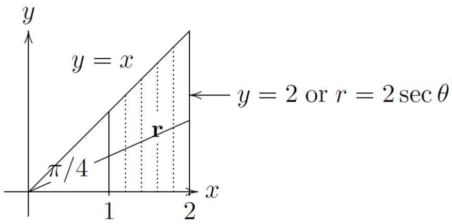
Added examples A4

Let $I = \int_1^2 \int_0^x \frac{1}{(x^2+y^2)^{3/2}} \, dy \, dx$.

Compute I using polar coordinates.



Solution.

$$I = \int_{\theta=0}^{\pi/4} \int_{r=\sec\theta}^{2\sec\theta} \frac{1}{r^3} r \, dr \, d\theta.$$

Compute the integral:

Inner: $\displaystyle\int_{\sec\theta}^{2\sec\theta} \frac{1}{r^2} \, dr = -\frac{1}{r}\Big|_{\sec\theta}^{2\sec\theta} = \frac{1}{2}\cos\theta.$

Outer: $\displaystyle I = \int_0^{\pi/4} \frac{1}{2}\cos\theta \, d\theta = \frac{1}{2}\sin\theta\Big|_0^{\pi/4} = \frac{\sqrt{2}}{4}.$

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Example A5

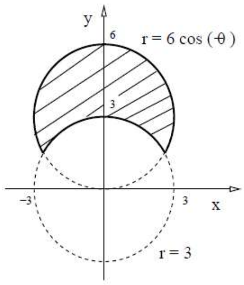
Find the area of the region in the plane inside the curve $r = 6\sin(\theta)$ and outside the circle $r = 3$, where r, θ are polar coordinates in the plane.

Solution: First sketch the integration region.

- $r = 6\sin(\theta)$ is a circle, since

$$r^2 = 6r\sin(\theta) \Leftrightarrow x^2 + y^2 = 6y$$

$$x^2 + (y-3)^2 = 3^2.$$



- The other curve is a circle $r = 3$ centered at the origin.

The condition $3 = r = 6\sin(\theta)$ determines the range in θ. Since $\sin(\theta) = 1/2$, we get $\theta_1 = 5\pi/6$ and $\theta_0 = \pi/6$.

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Example A5 (cont.)

$$A = \int_{\pi/6}^{5\pi/6} \int_{3}^{6\sin(\theta)} r dr\, d\theta = \int_{\pi/6}^{5\pi/6} \left(\frac{r^2}{2} \Big|_3^{6\sin(\theta)} \right) d\theta$$

$$A = \int_{\pi/6}^{5\pi/6} \left[\frac{6^2}{2} \sin^2(\theta) - \frac{3^2}{2} \right] d\theta = \int_{\pi/6}^{5\pi/6} \left[\frac{6^2}{2^2} (1 - \cos(2\theta)) - \frac{3^2}{2} \right] d\theta$$

$$A = 3^2 \left(\frac{5\pi}{6} - \frac{\pi}{6} \right) - \frac{3^2}{2} \left(\sin(2\theta) \Big|_{\pi/6}^{5\pi/6} \right) - \frac{3^2}{2} \left(\frac{5\pi}{6} - \frac{\pi}{6} \right).$$

$$A = 6\pi - 3\pi - \frac{3^2}{2} \left(-\frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2} \right), \text{ hence } A = 3\pi + 9\sqrt{3}/2. \quad \triangleleft$$

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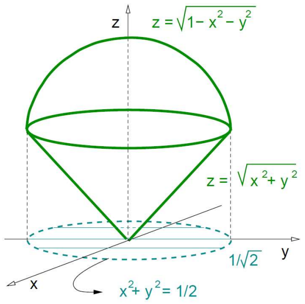
Example A6

Find the volume below the sphere

$$x^2 + y^2 + z^2 = 1$$

and above the cone $z = (x^2 + y^2)^{1/2}$.



Soln. In cylindrical coordinates, the two surfaces are

$$r^2 + z^2 = 1 \text{ and } z = r$$

Both eqs. do not depend on θ. The domain is axisymmetric about the z-axis.

The range of θ in the cone denoted as D is:

$$0 \leq \theta \leq 2\pi.$$

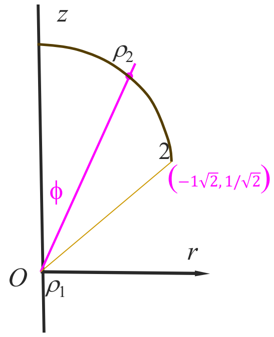
Denote the cross section of C with the plane θ = θ.

The range of ϕ in C is

$$0 \leq \phi \leq \pi/4.$$

The range of ρ for a given value of ϕ is

$$0 = \rho_1 \leq \rho \leq \rho_2 = 1$$



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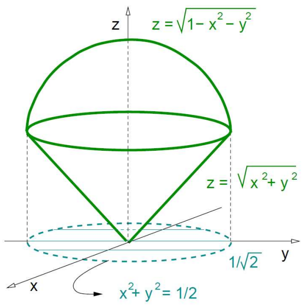
Example A6

$$0 \leq \theta \leq 2\pi.$$

$$0 \leq \phi \leq \pi/4.$$

$$0 \leq \rho \leq 1$$



$$V = \int_0^{2\pi} \int_0^{\pi/4} \int_0^1 \rho^2 \sin(\phi)\, d\rho\, d\phi\, d\theta,$$

$$V = \left[\int_0^{2\pi} d\theta\right] \left[\int_0^{\pi/4} \sin(\phi)\, d\phi\right] \left[\int_0^1 \rho^2\, d\rho\right],$$

$$V = 2\pi \left[-\cos(\phi)\Big|_0^{\pi/4}\right] \left(\frac{\rho^3}{3}\Big|_0^1\right),$$

$$V = 2\pi \left[-\frac{\sqrt{2}}{2} + 1\right] \frac{1}{3} \quad \Rightarrow \quad V = \frac{\pi}{3}(2 - \sqrt{2}).$$

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Example A6

Expression of the integral in cylindrical coordinates:

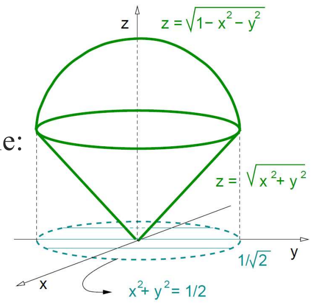
The projection P of the domain is within the circle:

$$x^2 + y^2 = 1/2$$

P can be described as

$$0 \le \theta \le 2\pi$$

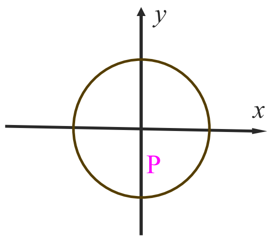
$$0 \le r \le 1/\sqrt{2}.$$



The lower surface of D is: $z = r$

The upper surface of D is: $z = \sqrt{1\text{-}r^2}$

$$V = \iiint_D 1\ r dz dr d\theta = \int_0^{2\pi} d\theta \int_0^{1/\sqrt{2}} dr \int_r^{\sqrt{1-r^2}} r\ dz$$



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Example A7

Use spherical coordinates to find the volume of the region outside the sphere $\rho = 2\cos(\phi)$ and inside the half sphere $\rho = 2$ with $\phi \in [0, \pi/2]$.

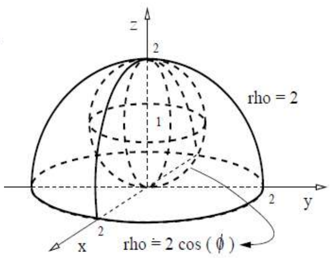
Solution: First sketch the integration region.

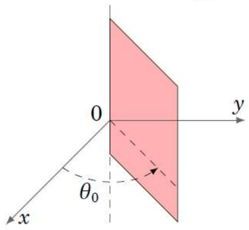
- $\rho = 2\cos(\phi)$ is a sphere, since

  $$\rho^2 = 2\rho\cos(\phi) \Leftrightarrow x^2 + y^2 + z^2 = 2z$$

  $$x^2 + y^2 + (z-1)^2 = 1.$$

- $\rho = 2$ is a sphere radius 2 and $\phi \in [0, \pi/2]$ says we only consider the upper half of the sphere.





(b) $\theta = \theta_0$

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Example A7 (cont.)

$$D=\{0\le\theta\le 2\pi,\ \ 0\le\phi\le\pi/2,\ \ 2\cos\phi\le\rho\le 2\}$$

$$V=\int_0^{2\pi}\int_0^{\pi/2}\int_{2\cos(\phi)}^{2}\rho^2\sin(\phi)\,d\rho\,d\phi\,d\theta.$$

$$V=\int_0^{2\pi}d\theta\int_0^{\pi/2}\sin\phi d\phi\int_{2\cos\phi}^{2}\rho^2 d\rho$$

$$\int_0^{\pi/2}\sin\phi d\phi\int_{2\cos\phi}^{2}\rho^2 d\rho=\int_0^{\pi/2}\sin\phi\left(\frac{2^3}{3}-\frac{(2\cos\phi)^3}{3}\right)d\phi$$

$$=\frac{2^3}{3}\int_0^{\pi/2}\sin\phi\left(1-\cos^3\phi\right)d\phi\qquad\left(u=\cos\phi,\ \ du=-\sin\phi d\phi\right)$$

$$=-\frac{2^3}{3}\int_1^0\left(1-u^3\right)du\ \ =\frac{2^3}{3}\frac{3}{4}=2$$

$$V=\int_0^{2\pi}2d\theta=4\pi$$

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IV. Applications of differential calculus

4.1 Taylor series of functions of several variables (Adams 12.9)

4.2 Extreme values of functions of several variables (Adams 13.1)

4.3 Constrained extrema and Lagrange multipliers (Adams 13.3)

This chapter is for 4 lectures for Week 5.

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4.1 Taylor series of functions of several variables 7687 3754

The Taylor series has several important applications, including:

- Analyzing the local behaviour of functions (such as finding local minima and maxima)
- Numerically solving partial differential equations

In most cases, we only utilize the first few terms of the Taylor series for practical applications.

Taylor series for one variable functions

A sufficinetly smooth function *f*(*x*) near a point $x = a$ can be approximated by a polynomial in terms of *x*-*a*:

$$f(x) = f(a) + f^{(1)}(a)(x-a) + \frac{f^{(2)}(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n + \cdots$$

The Taylor series has the same value and various orders of derivatives as the function at $x = a$.

The approximation improves with the order of expansion and the closeness of *x* to *a*.

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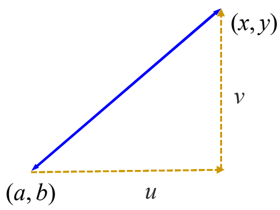

For a sufficiently smooth function *f*(*x, y*) near (*a*, *b*), denote

$$f(x, y) = f(a+u, b+v)$$

where $u = x - a, \ v = y - b.$

Introduce the function $F(t)$ as follows:

$$F(t) = f(a+tu, b+tv), \quad t \in [0,1].$$

We treat *a*, *b*, *u* and *v* as parameters and *t* as the only variable. Expand *F*(*t*) in *t* at *t* = 0:

$$F(t) = F(0) + F^{(1)}(0)t + \frac{F^{(2)}(0)}{2!}t^2 + \cdots + \frac{F^{(n)}(0)}{n!}t^n + \cdots$$

Let $t = 1$, we have

$$F(1) = F(0) + F^{(1)}(0) + \frac{F^{(2)}(0)}{2!} + \cdots + \frac{F^{(n)}(0)}{n!} + \cdots$$

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$$F(t)=f(a+tu,b+tv)$$

$$F(1)=F(0)+F^{(1)}(0)+\frac{F^{(2)}(0)}{2!}+\cdots+\frac{F^{(n)}(0)}{n!}+\cdots$$

We calculate the terms in the equation:

$$F(1)=f(a+u,b+v),\quad F(0)=f(a,b)$$

$$\frac{dF(t)}{dt}=\frac{d}{dt}f(a+tu,b+tv)=f_x(a+tu,b+tv)u+f_y(a+tu,b+tv)v$$

$$F^{(1)}(0)=f_x(a,b)u+f_y(a,b)v$$

$$F^{(2)}(t)=\frac{d}{dt}\{f_x(a+tu,b+tv)u+f_y(a+tu,b+tv)v\}$$

$$=\left(f_{xx}(a+tu,b+tv)u+f_{xy}(\cdots)v\right)u+\left(f_{yx}(\cdots)u+f_{yy}(\cdots)v\right)v$$

$$F^{(2)}(0)=\left(f_{xx}(a,b)u+f_{xy}(a,b)v\right)u+\left(f_{yx}(a,b)u+f_{yy}(a,b)v\right)v$$

$$=f_{xx}(a,b)u^2+2f_{xy}(a,b)uv+f_{yy}(a,b)v^2$$

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Taylor series for two variable functions (cont.)

The Taylor series for a sufficiently smooth function $f(x, y)$ at point (a, b) takes the form

$$f(x,y)=f(a,b)+f_x(a,b)(x-a)+f_y(a,b)(y-b)+$$

(Linear approximation)

$$+\frac{1}{2}\begin{bmatrix} f_{xx}(a,b)(x-a)^2+2f_{xy}(a,b)(x-a)(y-b) \\ +f_{yy}(a,b)(y-b)^2 \end{bmatrix}+\cdots.$$

(Quadratic approximation)

The first three terms are the linear approximation of the function.

The quadratic approximation has the same value, 1st and 2nd order partial derivatives as the function at the point considered.

As $a=b=0$, we have

$$f(x,y)=f(0,0)+xf_x(0,0)+yf_y(0,0)+$$

$$+\frac{1}{2}\left[x^2f_{xx}(0,0)+2xyf_{xy}(0,0)+y^2f_{yy}(0,0)\right]+\cdots.$$

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Taylor series for two variable functions

Theorem The Taylor series for a sufficiently smooth function $f(x, y)$ at point (a, b) is

$$f(x,y)=\sum_{l=0}^{\infty}\sum_{n=0}^{l}p_{ln}(x-a)^{l-n}(y-b)^{n},\quad p_{ln}=\frac{1}{(l-n)!n!}\frac{\partial^{l} f(a,b)}{\partial x^{l-n}\partial y^{n}}$$

Notice the convention is used

$$\frac{\partial^{0} f(a,b)}{\partial x^{0}\partial y^{0}}=f(a,b),\quad \frac{\partial^{1} f(a,b)}{\partial x^{1}\partial y^{0}}=\frac{\partial f(a,b)}{\partial x}.$$

$$f(x,y)=f(a,b)+f_x(a,b)(x-a)+f_y(a,b)(y-b)+$$

(Linear approximation)

$$+\frac{1}{2!}\left[f_{xx}(a,b)(x-a)^2+2f_{xy}(a,b)(x-a)(y-b)+f_{yy}(a,b)(y-b)^2\right]+\cdots$$

(Quadratic approximation)

$$+\frac{1}{3!}\begin{bmatrix} f_{xxx}(a,b)(x-a)^3+3f_{xxy}(a,b)(x-a)^2(y-b) \\ 3f_{xyy}(a,b)(x-a)(y-b)^2+f_{yyy}(a,b)(y-b)^3\end{bmatrix}+\cdots.$$

(Cubic approximation)

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Taylor series for three variable functions

The Taylor series for a sufficiently smooth function $f(x, y, z)$ at point $\boldsymbol{a}=(a,b,c)$ takes the form

$$f(x,y,z)=f(\boldsymbol{a})+f_x(\boldsymbol{a})(x-a)+f_y(\boldsymbol{a})(y-b)+f_z(\boldsymbol{a})(z-c)+$$

(Linear approximation)

$$+\frac{1}{2}\begin{bmatrix} f_{xx}(\boldsymbol{a})(x-a)^2+f_{yy}(\boldsymbol{a})(y-b)^2+f_{zz}(\boldsymbol{a})(z-c)^2 \\ 2f_{xy}(\boldsymbol{a})(x-a)(y-b)+ \\ 2f_{yz}(\boldsymbol{a})(y-b)(z-c)+ \\ 2f_{zx}(\boldsymbol{a})(x-a)(z-c)+\end{bmatrix}+\cdots.$$

(Quadratic approximation)

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Example 1

Find Taylor series for

$f(x, y) = \sin x \sin y$

near the origin up to quadratic terms.

Solution. $f(x,y) = f(0,0) + xf_x(0,0) + yf_y(0,0)$

$+\frac{1}{2}\left(x^2 f_{xx}(0,0) + 2xyf_{xy}(0,0) + y^2 f_{yy}(0,0)\right) + \cdots$

with $f(0,0) = \sin x \sin y|_{(0,0)} = 0,\quad f_x(0,0) = \cos x \sin y|_{(0,0)} = 0,$

$f_y(0,0) = \sin x \cos y|_{(0,0)} = 0,\quad f_{xx}(0,0) = -\sin x \sin y|_{(0,0)} = 0,$

$f_{xy}(0,0) = \cos x \cos y|_{(0,0)} = 1,\quad f_{yy}(0,0) = -\sin x \sin y|_{(0,0)} = 0,$

we have

$\sin x \sin y = xy + \cdots$

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Example 2

Find Taylor series for $f(x, y)=e^x\sin(y)$ about $(1, \pi/2)$ up to cubic terms.

Solution. We should use known Taylor series to calculate the Taylor expansions wherever possible and only calculate the coefficients directly using the Taylor series formula as a last resort.

To expand about $(1, \pi/2)$ we want an expression in $(x-1)$ and $(y- \pi/2)$. Denoting $u= x-1$, $v= y-\pi/2$, the expression will be in u and v.

$$f(x,y) = e^{u+1}\cos v = e\, e^u \cos v$$

To the cubic terms, we have

$$e^u = 1 + u + \frac{1}{2}u^2 + \frac{1}{6}u^3 + \cdots,$$

$$\cos v = 1 - \frac{1}{2}v^2 + \cdots,$$

$$f(x,y) = e\left(1 + u + \frac{1}{2}u^2 + \frac{1}{6}u^3 + \cdots\right)\left(1 - \frac{1}{2}v^2 + \cdots\right)$$

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Example 2 (cont.)

$$f(x,y)=e\left(1+u+\frac{1}{2}u^2+\frac{1}{6}u^3+\cdots\right)\left(1-\frac{1}{2}v^2+\cdots\right)$$

$$=e\begin{pmatrix}1+u+\frac{1}{2}u^2+\frac{1}{6}u^3+\cdots\\ -\frac{1}{2}v^2-\frac{1}{2}uv^2+\cdots\end{pmatrix}$$

$$=e\left(1+u+\frac{1}{2}u^2-\frac{1}{2}v^2+\frac{1}{6}u^3-\frac{1}{2}uv^2+\cdots\right)$$

We treat u and v as the same order small quantities, thus u^2v and uv^2 are cubic terms.

Express the result in (x, y):

$$f(x,y)=e\begin{pmatrix}1+(x-1)+\frac{1}{2}(x-1)^2-\frac{1}{2}(y-\pi/2)^2\\ +\frac{1}{6}(x-1)^3-\frac{1}{2}(x-1)(y-\pi/2)^2+\cdots\end{pmatrix}$$

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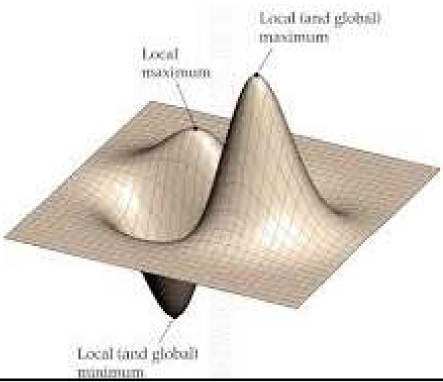
4.2 Extreme values of functions of several variables

Definition: Let $f: D\subseteq\mathbb{R}^n\to\mathbb{R}$. A point $\boldsymbol{a}\in D$ is said to be:

- A *local maximum* if $f(\boldsymbol{x})\le f(\boldsymbol{a})$ for all points $\boldsymbol{x}$ sufficiently close to $\boldsymbol{a}$;
- A *local minimum* if $f(\boldsymbol{x})\ge f(\boldsymbol{a})$ for all points $\boldsymbol{x}$ sufficiently close to $\boldsymbol{a}$;
- A *global maximum* if $f(\boldsymbol{x})\le f(\boldsymbol{a})$ for all points $\boldsymbol{x}\in D$;
- A *global minimum* if $f(\boldsymbol{x})\ge f(\boldsymbol{a})$ for all points $\boldsymbol{x}\in D$.
- *Extremum* means either maximum or minimum.



All points $\boldsymbol{x}$ sufficiently close to $\boldsymbol{a}$

For two variables, denote

$\mathbf{a}=(a, b)$ and $\mathbf{x}=(x, y)$.

$u=x-a$ and $v=y-b$.

All points $\boldsymbol{x}$ (excluding $\boldsymbol{a}$ itself) being sufficiently close to $\boldsymbol{a}\Leftrightarrow$ $|u|\ll1$, $|v|\ll1$ and not both being zero, but otherwise arbitrary.

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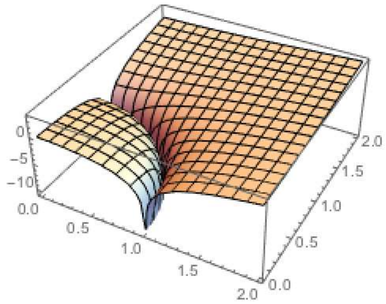
Extreme value theorem

Fact: If f is a continuous function of n variables on a closed, bounded domain of $\mathbb{R}^n$ then f is bounded and attains its global maximum and minimum.

A closed domain includes its boundary.

A bounded domain: the distance from any point in the domain to the origin is finite.



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Stationary, singular and saddle points

Definition: Let $f: D \subseteq \mathbb{R}^n \to \mathbb{R}$. The point $\boldsymbol{a} \in D$ is said to be

- a stationary or critical point if $\nabla f(\boldsymbol{a}) = 0$;
- a singular point if ∇f does not exist at $\boldsymbol{a}$;
- a saddle point if it is a stationary point but not a local extremum. Similar concept for single function is a reflection point.

Example 1. Show that the tangent plane at the stationary point (a, b) of a surface $z = f(x, y)$ is a horizontal plane.

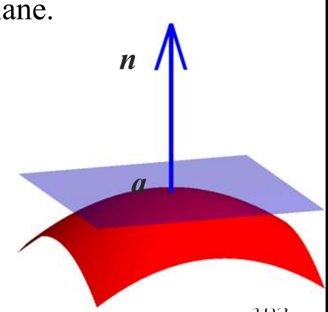
Denote the surface as a level surface:

$$F(x, y, z) = z - f(x, y) = 0.$$

$$\nabla F = (F_x, F_y, F_z) = (-f_x(x,y), -f_y(x,y), 1)$$

The normal vector for the surface at (a,b) is

$$\boldsymbol{n} = (-f_x(a,b), -f_y(a,b), 1) = (0,0,1)$$



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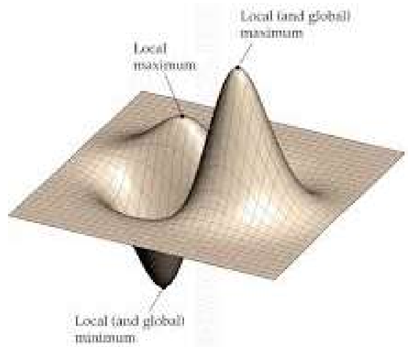
Necessary conditions for extrema

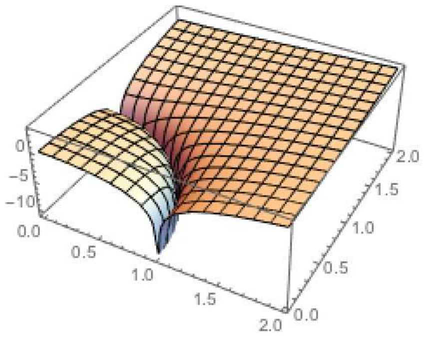
Theorem: Let $f : D \subseteq \mathbb{R}^n \to \mathbb{R}$. If f has a local extremum at the point $\boldsymbol{a} \in D$, then $\boldsymbol{a}$ must be:

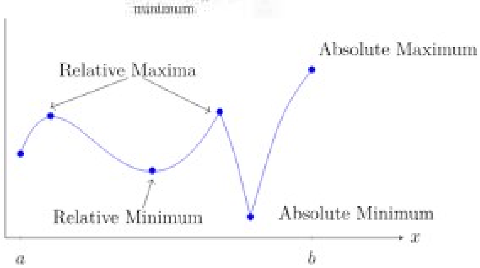
(1) a stationary point of f,

(2) a singular point of f, or

(3) a boundary point of D.







Necessary conditions for extrema (cont.)

A local extremum $\boldsymbol{a}$ must be:

(1) a stationary point of f, (2) a singular point of f, or

(3) a boundary point of D.

Proof: Examples showed that a local extremum $\boldsymbol{a}$ may be a boundary point, a singular point or a stationary point.

We need to show there are no other possibilities. For this purpose, we will show for non-singular interior point $\boldsymbol{a}$, if it is a local extremum, it must be a ***stationary*** point.

We will show this by contradiction: Assume that the extremum $\boldsymbol{a}$ is an interior, non-singular point but is not a stationary point, i.e., $\nabla f(\boldsymbol{a}) \neq 0$.

f increases in the direction of $\nabla f(\boldsymbol{a})$ and decreases in the direction of $-\nabla f(\boldsymbol{a})$ near $\boldsymbol{a}$ in the domain as $\boldsymbol{a}$ is an interior point.

$\boldsymbol{a}$ is thus neither a maximum nor a minimum. This contradicts with the assumption that $\boldsymbol{a}$ is a local extremum.

Second order derivative test for single variable functions

Suppose $f(x)$ is sufficient smooth and $f'(a) = 0$.
As x is near a or $|u| =| x-a |<<1$, we have

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots$$
$$= f(a) + \frac{f''(a)}{2!}u^2 + \cdots.$$

Therefore $\quad f(x) - f(a) = \frac{f''(a)}{2!}u^2 + \cdots.$

1. If $f''(a) > 0 \Rightarrow f(x) - f(a) > 0$

$\Leftrightarrow \quad f(x) > f(a)$ for any x near a.

$\Leftrightarrow \quad x = a$ is a local minimum.

2. If $f''(a) < 0 \Rightarrow f(x) - f(a) < 0$

$\Leftrightarrow \quad f(x) < f(a)$ for any x near a.

$\Leftrightarrow \quad x = a$ is a local maximum.

Hessian matrix and leading minors

Definition: Suppose $f(x, y)$ is a sufficiently smooth function then the Hessian matrix of f at (a, b) is defined as

$$H = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix},$$

where the derivatives are evaluated at (a, b).

Its leading minors are defined as follows:

$$\det H_1 \equiv \det(f_{xx}) = f_{xx},$$

$$\det H = \det\begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix} = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix} = f_{xx}f_{yy} - f_{xy}f_{yx} = f_{xx}f_{yy} - f_{xy}^2.$$

Leading minor test

Suppose $f(x, y)$ has a stationary point at (a, b), i.e.

$$f_x(a, b) = f_y(a, b) = 0,$$

For (x, y) near (a, b), we express $f(x, y)$ using Taylor series

$$f(x,y) = f(a,b) + uf_x(a,b) + vf_y(a,b)$$
$$+\frac{1}{2}\left(u^2 f_{xx}(a,b) + 2uvf_{xy}(a,b) + v^2 f_{yy}(a,b)\right) + \cdots,$$

where $u = x - a$ and $v = y - b$. As (x, y) is near (a, b), i.e. $|u| << 1$, $|v| << 1$, the higher order terms are neglected.

As $f_x(a, b) = f_y(a, b) = 0$,

$$f(x,y) = f(a,b) + \frac{1}{2}\left(u^2 f_{xx}(a,b) + 2uvf_{xy}(a,b) + v^2 f_{yy}(a,b)\right) + \cdots$$

$$\Leftrightarrow \quad \Delta f = f(x,y) - f(a,b) = \frac{1}{2}\left(u^2 f_{xx}(a,b) + 2uvf_{xy}(a,b) + v^2 f_{yy}(a,b)\right) + \cdots$$

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Leading minor test (cont.)

Denote $A = f_{xx}(a,b), B = f_{xy}(a,b), C = f_{yy}(a,b)$

$$\Delta f = f(x,y) - f(a,b)$$
$$= \frac{1}{2}\left(u^2 f_{xx}(a,b) + 2uvf_{xy}(a,b) + v^2 f_{yy}(a,b)\right) + \cdots$$
$$= \frac{1}{2}\left(Au^2 + 2Buv + Cv^2\right) + \cdots$$

$$\Delta f \approx \frac{1}{2}\left(Au^2 + 2Buv + Cv^2\right) = \frac{A}{2}\left(u^2 + 2\frac{B}{A}uv + \frac{C}{A}v^2\right)$$
$$= \frac{A}{2}\left(u^2 + 2\frac{B}{A}uv + \frac{B^2}{A^2}v^2 + \frac{C}{A}v^2 - \frac{B^2}{A^2}v^2\right)$$
$$= \frac{A}{2}\left(\left(u + \frac{B}{A}v\right)^2 + \frac{AC - B^2}{A^2}v^2\right)$$
$$= \frac{1}{2}f_{xx}(a,b)\left[\left(u + \frac{f_{xy}(a,b)}{f_{xx}(a,b)}v\right)^2 + \frac{\det(H)}{f_{xx}^2(a,b)}v^2\right]$$

As $\quad AC - B^2 = f_{xx}(a,b)f_{yy}(a,b) - f_{xy}^2(a,b) = \det(H).$

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Leading minor test (cont.)

For arbitrary u and v satisfying $|u|<<1$, $|v|<<1$ but not both being 0:

$$\Delta f = f(x,y) - f(a,b) = \frac{1}{2} f_{xx}(a,b) G + \text{Remainder},$$

where $$G = \left(u + \frac{f_{xy}(a,b)}{f_{xx}(a,b)} v \right)^2 + v^2 \frac{\det(H)}{f_{xx}^2(a,b)}.$$

(1) If $\det(H) > 0$, we have $G > 0$:

when $f_{xx} > 0 \Rightarrow \Delta f > 0 \Rightarrow f(x, y) > f(a, b)$:
(a, b) is a local minimum;

when $f_{xx} < 0, \Rightarrow \Delta f < 0 \Rightarrow f(x, y) < f(a, b)$:
(a, b) is a local maximum.

(2) If $\det(H) < 0$:

G is positive as $v = 0$ ($\therefore u \neq 0$) .

G is negative as $v \neq 0$ and $u + f_{xy}(a, b)/f_{xx}(a, b)v = 0$.

Δf takes different signs in the above two cases, so (a, b) is thus a saddle point.

(3) If $\det(H) = 0$, then further investigation is required.

Leading minor test

Theorem: Suppose $f(x, y)$ *is a* sufficiently smooth function with a stationary point at (a, b), i.e.

$$f_x(a, b) = f_y(a, b) = 0,$$

and the Hessian matrix H of f at (a, b) is denoted as follows

$$H = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix}, \qquad \det(H) = f_{xx}f_{yy} - (f_{xy})^2 ,$$

where the partial derivatives in H are evaluated at (a, b).

(1) If $\det(H) > 0$,

(a, b) is a local maximum if $\det(H_1) = f_{xx}(a, b) < 0$,

(a, b) is a local minimum if $\det(H_1) = f_{xx}(a, b) > 0$.

(2) If $\det(H) < 0$, it is a saddle point.

(3) If $\det(H) = 0$, then further investigation is required (after which the point (a, b) could turn into one of the above types).

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Example 2

Find & classify the stationary points of the function, $f(x,y)=x^2-y^2$

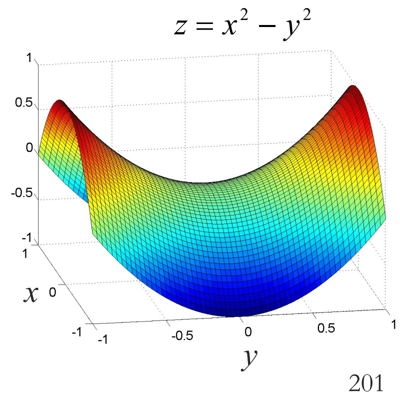
Solution. Find the stationary points:

$$f_x=2x,\quad f_x=0\ \Rightarrow\ x=0$$
$$f_y=-2y,\quad f_y=0\ \Rightarrow\ y=0$$

The stationary point is at (0, 0). The Hessian matrix is

$$H=\begin{pmatrix} f_{xx} & f_{xy}\\ f_{yx} & f_{yy}\end{pmatrix}=\begin{pmatrix} 2 & 0\\ 0 & -2\end{pmatrix},$$

As det(H) = -4 < 0, it is a saddle point.



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Example 3

Find and classify the stationary points of the following function:

$f(x, y) = 2x^3 - 6xy + 3y^2$.

Soln. Find the stationary points

$$f_x=6x^2-6y,\quad f_x=0\ \Rightarrow\ x^2=y$$
$$f_y=-6x+6y,\quad f_y=0\ \Rightarrow\ x=y$$

Two stationary points at (0, 0) and (1, 1).

The Hessian matrix is $H=\begin{pmatrix} f_{xx} & f_{xy}\\ f_{yx} & f_{yy}\end{pmatrix}=\begin{pmatrix} 12x & -6\\ -6 & 6\end{pmatrix},$

At (0, 0) the Hessian matrix is

$$H=\begin{pmatrix} 0 & -6\\ -6 & 6\end{pmatrix},$$

det(H) =-36< 0 it is a saddle point.

At (1, 1), the Hessian matrix is

$$H=\begin{pmatrix} 12 & -6\\ -6 & 6\end{pmatrix},$$

det(H) =36>0, det(H_1) = 12> 0, a local minimum.

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Leading minor test for functions of 3 variables

If $f(x, y, z)$ a sufficiently smooth function at the point (a, b, c), the Hessian matrix H of f at (a, b, c) is

$$H = \left.\begin{pmatrix} f_{xx} & f_{xy} & f_{xz} \\ f_{yx} & f_{yy} & f_{yz} \\ f_{zx} & f_{zy} & f_{zz} \end{pmatrix}\right|_{(a,b,c)},$$

Its leading minors are given as follows:

$$\det H_1 = \det(f_{xx}) = f_{xx}, \quad \det H_2 = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix}, \quad \det H = \begin{vmatrix} f_{xx} & f_{xy} & f_{xz} \\ f_{yx} & f_{yy} & f_{yz} \\ f_{zx} & f_{zy} & f_{zz} \end{vmatrix}.$$

For a function $f(x, y, z)$, a stationary point (a, b, c) is:

(1) a local maximum if $\det(H_1) < 0$, $\det(H_2) > 0$ and $\det(H_3) < 0$;

(2) a local minimum if $\det(H_1) > 0$, $\det(H_2) > 0$ and $\det(H_3) > 0$;

(3) a saddle point if none of the above conditions hold.

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Example 4

Find and classify the stationary points of the following function:

$$f(x, y, z) = x^2 + 2y^2 + 3z^2 + 2xy + 2xz + 3.$$

Solution. Find the stationary points

(1) $f_x = 2x + 2y + 2z = 0$

(2) $f_y = 4y + 2x = 0 \quad \Rightarrow \quad y = -x/2$

(3) $f_z = 6z + 2x = 0 \quad \Rightarrow \quad z = -x/3$

Using (2) & (3) to eliminate y and z from (1), we see that

$$x = 0 \quad \Rightarrow \quad y = z = 0.$$

Hence we have a stationary point (0, 0, 0).

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Example 4 (cont.)

Since $f_x = 2x + 2y + 2z$, $f_y = 2x+4y$, $f_z = 2x + 6z$, the Hessian matrix

$$H = \begin{pmatrix} f_{xx} & f_{xy} & f_{xz} \\ f_{yx} & f_{yy} & f_{yz} \\ f_{zx} & f_{zy} & f_{zz} \end{pmatrix} = \begin{pmatrix} 2 & 2 & 2 \\ 2 & 4 & 0 \\ 2 & 0 & 6 \end{pmatrix},$$

$$\text{As } \det H = \det \begin{pmatrix} 2 & 2 & 2 \\ 2 & 4 & 0 \\ 2 & 0 & 6 \end{pmatrix} = 8 > 0,$$

and $\det H_1 = 2 > 0$ and $\det H_2 = \det \begin{pmatrix} 2 & 2 \\ 2 & 4 \end{pmatrix} = 4 > 0$. It is a local minimum.

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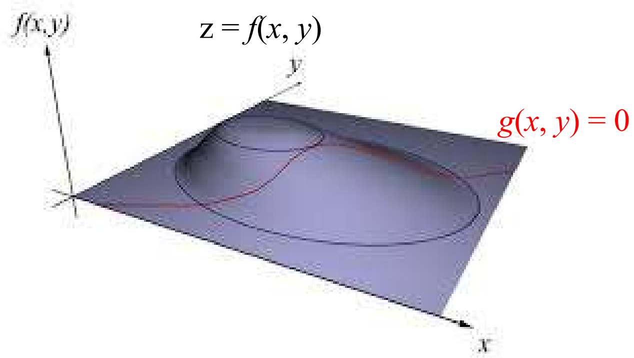
4.3 Constrained extrema and Lagrange multipliers

To find the extrema of a function $z = f(x, y)$

subject to a constraint $g(x, y) = 0$.

Here $f(x, y)$ is called the objective function and $g(x, y) = 0$ is the constraint. The whole problem is called a constrained extreme problem.



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Constrained extrema and Lagrange multipliers

To find the extrema of a function $z = f(x, y)$

subject to a constraint $g(x, y) = 0$.

Here $f(x, y)$ is called the objective function and $g(x, y) = 0$ is the constraint. The whole problem is called a constrained extremum problem.

Introduce the Lagrange function:

$$L(x, y, \lambda) = f(x, y) + \lambda\, g(x, y),$$

where λ is the Lagrange multiplier.

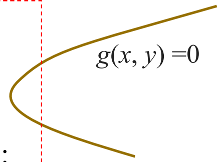
The stationary equations for the constrained problem are:

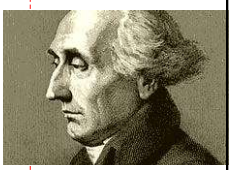
$$\left.\begin{aligned} \frac{\partial L}{\partial x} &= \frac{\partial f}{\partial x} + \lambda\frac{\partial g}{\partial x} = 0, \\ \frac{\partial L}{\partial y} &= \frac{\partial f}{\partial y} + \lambda\frac{\partial g}{\partial y} = 0, \end{aligned}\right\} \Leftrightarrow \quad \nabla L(x, y, \lambda) = 0,$$

combined with the constraint

$$g(x, y) = 0.$$

Here we assume that at the stationary points $\nabla g(x, y) \neq 0$.





Joseph-Louis Lagrange (1736–1813)

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Proof of Lagrange method

For the curve $g(x, y) = 0$, denote its tangent vector as $\mathbf{t}$ and its normal vector as $\mathbf{n}$. The normal $\mathbf{n}$ of the curve $g(x, y) = 0$ is given by

$$\mathbf{n} = \nabla g, \quad \text{provided that } \nabla g \neq 0 \text{ at the point considered.}$$

Since $\mathbf{n} \perp \mathbf{t}$, we know: $\nabla g \perp \mathbf{t}$.

The directional derivative of $f(x, y)$ along tangent $\mathbf{t}$ is $D_t f = \mathbf{t} \cdot \nabla f$.

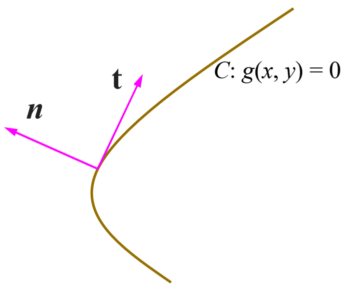
At the stationary point, $D_t f = 0$, thus

$$D_t f = \mathbf{t} \cdot \nabla f = 0 \quad \Rightarrow \quad \nabla f \perp \mathbf{t}.$$

As both ∇g and ∇f are perpendicular to $\mathbf{t}$, they are parallel, thus

$$\Rightarrow \quad \nabla f = -\lambda \nabla g,$$

or $\nabla(f + \lambda g) = 0$.



Example 1

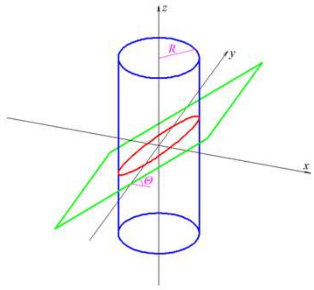
Find the maximum/minimum values for $f(x, y) = 3x+y$

subject to the constraint $g(x, y) = x^2 + y^2 - 10 = 0$

Solution. The Lagrange function is

$L = f + \lambda g = 3x+y + \lambda(x^2 + y^2 - 10)$.

The stationary equations are



$$\frac{\partial L}{\partial x} = 3 + 2\lambda x = 0, \quad \Rightarrow \quad x = -\frac{3}{2\lambda}$$

$$\frac{\partial L}{\partial y} = 1 + 2\lambda y = 0, \quad \Rightarrow \quad y = -\frac{1}{2\lambda}$$

$$x^2 + y^2 = 10. \quad \Rightarrow \quad \left(-\frac{3}{2\lambda}\right)^2 + \left(-\frac{1}{2\lambda}\right)^2 = 10 \quad \Rightarrow \quad 2\lambda = \pm 1$$

Thus there are two critical points: (3, 1), (-3, -1). The function at the points: $f(3, 1) = 10$ and $f(-3, -1) = -10$.

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Example 1 (cont.)

$$\nabla g(\pm 3, \pm 1) = (\pm 6, \pm 2) \neq 0$$

The function $f = 3x+y$ is continuous and defined on the circle $x^2+y^2 = 10$, a closed bounded set. Therefore, the function f is bounded and attains its maximum and minimum values.

Since there are no singular points, f must achieve its extrema at the two stationary points.

The maximum and minimum are $f(3, 1) = 10$ and $f(-3, -1) = -10$, respectively.

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Lagrange multiplier for functions with 3 variables

To find the extrema of a function $f(x, y, z)$
subject to a constraint $g(x, y, z) = 0$.

Introduce the Lagrange function

$$L(x, y, z, \lambda) = f(x, y, z) + \lambda g(x, y, z) .$$

Its stationary equations are as follows:

$$\left.\begin{array}{l} L_x = f_x + \lambda g_x = 0 \\ L_y = f_y + \lambda g_y = 0 \\ L_z = f_z + \lambda g_z = 0 \end{array}\right\} \Leftrightarrow \quad \nabla L(x,y,z,\lambda) = 0$$

$$g(x, y, z) = 0 .$$

Here we assume that at the stationary points

$$\nabla g(x, y, z) \neq 0.$$

Example 2

Using Lagrange multipliers, find the maximum and minimum distances from the point (2, 1, -2) to the spherical surface:

$$x^2+y^2+z^2 = 1.$$

Solution. The distance from (x, y, z) to (2, 1, -2) is given as

$$d = \sqrt{(x-2)^2 + (y-1)^2 + (z+2)^2}$$

d and d^2 attain extrema at the same points. We thus choose

$$f = d^2 = (x-2)^2 + (y-1)^2 + (z+2)^2$$

With the constraint $g(x, y, z) = x^2+y^2+z^2-1 = 0$.

Introduce the Lagrange function

$$L = f + \lambda g = (x-2)^2 + (y-1)^2 + (z+2)^2 + \lambda\left(x^2 + y^2 + z^2 - 1\right)$$

Example 2 (cont.) $L = f + \lambda g = (x-2)^2 + (y-1)^2 + (z+2)^2 + \lambda(x^2 + y^2 + z^2 - 1)$

The stationary equations are

$$L_x = 2(x-2) + 2\lambda x = 0 \;\Rightarrow\; (x-2) + \lambda x = 0 \;\Rightarrow\; x = 2/(1+\lambda)$$
$$L_y = 2(y-1) + 2\lambda y = 0 \;\Rightarrow\; (y-1) + \lambda y = 0 \;\Rightarrow\; y = 1/(1+\lambda)$$
$$L_z = 2(z+2) + 2\lambda z = 0 \;\Rightarrow\; (z+2) + \lambda z = 0 \;\Rightarrow\; z = -2/(1+\lambda)$$

As $x^2+y^2+z^2 = 1$, we have

$$\frac{4}{(1+\lambda)^2} + \frac{1}{(1+\lambda)^2} + \frac{4}{(1+\lambda)^2} = 1 \;\Rightarrow\; \frac{9}{(1+\lambda)^2} = 1 \;\Rightarrow\; \lambda_{1,2} = -4,\ 2$$

As $\lambda_1 = -4$, $(x_1, y_1, z_1) = (-2/3, -1/3, 2/3)$
As $\lambda_2 = 2$, $(x_2, y_2, z_2) = (2/3, 1/3, -2/3)$

$$\nabla g\left(\pm 3/2, \pm 1/3, \pm 2/3\right) = 2\left(\pm 3/2, \pm 1/3, \pm 2/3\right) \neq 0$$

$d_1 = \sqrt{(x_1-2)^2 + (y_1-1)^2 + (z_1+2)^2} = 4$ The maximum distance.
$d_2 = \sqrt{(x_2-2)^2 + (y_2-1)^2 + (z_2+2)^2} = 2$ The minimum distance.

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Comment

The function is continuous and defined on the sphere surface

$x^2+y^2+z^2 = 1$,

which is a closed bounded set. Therefore the function f is bounded and achieves its bounds.

Since there are neither singular points nor boundary points, f must achieve its extrema at the two stationary points.

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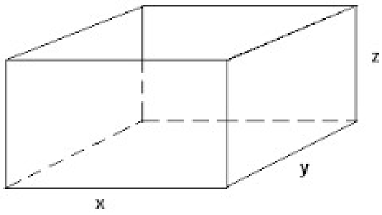
Example A1



We seek to construct a cardboard box of maximum volume, given a fixed area of cardboard.

Solution. Denoting the dimensions of the box by x, y, z, the problem can be expressed as maximize xyz subject to

$$xy + yz + xz = c/2,$$

where $c > 0$ is the given area of cardboard.

Consider the Lagrangian

$$L = xyz + \lambda(xy + yz + xz - c/2\ ).$$

The first-order necessary conditions are easily found to be

$$yz + \lambda(y + z) = 0, \qquad (1)$$
$$xz + \lambda(x + z) = 0, \qquad (2)$$
$$xy + \lambda(x + y) = 0, \qquad (3)$$

together with the original constraint.

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Example A1 (cont.)

$$yz + \lambda(y + z) = 0, \qquad (1)$$
$$xz + \lambda(x + z) = 0, \qquad (2)$$
$$xy + \lambda(x + y) = 0, \qquad (3)$$
$$xy + yz + xz = c/2. \qquad (4)$$

Before solving the above equations, we note that the sum of (1, 2,3) is

$$(xy + yz + xz) + 2\lambda(x + y + z) = 0,$$

which, under the original constraint, becomes

$$c/2 + 2\lambda(x + y + z) = 0.$$

Hence, it is clear that $\lambda \neq 0$. Neither of x, y, z can be zero since if any is zero, all must be so according to (1)–(3).

To solve the equations (1)–(3), multiply (1) by x and (2) by y, and then subtract the two to obtain $\lambda(x - y)z = 0$.

Operate similarly on the second and the third to obtain $\lambda(y - z)x = 0$. It follows that $x = y = z = (c/6)^{0.5}$ is the unique solution to the necessary conditions. The box must be a cube.

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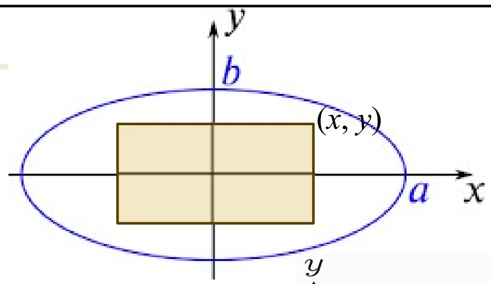
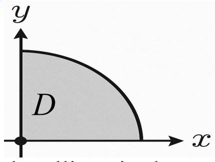
Example A2

Find the rectangle with the largest area that fits inside the ellipse $x^2/a^2 + y^2/b^2 = 1$, given that the rectangle is symmetric to the coordinate axes.





Soln. Let (x, y) be the corner in the first quadrant, then the rectangle has corners $(\pm x, \pm y)$ and its area A is

$$A(x, y) = 4xy, \ (x, y) \in D.$$

Here D is in the first quadrant: bounded by the x and y-axes and the ellipse in the first quadrant. $A(x, y)$ is continuous on a bounded and closed domain D.

It must attain its global extrema at (a) interior stationary points, (b) on the x- and y-axes, and/or (c) on the ellipse in the first quadrant.

(a) Interior stationary points (x, y) of D: the stationary equations are

$$A_x = 4y = 0, \quad A_y = 4x = 0,$$

In the inner of D, $x, y > 0$, thus there are no interior stationary points.

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Example A2 (cont.)

(b) At the x and y-axes, y or $x = 0$, $A = 0$, the minimum value of A.

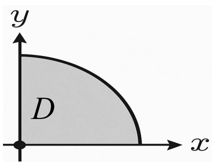
(c) The ellipse in the positive quarter satisfying the constraint:

$$x^2/a^2 + y^2/b^2 = 1$$

The Lagrange function is $L(x, y; \lambda) = 4xy + \lambda\left(\frac{x^2}{a^2} + \frac{y^2}{b^2} - 1\right)$

$$\frac{\partial L}{\partial x} = 4y + 2\lambda\frac{x}{a^2} = 0 \iff \frac{x}{a^2} = -\frac{2y}{\lambda} \iff \frac{x^2}{a^2} = -\frac{2xy}{\lambda}$$

$$\frac{\partial L}{\partial y} = 4x + 2\lambda\frac{y}{b^2} = 0 \iff \frac{y}{b^2} = -\frac{2x}{\lambda} \iff \frac{y^2}{b^2} = -\frac{2xy}{\lambda}$$



Thus: $\frac{x^2}{a^2} = \frac{y^2}{b^2}$.

Substituting this into the constrain eq. yields

$$\frac{x^2}{a^2} = \frac{y^2}{b^2} = \frac{1}{2} \Rightarrow \text{stationary point: } \left(\frac{a}{\sqrt{2}}, \frac{b}{\sqrt{2}}\right) \Rightarrow A = 2ab.$$

The maximum area is $2ab$.

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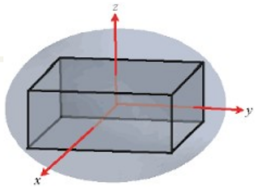
Example A3



Find the rectangular box with the largest volume that fits inside the ellipsoid $x^2/a^2 + y^2/b^2 + z^2/c^2 = 1$, given that the box is symmetric to the coordinate planes.

Soln. Let (x, y, z) be the corner in the positive octant, then the box has corners $(\pm x, \pm y, \pm z)$ and its volume V is

$$V(x, y, z) = 8xyz, \ (x, y, z) \in D.$$

Here D is in the positive octant bounded by the three coordinates planes and the ellipsoid surface in the positive octant. V is continuous on a bounded and closed domain D.

It must attain its global extrema at (a) the interior stationary points, (b) at the coordinate planes and/or (c) the ellipsoid surface in the positive octant.

(a) Interior stationary points (x, y, z) of D: the stationary equations are

$$V_x = 8yz = 0, \quad V_y = 8xz = 0, \quad V_z = 8xy = 0,$$

In the interior of D, $x, y, z > 0$, thus there are no interior stationary points.

Example A3 (cont.)

(b) At the coordinate planes, $x, y, z = 0$, $V = 0$, the minimum value of V.

(c) The ellipsoid surface in the positive octant satisfying the constrain:

$$x^2/a^2 + y^2/b^2 + z^2/c^2 = 1$$

The Lagrange function is $L(x, y, z; \lambda) = 8xyz + \lambda\left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1\right)$

$$\frac{\partial L}{\partial x} = 8yz + 2\lambda\frac{x}{a^2} = 0 \iff \frac{x}{a^2} = -\frac{4yz}{\lambda} \iff \frac{x^2}{a^2} = -\frac{4xyz}{\lambda}$$

$$\frac{\partial L}{\partial y} = 8zx + 2\lambda\frac{y}{b^2} = 0 \iff \frac{y}{b^2} = -\frac{4xz}{\lambda} \iff \frac{y^2}{b^2} = -\frac{4xyz}{\lambda}$$

$$\frac{\partial L}{\partial z} = 8xy + 2\lambda\frac{z}{c^2} = 0 \iff \frac{z}{c^2} = -\frac{4xy}{\lambda} \iff \frac{z^2}{c^2} = -\frac{4xyz}{\lambda}, \quad \text{Thus: } \frac{x^2}{a^2} = \frac{y^2}{b^2} = \frac{z^2}{c^2}.$$

Substituting this into the constrain eq. yields

$$\frac{x^2}{a^2} = \frac{y^2}{b^2} = \frac{z^2}{c^2} = \frac{1}{3} \Rightarrow \text{stationary point: } \left(\frac{a}{\sqrt{3}}, \frac{b}{\sqrt{3}}, \frac{c}{\sqrt{3}}\right) \Rightarrow V = \frac{8abc}{3\sqrt{3}}.$$

It is the maximum volume the box.

Example A4

Find Taylor series for $f(x, y)=1/(1-xy^2+2y)$ about $(1, 0)$ up to cubic terms.

Solution. $u=x-1$, $x = u+1$

$$f = \frac{1}{1-\left(xy^2-2y\right)} = \frac{1}{1-\left(\left(u+1\right)y^2-2y\right)} = \frac{1}{1-\left(-2y+y^2+uy^2\right)}$$

Let $t=-2y+y^2+uy^2$,

We have $f=\dfrac{1}{1-t}=1+t+t^2+t^3+\cdots$

$$t^2=\left(-2y+y^2+uy^2\right)\left(-2y+y^2+uy^2\right)$$

$$=4y^2-2y^3-2y^3+\cdots=4y^2-4y^3+\cdots$$

Here we neglecting all terms higher than cubic terms, such y^4, uy^3, etc., because we treat u and y as the same order small quantities.

Example A4 (cont.)

$$f=\frac{1}{1-t}=1+t+t^2+t^3+\cdots$$

$$t=-2y+y^2+uy^2,$$

$$t^2=4y^2-4y^3+\cdots$$

$$t^3=t\ t^2=\left(-2y+y^2+uy^2\right)\left(4y^2-4y^3+\cdots\right)=-8y^3+\cdots$$

$$f=1+t+t^2+t^3+\cdots$$

$$=1+\left(-2y+y^2+uy^2\right)+\left(4y^2-4y^3\right)-8y^3+\cdots$$

$$=1-2y+5y^2+uy^2-12y^3+\cdots$$

$$=1-2y+5y^2+\left(x-1\right)y^2-12y^3+\cdots$$